T-L IRRIGATION CO.

Hastings, Nebraska

GPS NAVIGATION LINEAR
With PLC III Control

Installation
Operation
Troubleshooting

CD90480

June 7, 2024
Corrected Fig. 27
PLC III Firmware = v267 on
Rover GPS Nav Bd Firmware = HL59 on

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<u>HL59 RELEASE NOTES (for Manual or PLC Control Linears):</u>

- Must “Restore Settings” if updating from any previous version prior to HL54.
- If PLC, for use in conjunction with PLC III firmware **v267**.

<u>HL59 Update Features (March 2023)</u>

- Added Rover Lat Long to $STA string being sent from Rover to PLC on Com 1. Not used by PLC. String can be used by 3rd party to access RTK location of Rover.

<u>HL58 Update Features (June 2022) **Use with PLC v264 or v265**</u>

- Firmware changes and additions to add compatibility for new Hemisphere P34 RTK Receiver Board when it is put into our production. P326 is now end of life but we still have inventory. Anticipate late 2022 we will change to the P34 per inventory and shipping volume. Firmware is backwards compatible to work with previous versions P300, P306 & P326.

<u>HL57 Update Features (April 2021) **Use with PLC v264 or v265**</u>

- If Manual Linear use - Fixed bug causing Aux Relay output to stay ON when switched to Stop.

<u>HL56 Update Features (Dec 2020) Use with PLC v264</u>

- Base operation – Correction Type default is now RTCM. Galileo = No so it is backwards compatible. Eliminated ROX option. This fixes an issue with P326 green boards with Base set to ROX and losing a fix if seeing more than 30 satellites. See Base Operation in the manual for more information.
- Base operation – fixed several issues
- Added operation that if Datalogging is on and power is cycled and then “Error Creating File” for more than 1 minute like it can’t find the USB when powered back up, then Datalogging is set to Stop and the Error message no longer appears.

<u>HL55 Update Features (June 2020) Use with PLC v263</u>

- Manual Control Linear Fix for Auxiliary Relay activation areas. HL55 is same as HL54 for PLC Linears.

<u>HL54 Update Features (April 2020) Use with PLC v263</u>

- Fixed Path issue with Interm. or Forward points not updating distance when changing Home location.
- NOTE: CURRENT DISTANCE IS NOW THE CENTER OF THE TRACTOR/TOWER BASE.
  - The “Current Distance” should not change now when changing direction.
  - **<u>Note that the Tractor will travel 12.6 ft farther at the field ends or Intermediate Points if using the Current Distance</u>** to stop/reverse at the Reverse or Home or Intermediate waypoints.
- NOTE: BASE OPERATION HAS BEEN REVISED. SEE CD90479 or CD90480 FOR INFORMATION.
- etc.

<u>PLC III - v267 RELEASE NOTES:</u>

- **v267** Firmware to be used for PLC III GPS Navigation Linears
- in conjunction **with** GPS Nav Board **HL58 or HL59**.
- Do not need to “Restore Settings” if updating from v263 or v264 or v265 or v266. But after programming, make sure to enter Distance settings and verify Path to use in 1=RUN SETUP before operating system.
- Must “Restore Settings” if updating from any version prior to v263.

<u>v267 Update Features (March 2023) Use with Rover Nav Board Firmware HL58 or HL59</u>

- Fixed issue in 1=Run Setup when set to Metric units so Rev AS/AR Distance minimum is 4 meters (14 ft)

<u>v266 Update Features (March 2023) Use with Rover Nav Board Firmware HL58 or HL59</u>

- Fixed issue in 1=Run Setup when set to Metric units so Rev AS/AR Distance can be set 4 to 11 meters.

<u>v265 Update Features (April 2021) Use with Rover Nav Board Firmware HL54 or HL55 or HL56 or HL57</u>

- Fixed incorrect upper and lower limits for fields in 2=GUID SETUP.

<u>v264 Update Features (Dec 2020) Use with Rover Nav Board Firmware HL54 or HL55 or HL56</u>

- Updates for changing Current Distance to be the center of the Tractor.

1. INTRODUCTION

Use this manual in conjunction:

Linear Hose Drag Installation Manual	or	Linear Ditch Feed Installation Manual
PLC II Operators Manual	or	Ultra G2 Operators Manual
Pivot Operators Manual.		

1.1. General Requirements & Guidelines

The following guidelines have been established from experience with T-L linear systems. These guidelines must be followed for satisfactory linear performance.

1. A ***STRAIGHT Guidance Path*** is the most important single item for successful operation. If intermediate waypoints are used, they must not placed so as to exceed the steering limitations of the linear system.

2. A ***Flat, Uniform, Dry, Hard Roadway*** for the tractor is necessary for proper guidance and traction. The roadway must be maintained free of wheel tracks or ruts and must be solid to minimize wheel slippage. For hose drag systems, the roadway must be maintained to minimize the dirt pulled in the hose loop.

3. ***Wheel tracks*** must be controlled on all towers. A dry pass followed by a light application on the first irrigation is required to establish a track base and reduce the possibility of deep rutting on subsequent irrigations. Wheel track depth should not exceed 4-6" and should be filled if they cause guidance shut downs. Special caution should be taken with pivoting linears to reduce wheel tracks.

4. ***Deep Furrows*** parallel to the tracks of the system MUST be avoided, as they can put a side load on the system and cause guidance and structural problems. *Row Crops* can be planted parallel to the tower tracks BUT the rows MUST be planted perfectly parallel to the wheel tracks. Planting perpendicular to wheel tracks may be done on shallow furrows.

5. The ***Maximum Side Slope is 3%*** (all linears except ULTRA) measured perpendicular to the direction of the travel.

6. The ***Maximum Slope of 5%*** (all linears except ULTRA) should not be exceeded along the travel path of the system.

7. **ULTRA Linear – Maximum Side Slope of 2%, Maximum Slope along Travel Path = 4%. AVOID COMBINATION SLOPES, especially on longer machines.**

8. ***Sudden or Extreme Terrain Changes*** should be avoided because they can cause excessive compression or tension and/or guidance irregularity.

9. ***Minimum Travel Speed*** on all T-L Linears is <u>8 Inches per Minute</u>.

10. Install ***Physical Barricade(s)*** for the lead tower(s) as needed to ensure the linear will not travel into an obstruction.

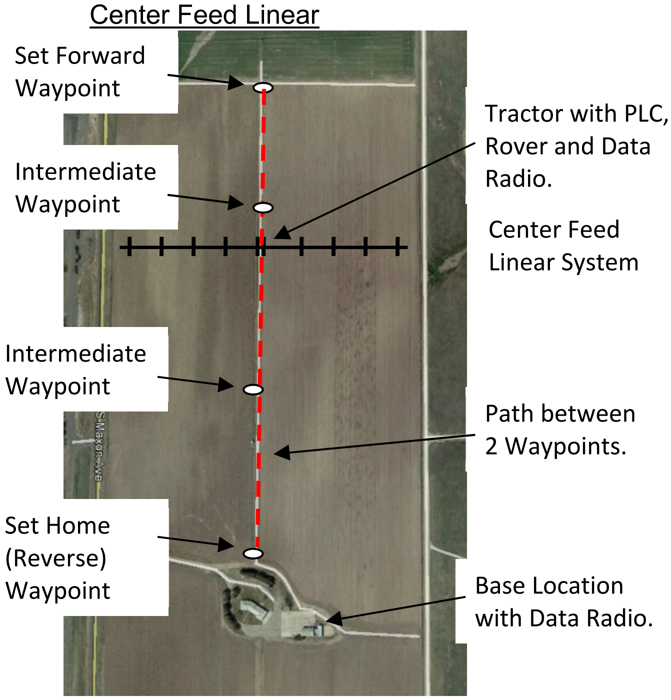
1.2. Initial Installation Steps

Step 1.	Be familiar with the Components and Overview	Sec. 1
Step 2.	Install Base	Sec. 2
Step 3.	Install Rover	Sec. 3
Step 4.	Wire Base & Rover	Sec. 4
Step 5.	Be familiar with the Navigation Board & USB	Sec. 5
Step 6 & 7.	Reprogram Firmware if / when needed	Sec. 6 & 7
Step 8.	Make sure Base is Operational	Sec. 8
Step 9.	Perform Initial Setup of PLC III and Rover	Sec. 9
Step 10.	Review Rover Operation	Sec. 10
Step 11.	PLC III Screens & Menus	Sec. 11
Step 12.	Review Steering Operation - STT	Sec. 12
Step 13.	Review Steering Operation – ULTRA G2	Sec. 13
Step 14.	Review Data Radio Operation	Sec. 14
Step 15&16.	Review Troubleshooting Sections	Sec. 15 & 16
Step 17&18	Record PLC & Base Information	Sec. 17 & 18

1.3. Overview of Operation

- Home (Reverse) Lat-Long End Point Set.
- Forward Lat-Long End Point Set.
- Up to 9 Intermediate Lat-Long Points can be Set.
- PLC & Rover Box at Tractor keep linear moving in straight line between 2 end points.
- Base Station sends corrections via Data Radio to Rover Box for sub-inch accuracy.

Center Feed Linear



End Feed Linear

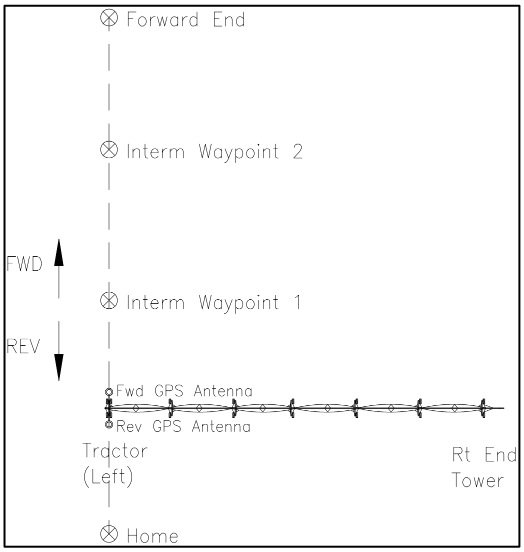


Figure 01. Overview of Operation.

1.4. Overview of Main Components

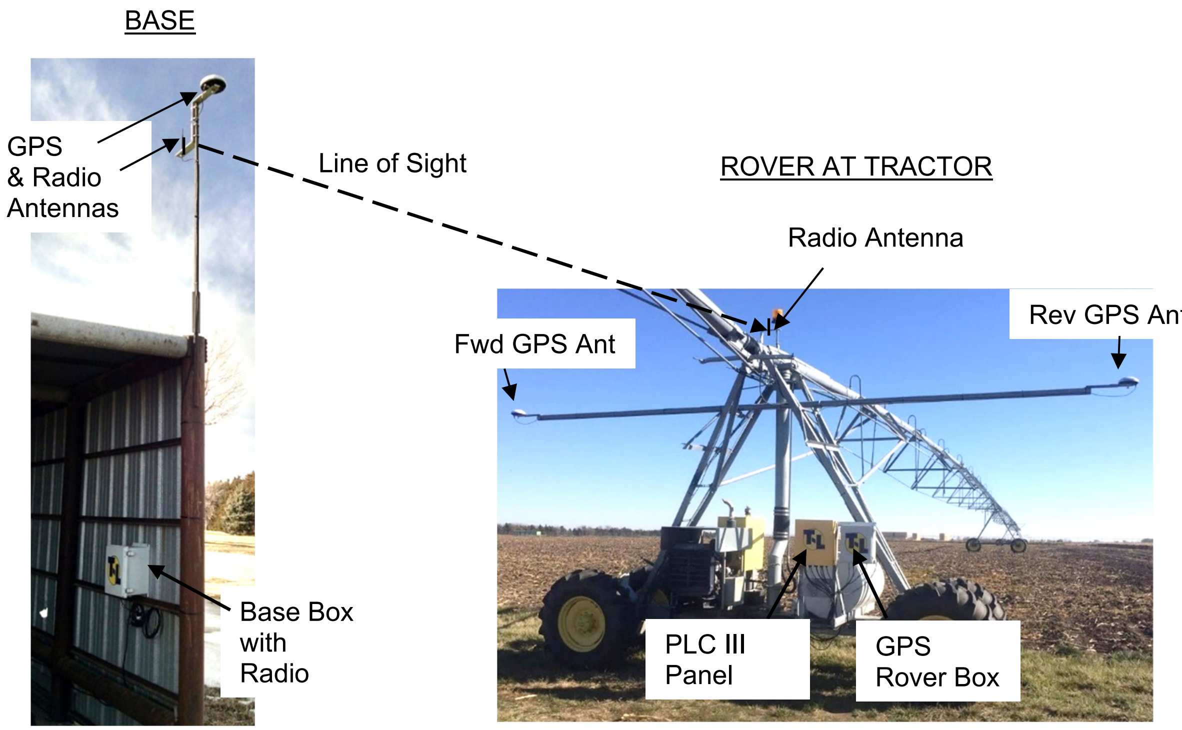


Figure 02. Overview of Main Components.

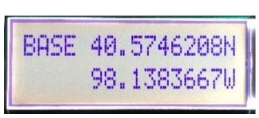
1.5. Record System Information Once Installed – PLC III & ROVER

A. Record PLC III Settings in PLC III SUMMARY SCREENS Section at the end of this manual for future reference.
B. Write down or at the Rover "Save" to USB the Path Lat-Long coordinates.

1.6. Record System Information Once Installed – GPS BASE

A. Record Base coordinates in GPS BASE INFORMATION Section at the end of this manual for future reference.
B. Record Base coordinates with a marker on the inside of the Door on the Base Box.
C. Record Base coordinates with a marker on laminated sheet in Base Box.

Write Base Coordinates Inside Base Door & on Laminated Sheet.



GPS Linear Navigation
Base Information & Wiring
CD91206 01/08/16

DDD.DDDDDDDD°

BASE Location _____.___________° N or S

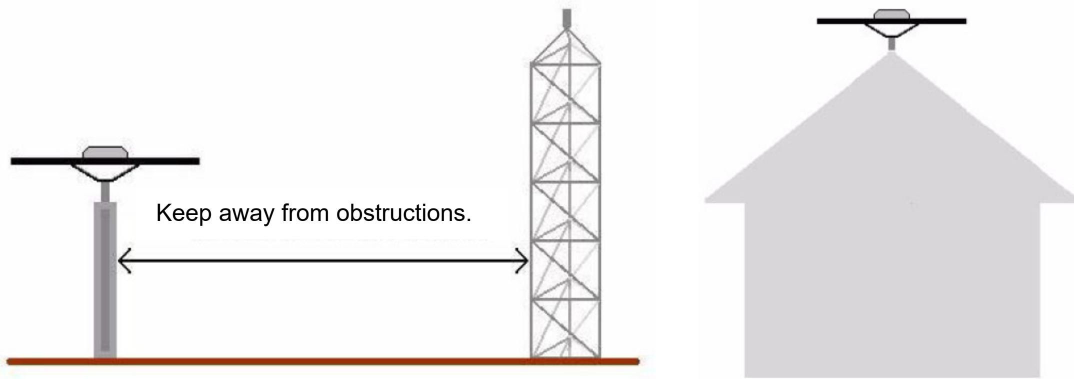
_____.___________° E or W

2. BASE INSTALLATION

2.1. Base Station GPS Antenna Position

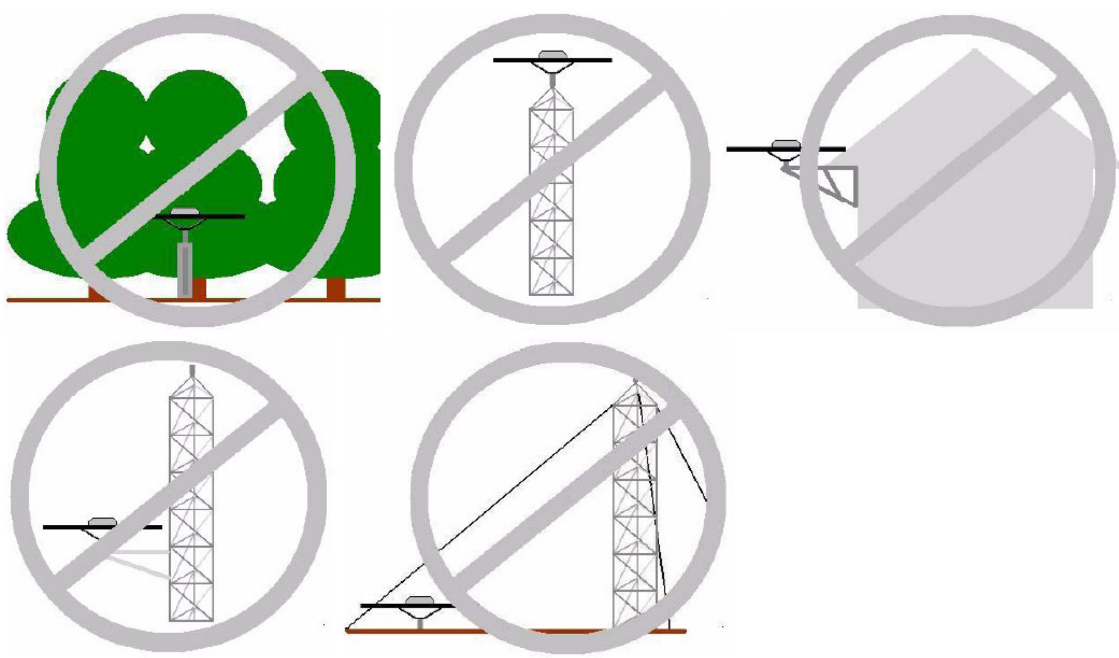
<u>Correct:</u>

- Utilize the 8m (26 ft) long GPS Coax Cable. That should get the GPS Antenna as high as needed.
- Mount on a sturdy, unmovable object.
- GPS Antenna must have a clear view of the sky.
- If there is an obstruction that cannot be avoided, mount on the south side if in the northern hemisphere and on the north side if in the southern hemisphere. Mount as far away from the obstruction as possible.



<u>Incorrect:</u>

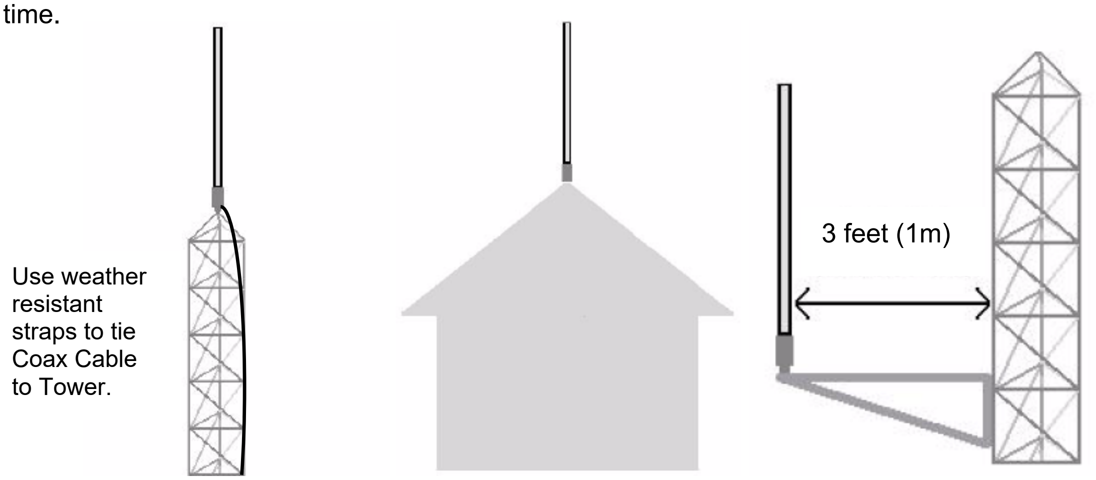
- Do not mount next to trees or large metal structures or towers.
- Do not mount on a tower or grain leg that can sway in the wind.



2.2. Base Station RADIO Antenna Position

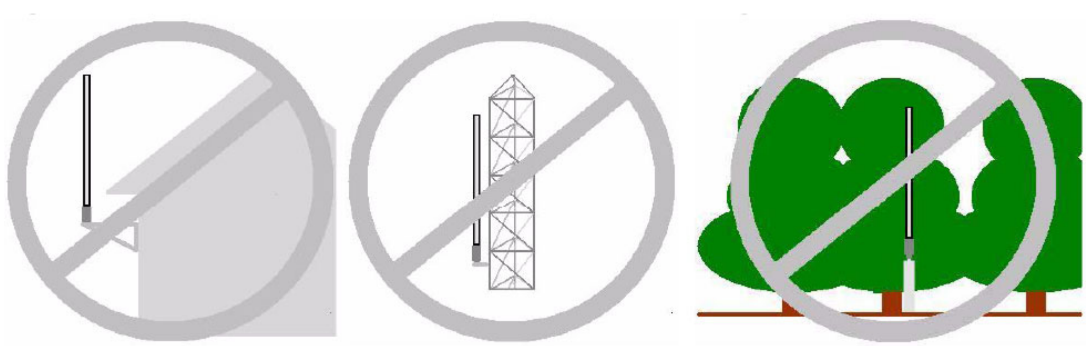
Correct:

- The Radio Antenna can move slightly.
- Mount at least 20 ft in the air. Mount higher if necessary to gain line of sight to target.
- Mounting on top of a radio tower or grain leg is acceptable, but the longer the Radio Coax Cable, the more signal loss you will have in the cable.
- If mounting on the side of a radio tower, keep the antenna at least 3 ft away on the side nearest the systems to be served.
- Use weather resistant straps to tie Radio Coax Cable to the structure so it will not come loose over time.



Incorrect:

- Do not mount next to buildings or right next to a tower or next to trees if the system to be served is on the opposite side.

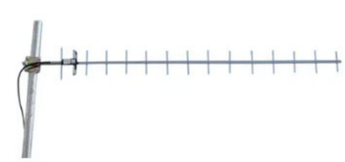


Omni Antenna:

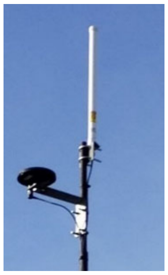
- An Omni antenna will broadcast or receive a 360 degree pattern.
- This is the standard antenna provided by T-L.

Directional (Yagi) Antenna

- A directional (Yagi) antenna will broadcast or receive with a 60 degree pattern.
- If the system(s) to be served is within the 60 degree pattern, a directional antenna could be used to increase distance.
- This could also be used on the Rover to point back to the Base.
- Contact T-L for assistance.



2.3. Base Information Overview

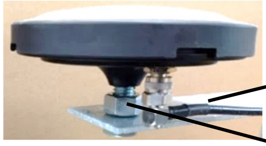


- Base required for RTK sub-inch accuracy, sends corrections to Rover via Data Radio.
- Dealer / Customer to field install the Base:
- It is IMPORTANT to install the GPS Antenna on a stationary mast **that will not move**.
- GPS Antenna should have clear view of entire sky, can be lower than Radio Antenna.
- **Radio Antenna to be located high enough for line of sight to Rover Antenna. Minimum height of the Base Radio Antenna is 20 ft (6m).**
- **Radio Antenna 12" <u>higher or lower</u> than GPS Antenna, NOT on the same plane.**
- One Base can serve several Rovers if within a couple mile range and Radio Antenna line of sight.
  - only way to be sure is to field test the range.
- Items included with the Base Station kit from T-L.
  - 120V/240AC to 12VDC Adapter.   MUST HAVE FULL TIME 120/240VAC POWER.
  - Optionally could be solar powered, contact T-L. (100 Watt Solar panel & 50 A-Hr Battery)
  - 26 ft (8m) of Radio Antenna Coax Cable.
  - 26 ft (8m) of GPS Antenna Coax Cable.
  - **Longer Coax Cables are available upon request at an additional charge**.

2.4. Base Installation

See next page for part listing.

<u>Nov. 1, 2019 GPS Antenna & Cable Change:</u>
#3 A45 GPS Antenna with TNC Connector.
#4 Coax Cable Right Angle TNC both ends.



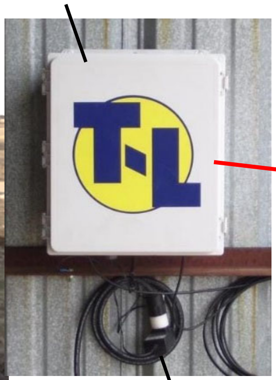


Figure 3. Base Installation.

Parts in MD90806 GPS RTK BASE STATION USA
or MD90806-AU (Australia) or MD90806-NZ (New Zealand) or MD90806-EU (Euro)

REF.	PART NO.	DESCRIPTION	# REQ'D.	
1	AP3286	GPS ANT MNT L BRKT	1	
2	**EA50709**	**GPS LIN NAV BASE BOX ASSY**	**1**	**see below**
Kit	**EA90807**	**BASE RADIO W/ ADPT ANT US**	**1**	**see below**
or	**EA90807-AU**	**BASE RADIO W/ ADPT ANT AUS**		
or	**EA90807-NZ**	**BASE RADIO W/ ADPT ANT NZ**		
or	**EA90807-EU**	**BASE RADIO W/ ADPT ANT EURO**		
3	ED50797	A45 GPS Antenna with TNC Connector (11/19 -)	1	
	ED50791	A50 GPS Antenna with Type N Connector (2/14 – 11/19)	1	
4	EW53526	GPS Coax Cable 26 ft long TNC X TNC (11/19 -)	1	
	EW53523	GPS Coax Cable 26 ft long TNC X N (available thru Parts)	1	
Box	**MB45260**	**GPS LIN NAV BAS HRDWR KIT**	**1**	**see below**

Parts in EA50709 GPS Linear Nav Base Box Assy

REF.	PART NO.	DESCRIPTION	# REQ'D.
6	EH52401	SPLIT STRAIN RELIEF 20mm	2
7	EH5244	1/2" CORD CONN .100-.300"	1
8	EW5383	WIRE 16AWG TFFN	4 ft
		(2 ft. Black, 1 ft Red, 1 ft Brown for field wiring to Radio)	

Parts in EA90807 Base Radio with Adapter & Antenna USA
or EA90807-AU (Australia) or EA90807-NZ (New Zealand) or EA90807-EU (Euro)

REF.	PART NO.	DESCRIPTION	# REQ'D.
5	EW53575	RADIO COAX CBL 8m FOR HI GAIN RP SMA M X N M	1
9	ED50826	900 MHz 6dBi OMNI ANTENNA N FEM	1
or	ED50824	2.4 GHz 6dBi OMNI ANTENNA N FEM	
		(includes mounting hardware)	
		Dec. 2018 changed from Rubber Ducky to High Gain Ant.	
10	EH5294	WALL ADPTR 120/240 TO 12	1
11	ER90809	900 MHZ DATA RADIO USA	1
or	ER90809-AU	900 MHZ DATA RADIO AUS	
or	ER90809-NZ	900 MHZ DATA RADIO NZ	
or	ER90809-EU	900 MHZ DATA RADIO EURO	

Parts in MB45260 GPS LIN NAV BASE HRDWR KIT

REF.	PART NO.	DESCRIPTION	# REQ'D.
12	EH52413	1/2" CORD CONN .230-.546"	1
13	FB5455	HHCS 5/8 X 1-1/2 GR5	1
14	FB5725	U-BOLT 5/16x1-3/8x2-3/16	4
15	FN5400	NUT HEX 5/16-18	8
16	FN5406	NUT HEX 5/8-11	1
17	FT5702	WIRE TIE 0.27" X 13" BK	10
18	FW5649	FW 5/16"	8

Longer Radio Antenna Coax Cable with Jumper to Radio (Available through Engr.)

PART NO.	DESCRIPTION	# REQ'D.
EW53576	Radio Coax Cable 65 ft Long, LMR400, N Male X N Male	1
EW53577	Radio Coax Jumper 12" Long, RP SMA Male X N Female	1

2.5. Write Down Base Coordinates

After the Base has been wired and allowed to "warm up" and find its location, make sure to write the Base Lat – Long down, for instance inside the door of the Base with a permanent marker.

3. ROVER INSTALLATION

Possible Combinations on PLC Linears:

Rover Location	Tractor Type	Water Feed	Overview Section
Rover at Tractor	Single Tower Tractor	Hose Drag	3.1
Rover at Center of End Feed	Single Tower Tractor	Hose Drag	3.2
Rover at Tractor	Single Tower Tractor	Ditch Feed	3.3
Rover at Center of End Feed	Single Tower Tractor	Ditch Feed	3.4
Rover at Tractor	Ultra G2 Tractor	Hose or Ditch	3.5
Rover at Center	Ultra G2 Tractor	Hose or Ditch	3.6

3.1. Rover at Tractor – Single Tower Tractor Hose Drag Overview

- PLC Panel at Tractor
- Rover Box & Antennas at Tractor, Rover Box Mounted next to PLC Panel

NOTE: Keep separation between Radio Antenna and Run Light.

Figure 04. Rover Installation at Tractor Overview – STT Hose Drag.

3.2. Rover at Center – End Feed Single Tower Tractor Hose Drag Overview

- PLC Panel at Tractor
- GPS Antennas and Radio Antenna located at a Center Tower. Rover Box mounted on Tower Step.
    - See Sec. 3.12 for Rover Box installation on Tower Step.
- 20Ga 3Pr Comm Cable on Spans.

<u>End Feed Tractor – Rover at Center</u>
GPS & Radio Antennas move to Center.
Rover Box Mounted on Tower Step.
- Comm Span Cable from Rover to PLC.
- PLC Panel at Tractor.

Figure 05. Rover Installation at Center Overview – STT End Feed Hose Drag.

3.3. *Rover at Tractor – Single Tower Tractor Ditch Feed Overview*

- Rover Box & Antennas at Tractor. See Sec. 3.12 for Rover Box installation on Tower Step.
- PLC Panel at Tractor.

Figure 06. Rover Installation at Tractor Overview – STT Ditch Feed.

3.4. *Rover at Center – End Feed Single Tower Tractor Ditch Feed Overview*

- PLC Panel at Tractor.
- GPS Antennas and Radio Antenna located at a Center Tower. Rover Box mounted on Tower Step.
    - See Sec. 3.12 for Rover Box installation on Tower Step.
- 20Ga 3Pr Comm Cable on Spans.

<u>End Feed Ditch Tractor – Rover at Center</u>
GPS & Radio Antennas move to Center.
Rover Box Mounted on Tower Step.
- Comm Span Cable from Rover to PLC.
- PLC Panel at Tractor (not shown).

Figure 07. Rover Installation at Center Overview – STT End Feed Ditch Feed.

3.5. *Rover at Tractor – ULTRA G2 Tractor Hose Drag or Ditch Feed Overview*

- PLC Panel at Tractor
- Rover Box & Antennas at Tractor. See Sec. 3.12 for Rover Box installation on Tower Step.

Figure 08. Rover Installation at Tractor Overview – ULTRA G2.

3.6. Rover at Center – ULTRA G2 Tractor Hose Drag or Ditch Feed Overview

- PLC Panel at Tractor
- GPS Antennas and Radio Antenna located at a Center Tower. Rover Box mounted on Tower Step.
    - See Sec. 3.12 for Rover Box installation on Tower Step.
- 20Ga 3Pr Comm Cable on Spans.
- Comm Cable wires route through Collector Ring.

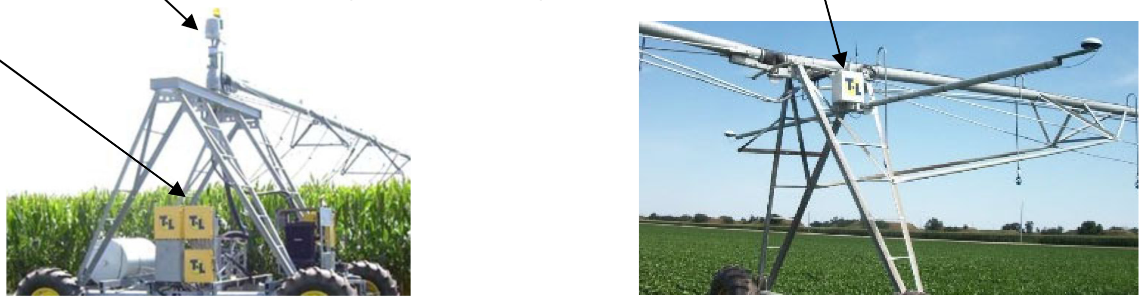
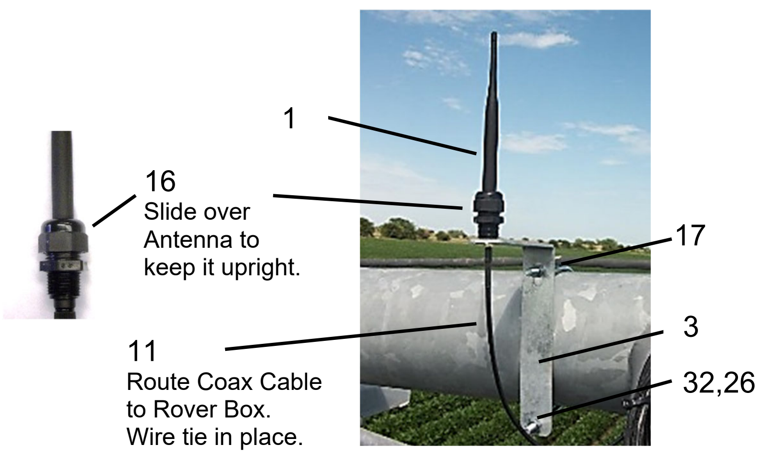


Figure 09. Rover Installation at Center – ULTRA G2.

3.7. Rover Radio Antenna Installation – 6-5/8” Pipe

- U-Bolt & Bracket on 6-5/8” Pipe, all Tractor styles and locations except ULTRA G2 at Tractor.
- See Sec. 3.13 that follows for complete part listing and where the parts originate from on the order.



REF.	PART NO.	DESCRIPTION	# REQ'D.
1	ED5082	900 MHz RUBBER DUCKY ANT RP SMA M	1
or	ED50820	2.4 GHz RUBBER DUCKY ANT RP SMA M	
3	AB3288	RF ANT MNT BRKT	1
11	EW5357	RADIO COAX 8M FOR RUBBER DUCKY	1
16	EH52413	1/2" CORD CONN .230-.546"	1
17	FB383	ET CNTRL PLT U-BOLT	1
26	FN5401	NUT HEX 3/8-16	2
32	FW5651	FW 3/8"	2

Figure 10. Rubber Ducky Radio Antenna Installation on 6-5/8” Pipe.

3.8. Rover Radio Antenna Installation –8" Pipe

- Install AB3288 Mount Bracket with one Bolt and field bend for fit.
- Keep Radio Antenna vertical.

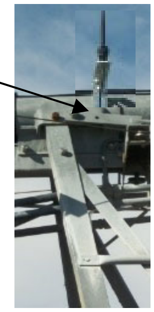
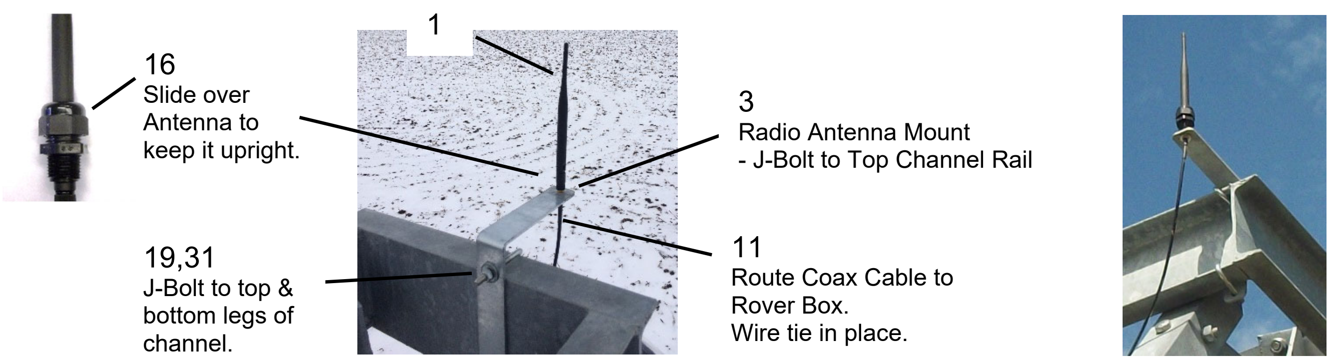


Figure 11. Rubber Ducky Radio Antenna Installation on 8" Pipe.

3.9. Rover Radio Antenna - Ultra G2 at Tractor

Antenna Mount attaches with J-Bolts to Upper Channel Rail.
See Sec. 3.13 that follows for complete part listing and where the parts originate from on the order.



REF.	PART NO.	DESCRIPTION	# REQ'D.
1	ED5082	900 MHz RUBBER DUCKY ANT RP SMA M	1
	or ED50820	2.4 GHz RUBBER DUCKY ANT RP SMA M	
3	AB3288	RF ANT MNT BRKT	1
11	EW5357	RADIO COAX 8M FOR RUBBER DUCKY	1
16	EH52413	1/2" CORD CONN .230-.546"	1
19	FB54385	J-BOLT 5/16 X 3 PLTD WITH NUT	2
31	FW5649	FW 5/16"	4

Figure 12. Rubber Ducky Radio Antenna Installation at ULTRA G2 Tractor.

3.10. Rover GPS Antennas Installation

See Sec. 3.13 that follows for complete part listing and where the parts originate from on the order.
- Once the system is operational, adjust Fwd & Rev Antennas so system runs in the same track.

ED50797 A45 Antenna
As of 11/01/19

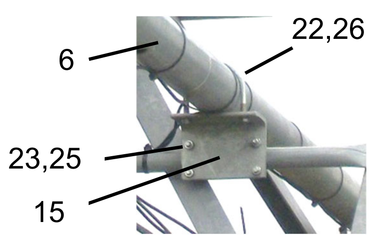
EW53526 Coax Cable
for A45 Antenna

EC50798 Adapter
- use when replacing A50 with A45 and keep existing Coax Cable.

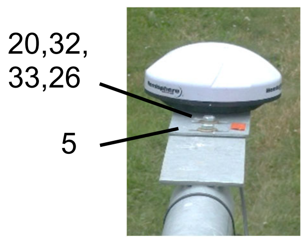
ED50791 A50 Antenna
Obsolete 11/01/19

EW53523 Coax Cable
for A50 Antenna
Still available from Parts.

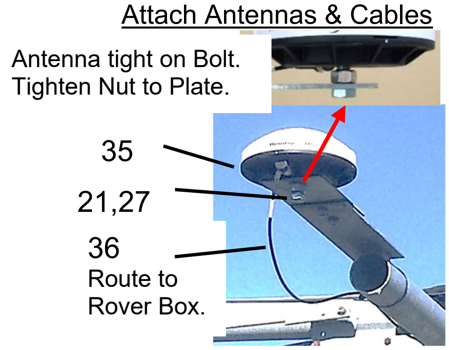
Install 3" OD Tube



Attach Mount Plate



Attach Antennas & Cables



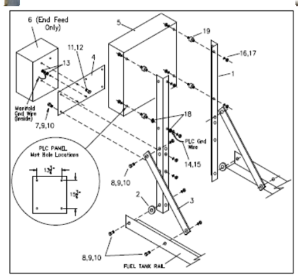
- Mount Tube on top step if possible.
- Center Tube Fwd & Rev.
- Align Tube Parallel to Tractor.
- Adjust Antennas for same track.

REF.	PART NO.	DESCRIPTION	# REQ'D.
5	AP15640	GPS LINEAR ANT MNT PLT	2
6	AW25863	GUID CONTROLLER TUBE 3" OD	1
15	AL2481	CNTRL TB MNT ANG	2
20	FB5439	HHCS 3/8 X 1 GR5	4
21	FB5455	HHCS 5/8 X 1-1/2 GR5	2
22	FB57102	U-BOLT 3/8 X 3 X 3-5/8	2
23	FB5725	U-BOLT 5/16x1-3/8x2-3/16	4
25	FN5400	NUT HEX 5/16-18	8
26	FN5401	NUT HEX 3/8-16	8
27	FN5406	NUT HEX 5/8-11	2
32	FW5651	FW 3/8"	4
33	FW5652	LW 3/8"	4
35	ED50797	A45 GPS Antenna with TNC Connector (11/19 -)	2
	ED50791	A50 GPS Antenna with Type N Connector (2/14 – 11/19)	2
36	EW53526	GPS Coax Cable 26 ft long TNC X TNC (11/19 -)	2
	EW53523	GPS Coax Cable 26 ft long TNC X N (available thru Parts)	2

Figure 13. GPS Antenna Installation.

3.11. Rover Box Installation – Rover at Tractor – Single Tower Hose Drag

See Sec. 3.13 that follows for complete part listing and where the parts originate from on the order.



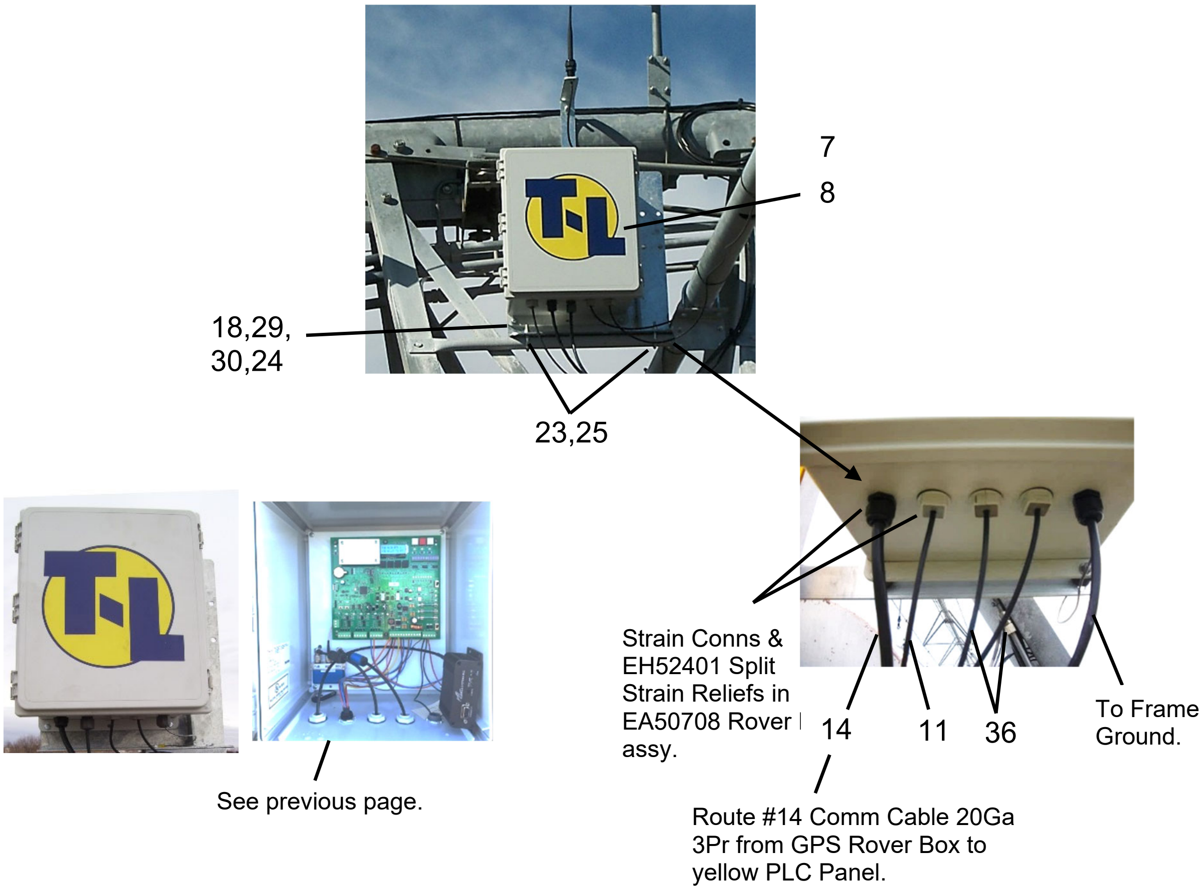
REF.	PART NO.	DESCRIPTION	# REQ'D.
2	ER90809	900 MHZ DATA RADIO USA	1
	(or ER90809-AU or ER90809-NZ or ER90809-RU)		
4	AL3287	GPS LIN NAV BOX MNT ANG	2
8	EA50708	GPS LIN NAV RVR BOX ASSY	1
11	EW5357	RADIO COAX 8M FOR RUBBER DUCKY	1
14	EW53635	Comm Cable 20Ga 3Pr	25 ft
24	FN5399	NUT HEX 1/4-20	8
28	FT5702	WIRE TIE 0.27" X 13" BK	20
29	FW5647	FW 1/4"	8
30	FW5648	LW 1/4"	8
31	FW5649	FW 5/16"	4
34	UD57800	RUBBER RADIATOR MNT 1/4"	4
36	EW53523	GPS COAX CABLE RG142 8m for GPS ANT	2

Figure 14. Rover Box Installation at STT Hose Drag Tractor.

3.12. Rover Box Installation on Tower Step

Rover at Center - End Feed Single Tower Hose Drag or Ditch Feed
Rover at Tractor – ULTRA G2 Hose Drag or Ditch Feed
Rover at Center - ULTRA G2 Hose Drag or Ditch Feed

See Sec. 3.13 that follows for complete part listing and where the parts originate from on the order.



REF.	PART NO.	DESCRIPTION	# REQ'D.
7	AW3280	GPS NAV BOX WLDMNT	1
8	EA50708	GPS LIN NAV RVR BOX ASSY	1
11	EW5357	RADIO COAX 8M FOR RUBBER DUCKY	1
14	EW53635	Comm Cable 20Ga 3Pr	25 ft
18	FB5435	HHCS 1/4 X 1 GR5	4
23	FB5725	U-BOLT 5/16x1-3/8x2-3/16	2
24	FN5399	NUT HEX 1/4-20	8
25	FN5400	NUT HEX 5/16-18	4
29	FW5647	FW 1/4"	8
30	FW5648	LW 1/4"	8

Figure 15. Rover Box Installation on Tower Step.

3.13. Complete Parts Listing for Rover

Parts in MD90808 Rover Slave Radio Kit GPS Linear USA
or MD90808-AU Australia or MD90808-NZ New Zealand or MD90808-EU Europe

REF.	PART NO.	DESCRIPTION	# REQ'D.
1	ED5082	900 MHz RUBBER DUCKY ANT RP SMA M	1
or	ED50820	2.4 GHz RUBBER DUCKY ANT RP SMA M	
2	ER90809	900 MHZ DATA RADIO USA	1
	or ER90809- AU Radio Australia or ER90809-NZ Radio New Zealand or ER90809-RU Euro		
11	EW5357	RADIO COAX 8M FOR RUBBER DUCKY	1

Other Parts Included with GPS Navigation for Manual Control Linear

REF.	PART NO.	DESCRIPTION	# REQ'D.
3	AB3288	RF ANT MNT BRKT	1
4	AL3287	GPS LIN NAV BOX MNT ANG	2
5	AP15640	GPS LINEAR ANT MNT PLT	2
6	AW25863	GUID CONTROLLER TUBE 3” OD	1
7	AW3280	GPS NAV BOX WLDMNT	1
		- AW3280 used instead of AL3287 for Ditch Feed and for EF with GPS at Center.	
	CD90480	MAN GPS NAV LIN PLC	1
8	EA50708	GPS LIN NAV RVR BOX ASSY	1
10	EC50792	USB FLASH DRIVE	1
14	EW53635	COMM CABLE 20GA 3PR	25 ft
Box	**MB45261**	**GPS LIN NAV RVR HRDWR KIT**	**1 see below**
35	ED50797	A45 GPS Antenna with TNC Connector (11/19 -)	2
	ED50791	A50 GPS Antenna with Type N Connector (2/14 – 11/19)	2
36	EW53526	GPS Coax Cable 26 ft long TNC X TNC (11/19 -)	2
	EW53523	GPS Coax Cable 26 ft long TNC X N (available thru Parts)	2

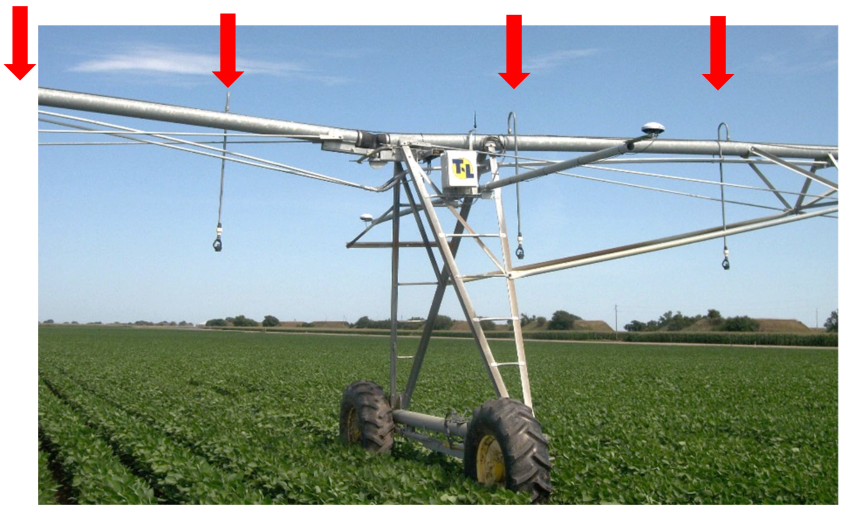
Parts in MB45261 GPS LIN NAV RVR HRDWR KIT

REF.	PART NO.	DESCRIPTION	# REQ'D.
15	AL2481	CNTRL TB MNT ANG	2
16	EH52413	1/2" CORD CONN .230-.546"	1
17	FB383	ET CNTRL PLT U-BOLT	1
18	FB5435	HHCS 1/4 X 1 GR5	4
19	FB54385	J-BOLT 5/16 X 3 PLTD /NUT	2
20	FB5439	HHCS 3/8 X 1 GR5	5
21	FB5455	HHCS 5/8 X 1-1/2 GR5	2
22	FB57102	U-BOLT 3/8 X 3 X 3-5/8	2
23	FB5725	U-BOLT 5/16x1-3/8x2-3/16	6
24	FN5399	NUT HEX 1/4-20	8
25	FN5400	NUT HEX 5/16-18	12
26	FN5401	NUT HEX 3/8-16	9
27	FN5406	NUT HEX 5/8-11	2
28	FT5702	WIRE TIE 0.27" X 13" BK	20
29	FW5647	FW 1/4"	8
30	FW5648	LW 1/4"	8
31	FW5649	FW 5/16"	4
32	FW5651	FW 3/8"	6
33	FW5652	LW 3/8"	6
34	UD57800	RUBBER RADIATOR MNT 1/4"	4

3.14. Sprinkler Installation at Rover

It is IMPORTANT to divert water spray away from the Rover Box and Antennas.

- Install Drops at least 2 outlets either side of the Rover Box.
- Make sure the sprinklers are low enough that they will not spray on the Rover Box or Antennas, especially in a strong wind.



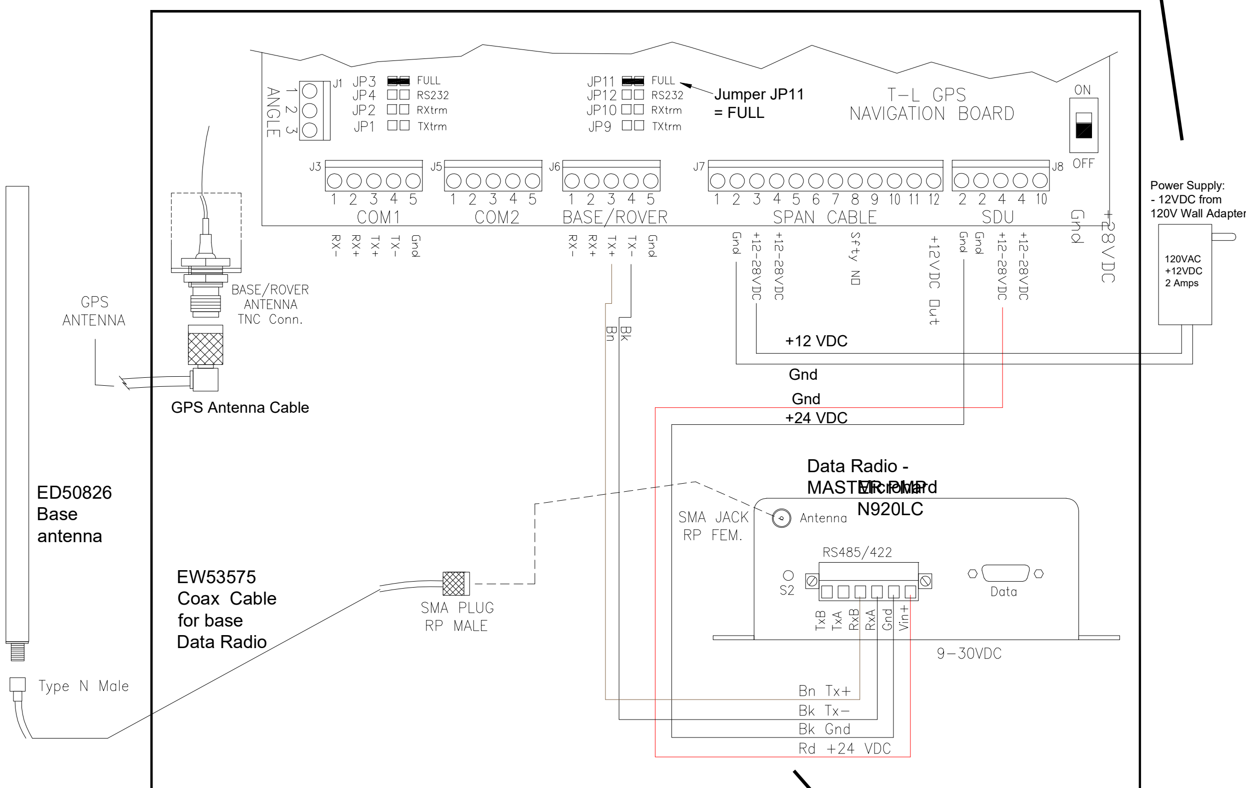
4. WIRING

4.1. Wiring of Base Box EA50709

See Section 2 for Base Installation information.

NOTE: Base can be powered by 100 Watt Solar Panel with 50 amp-hr battery.

Wall Adapter can be powered from 120 or 240 VAC.



16 Ga wires in a bag inside Base Box.

Figure 16. Wiring of Base.

4.2. Wiring of Rover Box EA50708, PLC Linear

See Section 3 for Rover Installation information.

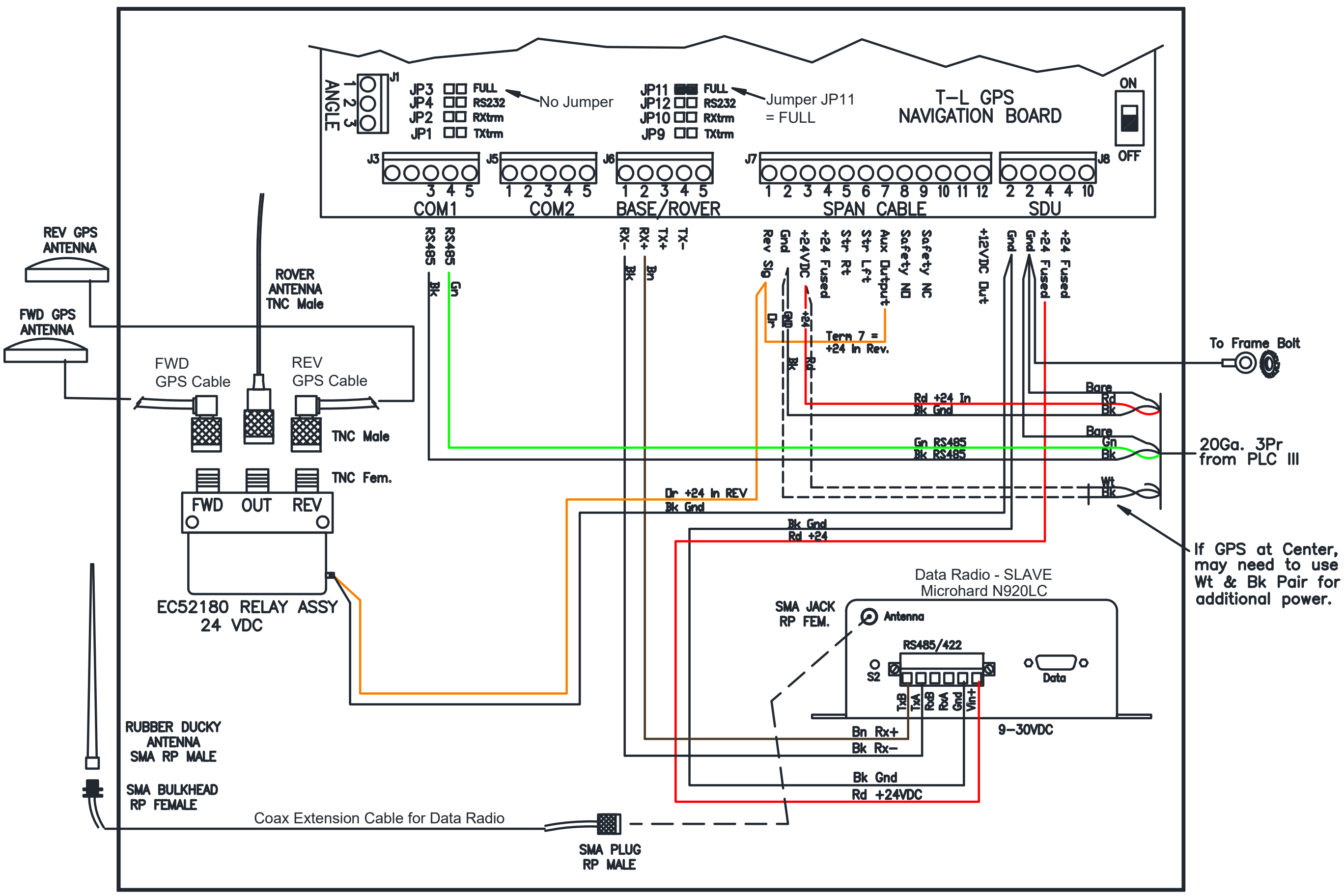


Figure 17. Wiring of Rover Box.

4.3. Wiring of PLC Panel to Twisted Pair Rover Cable

See Section 3 for Rover Installation information.

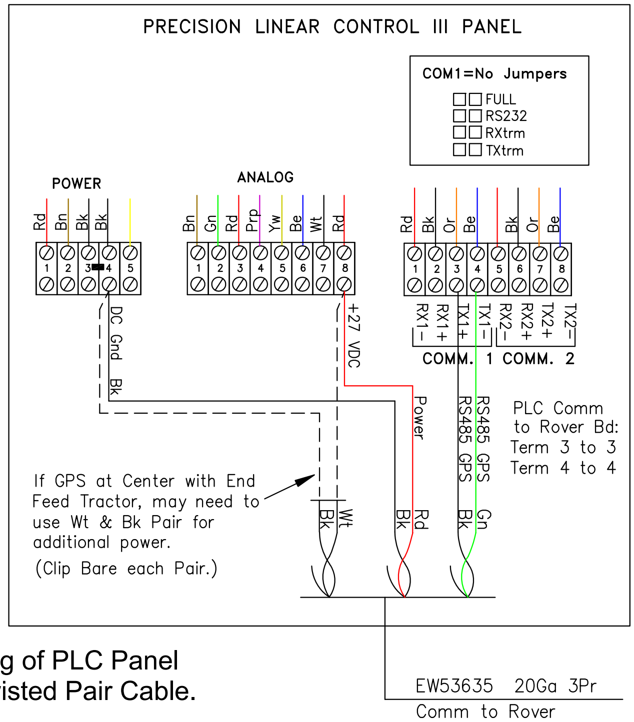


Figure 18. Wiring of PLC Panel
to Twisted Pair Cable.

4.4. Wiring of PLC 18/15 Swivel Cable – ULTRA G2 with GPS Nav at Center

See Section 3 for Rover Installation information.

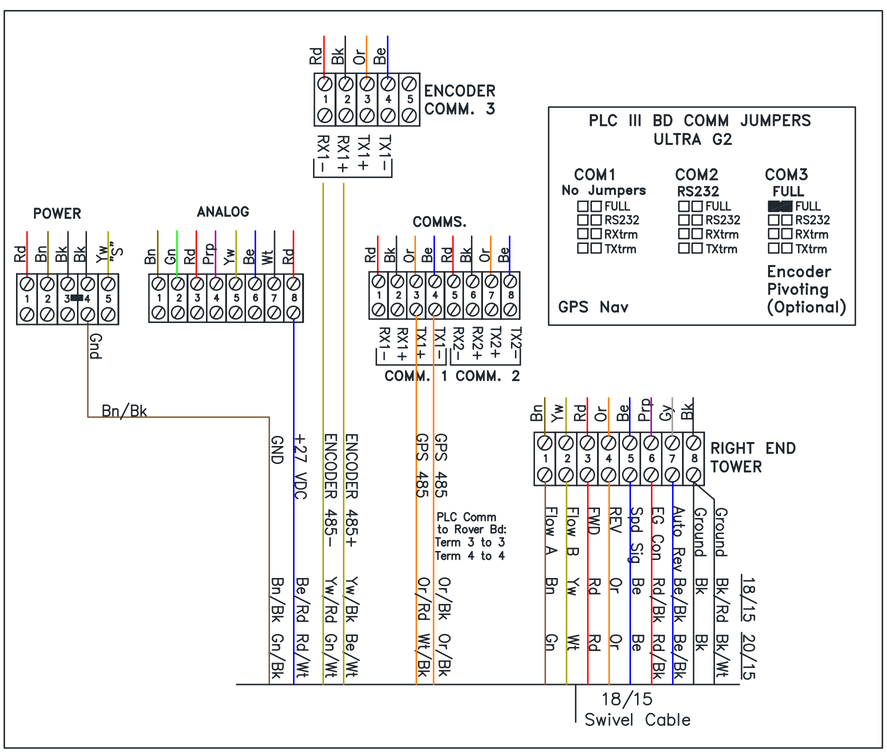


Figure 19. Wiring of PLC Panel to Swivel Cable Ultra G2 GPS Nav at Center. (change to 18/15 08/23)

4.5. Wiring of Collector Ring – ULTRA G2 with GPS Nav at Center

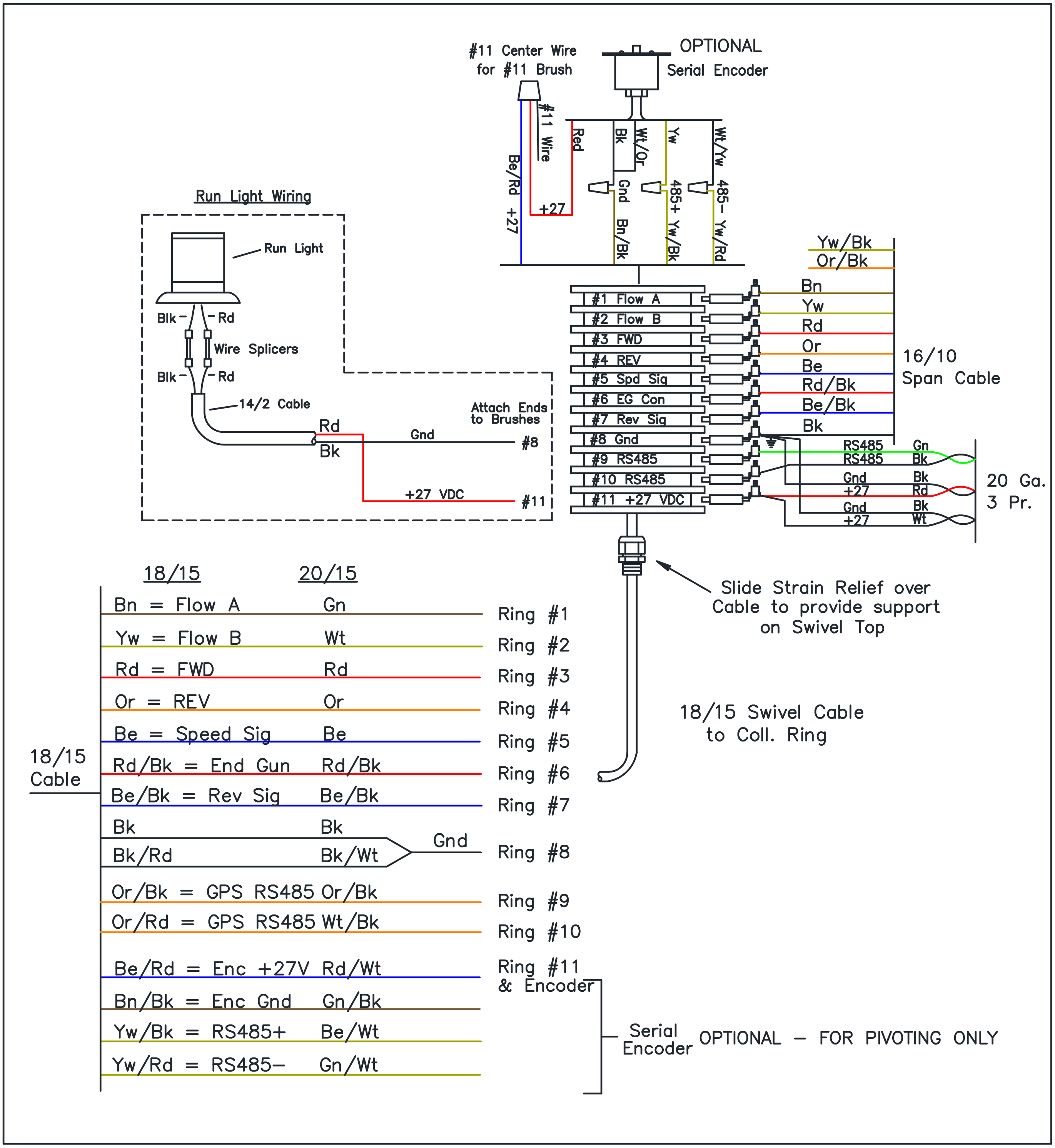


Figure 20. Wiring of ULTRA G2 with GPS Navigation at Center. (change to 18/15 Swivel Cable 08/23)

5. NAVIGATION PCB & USB DRIVE

5.1. Navigation Board Overview - Version A05 Green Board

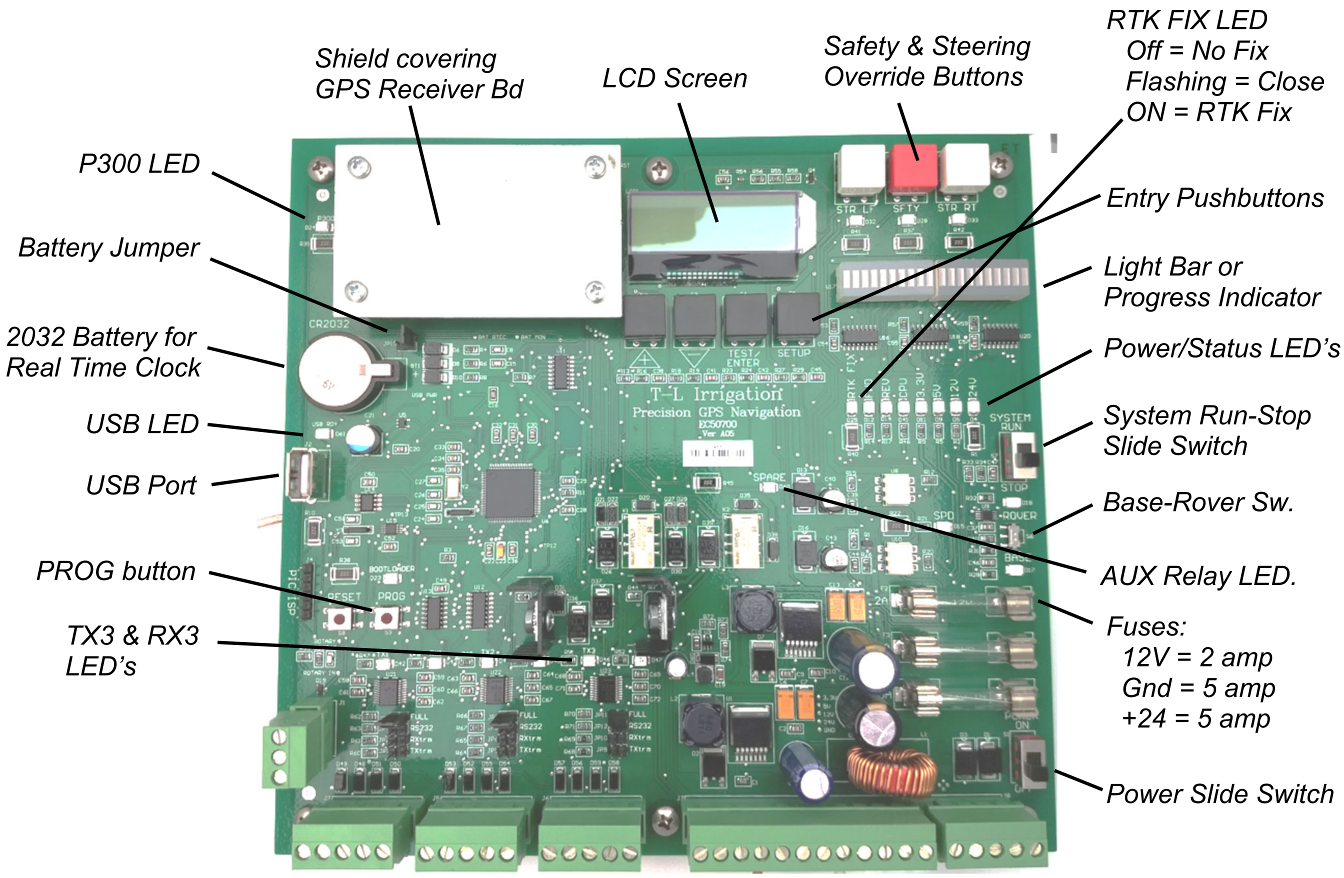


Figure 21. Navigation Board Overview A05.

5.2. T-L Supplied USB Flash Drive EC50792

- Use only the T-L Supplied USB Flash Drive. (Shipped in a plastic bag with manuals/documents.)
- The USB Drive must use the FAT32 File System.

Figure 22. T-L Supplied USB Flash Drive.

5.3. Files recognized by the T-L Navigation Board

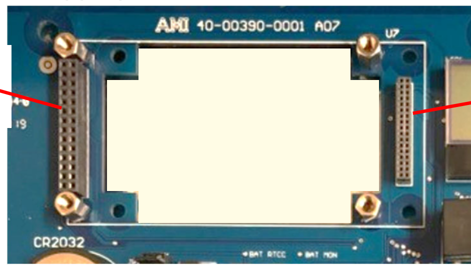
- "image.hex" = firmware file, used to reprogram new firmware into the Navigation Board.
- ".log" file = Data Logging file written to the USB from the board. Open with Notepad or Excel.
- ".saf" file = Safety History file saved to the USB from the board. Open with Notepad or Word or Excel.

5.4. Navigation Board Overview - Version A07 Blue Board

- Released March 2020
- Additional overvoltage protection on Com Ports
- Addition of socket for Trimble GPS RTK Receiver Board
- Load appropriate firmware on T-L GPS Navigation Board for respective GPS RTK Receiver Bd.

Hemisphere Corner firmware =	HC48 or after
Hemisphere Linear firmware =	HL58 or after
Trimble Corner firmware =	TC25 or after
Trimble Linear firmware =	TL17 or after

<u>A07 Board will work with either Trimble or Hemisphere RTK Board.</u>
Load appropriate firmware on T-L GPS Nav Board.



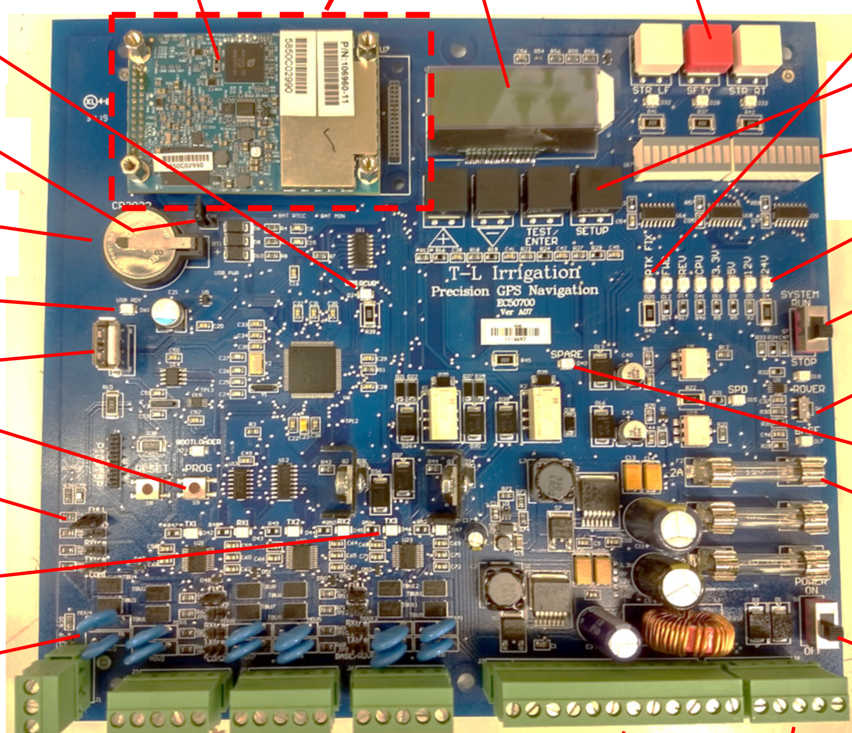


Figure 22.1. Navigation Board Overview A07 (Blue Board).

6. REPROGRAMMING FIRMWARE IN GPS NAV BOARD

6.1. Save Program File from E-mail or T-L Dealer Point to USB

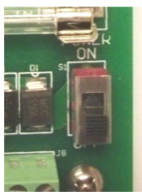
Program file name will be similar to: “GPS_NAV_vHL59.linear.hex”
- save this file direct to the USB so you can reference the firmware version.

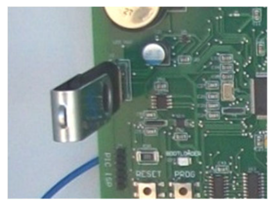
6.2. Rename the File to “image.hex”

- Create a copy and rename the copied file on your USB to “**image.hex**”.
- “**image.hex**” is the firmware file utilized to reprogram new firmware into the GPS Navigation Board.
- New firmware MUST be on the USB with the filename of “**image.hex**” or it will not load.
- It must be on the root directory and cannot be stored in a file folder on the USB.

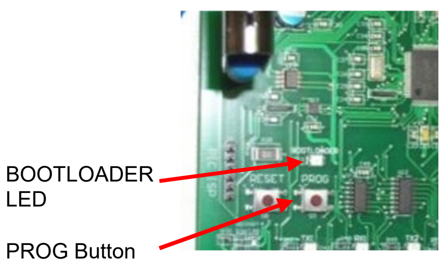
6.3. Reprogramming with New Firmware if Needed

- Power OFF the Navigation Board.
- Run-Stop switch to STOP.
- Insert USB with “image.hex” file into USB Port on Board.





- **Press & Hold the PROG button while moving the Power switch to On. Hold the PROG button until the word “Bootloader” appears on the screen, then release.**
  - The Bootloader LED on the Navigation Board will begin flashing.
  - If the “image.hex” file is found on the USB Flash Drive, the Bootloader LED will flash very rapidly and the screen will change to “Bootloader Erasing Flash”, then “Bootloader Programming”.
  - If the “image.hex” file is NOT found, the Bootloader LED will remain flashing at the same rate and the screen will remain on “Bootloader”. Then you will need to verify that the “image.hex” file is actually on the USB Flash Drive being used.



BOOTLOADER

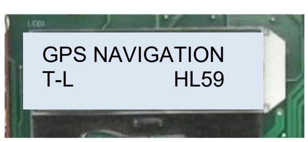
↓

BOOTLOADER
ERASING FLASH

↓

BOOTLOADER
PROGRAMMING

- **Once programming is complete, the title screen will appear for 3 seconds, and then go to the respective Base or Rover operation after establishing communication to the GPS Receiver.**



6.4. Reset Power to the Nav Board after New Firmware Loaded

Once the new firmware is loaded, power the Nav Bd off and back on.

6.5. Restore Settings <u>Only if told to do so</u>, (may not be required)

- "Restore Settings" will erase all saved information and settings for either the Base or the Rover.
- This includes the path waypoints for the Rover, they will have to be reloaded.
- HELPFUL HINT: You can "Save to USB" each Path and then "Load" the respective Path back from the USB after Restore Settings used.

IMPORTANT – BASE to do before Restore:

Before Restoring the Settings, make sure to write down:

- the Base Lat-Long from Base Test Screen.
- use the NOTES Section at the end of this manual.
- then afterwards you can set the Base coordinates to the exact same.

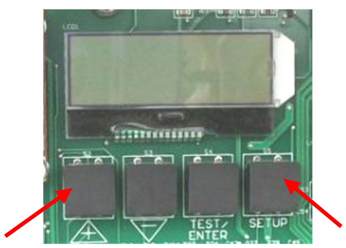
IMPORTANT – ROVER to do before Restore:

Before Restoring the Settings, make sure to note the following:

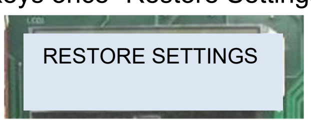
- Record Reverse and Forward End Point Points, Lat-Long and any Intermediate Points.
- HELPFUL HINT: You can "Save to USB" each Path and then "Load" the respective Path back from the USB after Restore Settings used.
- or use PLC III to Review each Path and record the Lat-Longs.
  - use the NOTES Section at the end of this manual.

Procedure to Restore Settings

Press & Hold "+" and "Setup" keys, then switch Rover Power On.



Release Keys once "Restore Settings" appears on the screen.



6.6. Enter Settings back if "Restore Settings" used:

BASE:

- Allow the Base to go through the full warmup and setting procedure and get to Base Running.
- Then Set the Base Lat-Lon.
  - these should have been recorded before restoring the settings.

ROVER:

- If Path saved to USB, reload the Path using the USB.
- Otherwise, most settings entered using the PLC III panel.

6.7. Reset Power to the Rover Nav Board after all Settings have been Entered.

7. REPROGRAMMING FIRMWARE IN PLC III BOARD

7.1. Save Program File from E-mail or T-L Dealer Point to USB

Program file name will be similar to: "*PLC_III_v267_GPSNav.hex*"
- save this file direct to the USB so you can reference the firmware version.

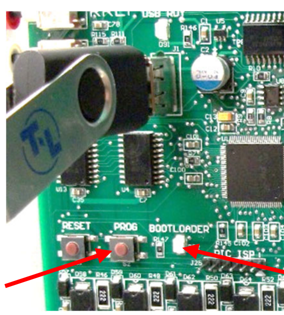
7.2. Rename the File to "plcimage.hex

- Create a copy and rename the copied file on your USB to "**plcimage.hex**".
- "**plcimage.hex**" is the firmware file used to reprogram new firmware into the PLC III Board.
- New firmware MUST be on the USB with the filename of "**plcimage.hex**" or it will not load.
- It must be on the root directory and cannot be stored in a file folder on the USB.

7.3. Reprogramming with New Firmware if Needed

- Power Switch on PLC Panel to OFF.
- Run-Stop System Toggle on PLC Panel to STOP.
- Insert USB with "plcimage.hex.
- Press & Hold PROG Button on PLC Board.
- Power Switch to PROGRAM (On).
- Release PROG Button when Bootloader appears on the screen.

- **Press & Hold the PROG button while moving the Power switch to PROGRAM. Hold the PROG button until the word "Bootloader" appears on the screen, then release.**
  - The Bootloader LED on the PLC III Board will begin flashing slowly for a time (25-30 secs).
  - If the "plcimage.hex" file is found on the USB Flash Drive, the Bootloader LED will flash very rapidly and the screen will change to "Bootloader Erasing Flash", then "Bootloader Programming".
  - If the "plcimage.hex" file is NOT found, the Bootloader LED will remain flashing at the same rate and the screen will remain on "Bootloader". Then you will need to verify that the "plcimage.hex" file is actually on the USB Flash Drive being used.



PROG Button

BOOTLOADER LED

BOOTLOADER

PLC III vA01

BOOTLOADER
ERASING FLASH

PLC III vA01

BOOTLOADER
PROGRAMMING

PLC III vA01

- **Once programming is complete, the Title Screen and then Main Status Screen will appear.**

T-L IRRIGATION

PRECISION LINEAR
CONTROL III v267

REV APPL 0.75"
AS

STEERING = L1:R1

7.4. Restore Settings <u>Only if told to do so</u>, (may not be required)

- "Restore Settings" may erase all the settings for the Menu items.

IMPORTANT – PLC III to do before Restore:

Before Restoring the Settings, make sure to write down all Setup Information:

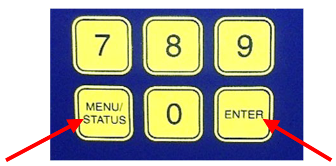
- Write down Menu Setup Information for each Menu Item.

1 = Run Setup	5 = Date/Times
2 = Guid Setup	6 = System Setup
3 = Aux Appl	
4 = EG Areas	

- use the "PLC III SUMMARY SCREENS" Section at the end of this manual to record settings.
- then afterwards you can set these back to the exact same.

Procedure to Restore Settings

- Power Switch to OFF.
- Run-Stop System Toggle to STOP.
- Press & Hold "Menu/Status" and "Enter" keys, then switch Power to PROGRAM.



Release Keys once "Restore Settings" appears on the screen.

RESTORE SETTINGS

↓

INITIALIZING BUFFERS

↓

UNITS
ENGLISH
IRRIGATED WIDTH
01073 FT

7.5. If only PLC updated, Enter Settings back:

- Menu Setup Information for each Menu Item.

1 = Run Setup	5 = Date/Times
2 = Guid Setup	6 = System Setup
3 = Aux Appl	
4 = EG Areas	

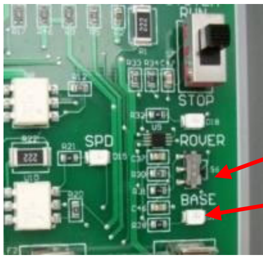
7.6. If both PLC III & Rover updated, Perform Sec 9.2 Initial Operation Steps:

- Refer to Section 9.2 Initial Field Operation Setup – Steps to Follow.

8. BASE OPERATION

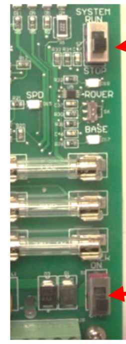
8.1. Set Base/Rover Switch to BASE and Run-Stop Switch to STOP

- With Power OFF, Set the Base/Rover Switch to BASE.
- Set Run-Stop Switch to STOP. (Has no effect on Base operation.)



Down = BASE

Base LED should be lit when powered on.



Run-Stop Sw. in STOP.

Power Sw. can be left ON.

8.2. Base Operation – Base Position (<u>HL47 and previous</u>)

Prior to April 2020 firmware

Screen	Description
GPS NAVIGATION T-L vHL47	Title screen appears for 5 seconds any time the board is powered on.
WAITNG FOR RECVR TIMER 300	Initial communication of the T-L Navigation Board to the GPS Receiver Board is established.
BASE WARMUP	Base Warmup Screen after title screen - appears for 3 minutes 15 seconds.

Base Warmup for 3 minutes 15 seconds:
Then it checks if there is a Saved base position. If not, a process is followed to set the current position.

- No Saved Position displayed for 15 secs.
- Get Current Position for another 3 minutes.
- Set Current Position for 15 secs.

If the Base Navigation Board is moved or changed, the existing Saved Position will be incorrect:

- Saved Position displayed for 15 secs.
- Current Position > 15 meters Different from the Saved Position.
- Get Current Position for 3 minutes.
- Set Current Position for 15 secs.

Once the Current Position is set and saved, the same Base Lat-Lon should be used each time after.
(checks to see that the Saved Position is within 15 meters of the current antenna position.)

- Saved Position for 15 secs.
- The screen will then show BASE READY with the Run-Stop switch in STOP.

BASE READY

- **Under normal conditions, there should not be any need to make any changes or settings to the Base. It should take care of itself.**
- **Power can be left ON, and the System switch left in RUN.**

8.3. Base Operation – Base Position (HL54 and after)

April 2020 and after firmware

Power Up

GPS NAVIGATION T-L vHL59	Title screen appears for 5 seconds any time the board is powered on.
WAITNG FOR RECVR TIMER 300	Initial communication of the T-L Navigation Board to the GPS RTK Receiver Board is established.

Initial Operation with NO SAVED POSITION, will Automatically set the Base Position.

- For First Time operation or after "Restore Settings".
- The Base will automatically find its Lat Long position with the following steps.

NO SAVED POSN (PRESS ENTER)	Base Lat Long coordinates not saved on T-L Nav Board. ENTER = continue on. SETUP = Setup Groups (use only if told to do so).

Press Enter – Allow the following to run completely, do NOT interrupt.

GET CURR POSN TIMER 360 Sec	Allow 6 minutes for the Base to acquire a good Lat Long. Timer will count down till 20 secs, then change to following.
SET CURR POSN TIMER 020 Sec	Base Lat Long is saved to T-L Nav Board. Base Lat Long is written into GPS RTK Board.
BASE WARMUP CHECK IF TX	Screen may appear briefly.
BASE RUNNING	Base is operational and sending corrections to the Rover. Base TX LED should be flashing.

Operation at Power Up after Base Position Saved on T-L Nav Board.

- The Base will automatically load the same Lat Long position at every subsequent power up.
- There is no check for within 15 meters.

SET CURR POSN TIMER 020 Sec	Base Lat Long saved on T-L Nav Board is written into GPS RTK Board to ensure the exact same Base location every time.
BASE WARMUP CHECK IF TX	Screen will appear for up to 30 secs at powerup. - If it stays on this screen, there may be a transmit problem.
BASE RUNNING	Base is operational and sending corrections to the Rover. - Base TX3 LED should be flashing.

Manually Set Base Position if needed.

- The Base Lat Long Position can be changed at this point in SET BASE POSN Setup Group.
    - The automatic Base location procedure above must be followed first.

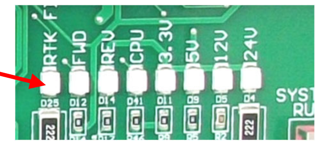
8.4. Write Down Base Coordinates

Make sure to write the Base Lat – Long down, for instance inside the door of the Base with a permanent marker.

- See “Base Test Screens” below to find the Base Lat Long.

8.5. RTK Fix LED on Base

Does NOT mean there is an RTK Fix at the ROVER.



Off = No Position yet from Base GPS Antenna.
Flashing = No WAAS correction used.
On Solid = Hemisphere firmware:
- SBAS (WAAS) correction & Transmitting to Rover.
- TX3 of Base should be flashing.

Trimble firmware:
- SBAS (WAAS) correction at Base.

8.6. Base Test Screens

Press “TEST/ENTER” to toggle into and out of Test Screens.
Use “+” or “-“ keys to scroll through Test Screens

Note: Eliminated screen “No Communication from Rover in HL56.

```
RTK      FIX
SATS.    17
```

Status of WAAS Fix (Blank, then Flash “FIX”, then solid “FIX” if a WAAS fix is established.)
Number of Satellites (GPS + Glonass). Galileo & Beidou added for Hemisphere P326 Board.

```
BASE  40.3484889N
     098.1654215W
```

Lat / Long of Base. Should not change.
RECORD THIS BASE LAT-LONG ONCE SET FOR FUTURE REF.

```
08/02/20   SUN
03:15:59   PM
```

Current Date & Time
- set in the Date/Time Setup Group

8.7. Base Setup Groups

Press “SETUP” key to toggle into and out of Setup Groups.
Use “+” or “-“ keys to scroll through the different Setup Groups.
Use “TEST/ENTER” key to enter into a specific Setup Group.

SETUP GROUP DATALOGGING	SETUP GROUP GPS RECVR INFO	SETUP GROUP DATE / TIME	SETUP GROUP SET BASE POSN	Added in HL45. Not used for Trimble. SETUP GROUP SET CORRECTION

8.8. Base Data Logging

This may be used to log information such as the Base Lat & Long, GGA Lat & Long, etc. to the USB.
Note in HL56, added operation that if Datalogging is on and power is cycled and then “Error Creating File” for more than 1 minute like it can’t find the USB when powered back up, then Datalogging is set to Stop and the Error message no longer appears.

Use “TEST/ENTER” key to enter through the Data Logging setup and Start or Stop Logging.
Use “+” or “-“ keys to scroll settings for each item to be set.

SETUP GROUP DATALOGGING	
DATA LOGGING START	Start or Stop Data Logging If Start, advance to next screen.
USB INSERTED? NO	Make sure USB is inserted and recognized.
NO USB DETECTED (PRESS ENTER)	If USB Inserted is Yes, but No USB detected, - press Enter. Will go back to Setup Group Data Logging.
LOGGING INTERVAL 01 SECS	Set data logging interval. (01 to 60 secs)
LOGGING ACTIVE (Enter)	Press Enter to exit to Main Dir Screen. **- Logging will start only after Enter is pressed.**

Start Logging – regardless of System Switch - Run Stop Position.

Press “SETUP” key to return to Setup Groups. Press “SETUP” key again to return to the main screen.

At Main Dir Screen, “Base” and the word “Log” will flash alternately when Data Logging.
During Startup – Warmup:

BASE WARMUP	LOG WARMUP

After Warmup complete:

BASE RUNNING	LOG RUNNING

File Format for Data Logging to USB:
- File will be saved with number & date for filename and ".log" ext
- Note the file extension will be ".log".
  - Example: 01032716.log = 1st file logged on March 27, 2016
- Transfer from the USB to a computer, open in Notepad to review.
  - or the file can opened in Excel (Delimited, Tab Delimited, General Column Format)

To Stop Logging, access the Data Logging Setup Group and Stop and press Enter.

SETUP GROUP
DATALOGGING

DATA LOGGING
STOP

Start or Stop Data Logging
Change to Stop and press Enter.

Base Log Parameters

Firmware HC46													
DATE	TIME	AM/PM	UTC	BASE LAT	BASE LONG	GGA LAT	GGA LON	SATS	RTK	STATE	SAVED LAT	SAVED LON	DIFF DIST
3/20/2020	5:33:11	PM	233301	40.58077759	-98.26856434	40.580779	-98.26856189	27	2	1	40.58077759	-98.26856434	0

8.9. GPS Recvr Info (below is for Hemisphere)

Use "TEST/ENTER" key to enter and scroll through the GPS Receiver Information.

	Ex.	
- GPS Engine		DF5r
- Bd Serial No.		111111
- Bd Fleet No.		1
- Bd Hdwre Version		07
- Bd Prodctn Code		02122009
- Subscrpt Begin		01/01/1900
- Subscrpt Expir		01/01/3100
- Software Version		1.2Qq06t
(Following added in HC38)		
- RTK Subscrpt		Yes
If Base - RTK TX Status		22
- P300 Correction		None or SBAS
- RTK Accuracy		0 — Accuracy Status
- L1,L2,L5,G1,G2		10,10,0,7,7 — Number of Satellites from GPS & Glonass
- B1,B2,E1B,E5A,E5B		0,0,2,2,2 — Number of Satellites from BeiDou & Galileo P326 & P34 (P300 & P306 = No Subscription)
- QL1, QL2		0,0 — Number of Satellites from Japan system P326 & P34 (P300 & P306 = No Subscription)

RTK Board	Engine	Software	Ship Dates
Red P300 =	DF3r	1.2Qe	04/14-11/15
Red P306 =	DF4r	1.2Qf9b	11/15-03/18
Green P326 =	DF5r	1.2Qq06t	04/18-
Black P34 =	DF5r	1.2Qr04ta	?? -

Subscrpt Expir last 4 digits:
3100 = Base or Rover
3104 = Base ONLY
3108 = Base or Rover
4104 = P326 Base or Rover
4140 = P34 Base or Rover

Press "SETUP" key to return to the Setup Groups.
Press "SETUP" again to return to the main screen.

8.10. Date / Time Setup

Use "TEST/ENTER" key to access the Date / Time settings.

08/03/20 MON
03:15 PM

- Use "+ and "-" keys to set each item,
  - then press Enter after each is set.

Press "SETUP" key to return to the Setup Groups.
Press "SETUP" key again to return to the main screen.

8.11. Set Base Position

"AUTO" operation added in HL54.

The Base Lat-Long can be manually set for a couple reasons:

- To match a previously saved Base position if for some reason during the Base Warmup it saves a different position. (Changes can be made within several feet, but not larger.)
- If the Base Nav board has a problem in season after a wheel track has been established and the board needs to be changed. The exact same Base Lat-Lon needs to be used for the SDU to follow the same wheel track.

Screen	Description
SETUP GROUP SET BASE POSN	Large position changes cannot be made. Only changes to the last 3 to 4 decimal places.
SET BASE LAT LON NO	AUTO, MANUAL or NO. If NO, return to Setup Group. "Auto" added in HC46 & TC21.

If AUTO: (added in HL54)

- Use Auto if the Base has been moved from another location.

Screen	Description
SET BASE LAT LON AUTO	Select "AUTO", press Enter.
SET TO AUTO? YES	Yes or No Confirmation Screen. "No" is default.

Press Enter – Allow the following to run completely, do NOT interrupt.

Screen	Description
GET CURR POSN TIMER 360 Sec	Allow 6 minutes for the Base to acquire a good Lat Long. Timer will count down till 20 secs, then change to following.
SET CURR POSN TIMER 020 Sec	Base Lat Long is saved to T-L Nav Board. Base Lat Long is written into GPS RTK Board.
BASE WARMUP CHECK IF TX	Screen may appear briefly.
BASE RUNNING	Base is operational and sending corrections to the Rover. Base TX LED should be flashing.

If MANUAL:

- Use to Manually set the Base Lat Long to match previous location.

IMPORTANT NOTE: The "AUTO" operation must be run first and get to "Base Running" on the main screen before manually changing Lat-Long. This will allow the Base to set its own altitude.

Screen	Description
SET BASE LAT LON MANUAL	Select "MANUAL", press Enter.
40.1234567N 098.1234567W	The existing Saved Base Position is displayed if there is one. Each digit or direction letter can be changed using "+" or "-", then Enter. Or Enter past each digit if correct.

If Trimble, do not power off the Base for 2 minutes or the position will not be saved.

Saving . . .	Do Not Power OFF For 2 Minutes

8.12. Set Correction (Added in HL45)

- Not available for Trimble firmware. Trimble is always set to RTCM3 corrections.

Following Changed in HL56. ROX no longer a choice, issue with GGA Age if too many satellites when used with the P326 RTK Receiver Boards Base & Rover (green boards).
Two choices for Correction Type – RTCM3 or CMR (CMR option added in HC43).

SETUP GROUP SET CORRECTION

SET CORRECTION RTCM3

RTCM3 or CMR.
See Caution below if setting to RTCM3 on Red P300/P306 Board.

RTCM3
- Default Correction Type in the firmware is RTCM3.
- CAUTION: If a Red RTK Board (P300 or P306) is used on either the Base or the Rover, then "NO" must be selected for "Include Galileo".

CMR
- For use with Trimble equipment that will accept a CMR type message. Typically RTCM3 will be used for Trimble equipment supplied by T-L Irrigation.

If RTCM3

INCLUDE GALILEO NO

Yes or No.
Default is NO, should work in all cases.
If more satellites needed with Green Boards, set to YES.

Caution: NO if a Red RTK Board (P300 or P306) is used on either Base or Rover.
Red P300 Board shipped 2014 - Nov. 2015
GPS Receiver Info, GPS Engine = DF3r
Red P306 Board shipped Nov. 2015 – March 2018
GPS Receiver Info, GPS Engine = DF4r
YES if Green RTK Boards on both Base and Rover
Green P326 Boards shipped March 2018 – current
GPS Receiver Info, GPS Engine = DF5r

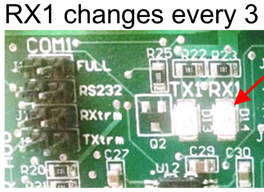
Setting * * *

Correction Type is being set.
Wait for this screen to change to the next.

SETUP GROUP SET CORRECTION

9. PLC III & GPS ROVER – INITIAL STARTUP STEPS

9.1. Overview of PLC III Panel and GPS Rover Functions

Item	PLC III Panel		GPS Rover Nav Bd
Menu / Setup Groups	1= RUN SET 5=DATE/TM 2= GUIDSET 6=SYS SET 3=AUX APP 7=SFTYHST 4=EG AREA 8= GPSPATH 9=TESTOUT 10=LOG *PLC III with GPS Nav firmware		SETUP GROUP STEERING PATH DATALOGGING SAFETY HISTORY GPS RECVR INFO DATE / TIME *GPS Rover with Linear Firmware
How Comm established	2=GuidSetup, set to GPS Navigation. - turn on Comm to Rover.		Steering Setup Group, System Type set to PLC Linear. - turn on Comm to PLC III.
Verify Comm	PLC Board TX1 & RX1 - RX1 changes every 3 secs.  		Rover Nav Board TX1 & RX1 - TX1 & RX1 blink every 3 secs. 
How Set Path(s) (if done at Rover)			Path Setup: Load from USB.
How Set Path(s) (if done at PLC)	Can use PLC III for "Set" Path in 8= GPS Path. Key In or use GPS Antenna.	→	PLC III transmits Key In Lat-Long to Rover and coordinates stored in Rover Board, or tells Rover to use GPS Antenna.
Direction of Travel	Set in PLC, transmitted to Rover.	→	Tells Rover which Antenna to Use. - In Rev, RF Relay energized.
XTE & Side Off To	Transmitted from Rover.	←	Calculated in Rover, transmitted to PLC. (example 0.6" To Left)
Steering of Linear	Determined by PLC, 2:1 & 3:1 per XTE Level, set in 2=GUIDSET		Performed by PLC III Panel. - per XTE & Side off To from Rover.
Distance Functions	Curr Distance from Rover.	←	Rover calculates Distance from Home.
Rover Not Setup	Won't Run if "Rover Not Setup", even if in Run Position.	←	Rover Not Setup determined by Rover. - need Path information.
Waiting for Fix	Won't Run at start if "Waiting For Fix", even if in Run Position.	←	Waiting for Fix determined by Rover. - need Fix for XTE accuracy.
Screens Following	PLC = Double Lines border of Text Box.		Nav Bd = Single Line

9.2. Initial Field Operation Setup - Steps to Follow

- see Sections 10 & 11 for complete operation and menu functions of the Rover & PLC III.

Step 1: Power up PLC III and GPS Rover.
- Power to the Rover will cycle any time the PLC III Power is cycled.

Step 2: GPS Rover Board

Note: System Slide Switch in Rover should ALWAYS be in STOP.
- The PLC will tell the Rover when to Run and when to Stop.

STEERING Setup Group in Rover, set System Type to "PLC Linear".
(turns on Rover Communication to PLC.)

SETUP GROUP
STEERING → SYSTEM TYPE
PLC LINEAR → SET WITH
PLC PANEL.

GPS Rover Board main direction screen will show PLC & Rover Not Setup.

FWD PLC
ROVER NOT SETUP

The Path could be loaded from USB into the Rover at this point (or set later in Step 5).

Step 3: PLC III Panel

6=SYSTEM SETUP, enter all settings. The last will be Tractor Type, set to STT or Ultra G2.

UNITS
English
IRRIGATED WIDTH
01285 Ft → LINEAR TYPE
Single Tower Tractor

Step 4: PLC III Panel

2=GUID SETUP, set Guidance Type to GPS Navigation and enter all settings.
- this will establish communication between the PLC III and the Rover Nav Board.

GUIDANCE TYPE
GPS NAVIGATION → COMMUNICATION OK
TO GPS ROVER.
(Enter to Continue) → Continue with Guid Setup.

Step 5: PLC III Panel

If Path not loaded into Rover Board in Step 2 above, load now from USB or set as follows:
8=GPSPATH, set Home & Forward Waypoints, and any Intermediate Points.
- will be serially communicated to the Rover Nav Bd to be saved.
Path1 must be used, cannot be left blank. Enter additional Paths 2 – 4 as required.

PATH TO SET
PATH1
ACTION
SET HOME (REV END) → Continue with Path Setup.

Then Cycle power.

Step 6: PLC III Panel

Enter "1 = Run Setup" information, including distances and Path to use.

FWD AS = 04488 ft.
REV AS= 00012 ft.
CURR DIST 02700 ft
AR DELAY 01 Min → GPS PATH TO USE
PATH1

Step 7: Cycle Power to PLC III and GPS Rover Nav Board.

- The Path to use as entered in the PLC III in Run Setup will be communicated to the Rover.
- Then the Rover will be setup and will continuously send "XTE" and "Side Off To" to the PLC III every 3 seconds, regardless of PLC III Run/Stop Switch position.
- Path being used is displayed in upper right of PLC III and Rover Nav Bd screen.

PLC III Status Screen 1: Path being used will be displayed on upper right of PLC III.

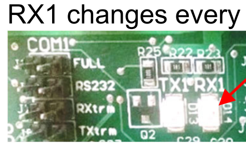
```
FWD APPL      0.75"  P1
AS

STEERING = L2:R1
```

Rover Nav Bd screen: Path being used should match the Path displayed in the PLC III.

```
FWD PLC       RTK  P1
XTE 00.9" TO LFT
```

Verify communication:

PLC Board:
- RX1 changes every 3 secs.



Rover Nav Board:
- TX1 & RX1 blink every 3 secs.

Step 8: PLC III Panel

Enter all other setup info: 3=AUX APP
4=EG AREA
5=DATE/TIME

Step 9: PLC III Panel – Run the System

RUN-STOP Switch of PLC III Panel set to RUN.

Note: Run/Stop System Switch on Rover Board will <u>not</u> be used, leave in STOP. The Run/Stop on the PLC will be in charge and tell the Rover Board when to Run.

Condition #1: Have RTK Fix and okay to Run.

- All okay and system starts to move. Status Screens display system information:

PLC Status Screen 1

```
FWD APPL      0.75"  P1
AS
EG L R
STEERING = L2:R1
```

Status Screen 2, etc.

```
CURR DIST =    02475 ft
TO FWD DIST= 02600 ft
SET IPM  018
     01:42:16 TO STOP
```

Status Screen 6

```
FWD XTE            01.5"

TOWARD LEFT END
SET018   L024   R012
```

Condition #2: No RTK Fix at Rover

- The Rover Board does not have an RTK Fix which is needed for sub-inch accuracy and Run is not allowed. The Rover will communicate to the PLC when it has a Fix.

PLC III Main Direction Screen:

```
WAITING FOR FIX
```

Rover Nav Board:

```
FWD PLC              P1
WAITING FOR FIX
```

Condition #3: Rover Not Setup

- This means the above steps have not been followed and Run is not allowed.

```
GPS ROVER NOT SETUP
RUN NOT ALLOWED
(SET RUN SWITCH BACK
TO STOP)
```

10. ROVER OPERATION

10.1. Rover Title and Main Direction Screens

If Rover has been Setup

GPS NAVIGATION
T-L vHL59

LOADING WAYPOINT

WAITING FOR RECVR
TIMER 300

FWD PLC
WAITING FOR FIX

FWD PLC RTK P1
XTE 00.4” TO LFT

If Rover Not Setup

GPS NAVIGATION
T-L vHL59

WAITING FOR RECVR
TIMER 300

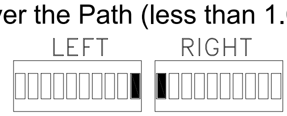
FWD PLC RTK
ROVER NOT SETUP

10.2. Rover Light Bars

Watch the Light Bars in conjunction with the XTE on the Main Direction Screen

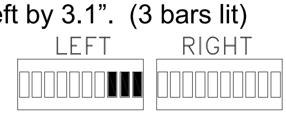
Example 1: Linear/Rover centered over the Path (less than 1.0” XTE).

FWD PLC RTK P1
XTE 00.4” TO LFT



Example 2: Linear/Rover off to the Left by 3.1”. (3 bars lit)

FWD PLC RTK P1
XTE 03.1” TO LFT



10.3. GPS Navigation Linear Example

Example of Linear system using GPS Navigation.
Use as reference for Test Screens of Rover
on next page.

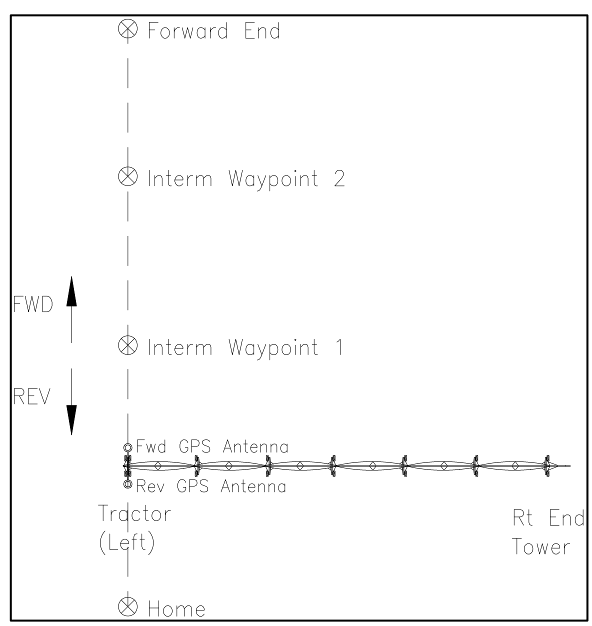


Figure 23. GPS Navigation Linear Example.

10.4. Rover Test Screens

Screen 1:

RTK	FIX
SATS	18

RTK Fix Status in the upper right:
- blank = starting to acquire Fix
- Float = close to getting an RTK Fix
- Fix = most accurate and okay to move.

Number of satellites being tracked.

Screen 2:

GGA Age 1	Sec

Age of the Correction information from the Base. **(added HL54)**
- should be 1 or 2 seconds most of the time.

Screen 3: Several possibilities (NOTE: Uses Antenna Location in Direction of Travel.)

Ex. 1

To Fwd	04100Ft
Frm Hom	00500Ft

Show distances between the 2 waypoints being used.
- no Intermediate Waypoints.
- To & From dependent on direction of travel.

Ex. 2

To Int1	00350Ft
Frm Hom	00250Ft

Intermediate Waypoint being used.

Ex. 3

Transition
Zone

If Intermediate Points, will need to transition the From & To.
- within 2.5 ft. of To, make the transition.

Screen 4: Current Distance from Home (changed in HL54)

CURRENT DISTANCE
00000Ft

Distance from the Home (Reverse End) Waypoint **to Center of Tractor or Tower Base if Guidance at Center**.
- Curr Dist is transmitted from the Rover to PLC III.

Int. Waypoint transitions occur at GPS Ant. distance from Home.

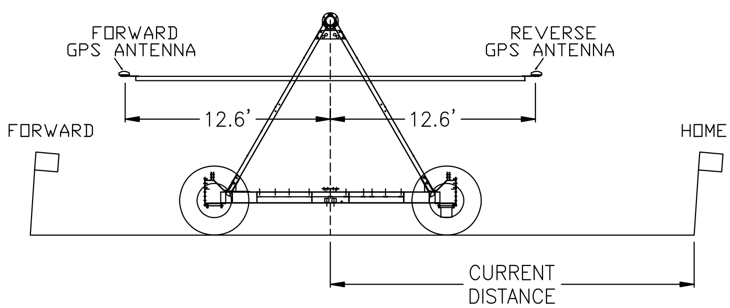


Figure 23.1.
Current Distance.

Screen 5 etc.

BASE	40.3484889N
	098.1654215W

Base Lat Long

RVR	40.3484889N
FWD	098.1654215W

Rover Lat Long (show direction antenna being used)

TO	40.3484889N
IN1	098.1654215W

Show the "TO" Waypoint
Hom, Fwd, In1, In2, etc. to In9.

FROM	40.3484889N
HOM	098.1654215W

Show the "FROM" Waypoint
Hom, Fwd, In1, In2, etc. to In9.

08/03/17	THU
03:15:59	PM

10.5. Rover SETUP Groups

Press “SETUP” key to toggle into and out of Setup Groups.
Use “+” or “-“ keys to scroll to the different Setup Groups.
Use “TEST/ENTER” key to enter and scroll through each Setup Group
Use “+” or “-“ keys to scroll settings for each item to be set.

SETUP GROUP STEERING	Sec 10.6
SETUP GROUP PATH	Sec. 10.7
SETUP GROUP DATALOGGING	Sec. 10.8
SETUP GROUP SAFETY HISTORY	Sec. 10.9
SETUP GROUP GPS RECVR INFO	Sec. 10.10
SETUP GROUP DATE / TIME	Sec. 10.11

10.6. STEERING Setup Group

SETUP GROUP
STEERING

SYSTEM TYPE
PLC LINEAR

Choices now are: Manual Linear or PLC Linear
Select “PLC Linear”

If PLC LINEAR, the Steering settings will be set, stored and used in the PLC Panel.
Alert the operator with this screen.

SET WITH
PLC PANEL.

Press Enter or Setup Key, return to Setup Groups.
No more screens for Steering Group.

10.7. PATH Setup Group

SETUP GROUP PATH

PATH OPTIONS SET PATH

SET PATH = use the PLC III to set the Waypoints.
ERASE = Erase a Path.
LOAD = Load Waypoints from USB
SAVE TO USB = save a Path in the Nav Bd to the USB.
REVIEW PATH = Review Waypoints (Added in HL54)

If SET PATH:

PATH OPTIONS SET PATH

SET WITH PLC PANEL.

Use the PLC III Panel to Set the Paths.
Press Enter or Setup Key, return to Setup Groups.

If ERASE:

PATH OPTIONS ERASE

ERASE = Erase an entire Path.

PATH TO ERASE PATH1

Path1, Path2, Path3 or Path4

PATH1 DOES NOT EXIST

Check if that Path really exists.
Press Enter or Setup Key, return to Setup Groups.

ERASE PATH1 NO

If No, do not erase and return to Setup Groups.
If Yes, erase and return to Setup Groups.

If LOAD:

- Load is used to load a path from the USB to the Navigation Board, file must be "*.way".
- Path could be one that was Saved to USB, or one created in a text, tab-delimited format.
- Every Path must have a Home Point and a Forward End Point. Intermediate points typically are not needed.

Format for .way File to Load for a Linear Path:

Home (Rev End)	Lat	Long
Intermediate Waypoints if any (up to 9 Interm)	Lat	Long
Forward End	Lat	Long

Example Path with 2 Intermediate Waypoints: (8 decimal points maximum)

40.58040789	-98.26855550
40.58060412	-98.26855570
40.58090784	-98.26855590
40.58110298	-98.26856052

PATH OPTIONS LOAD

LOAD = Load Waypoints from USB

PATH TO LOAD PATH1

Path1, Path2, Path3 or Path4
- Ability to Store up to 4 Paths, specify which is being Loaded.

ERASE EXISTING NO

If that Path already exists, give option to say No.
- If "No", to Main Menu and keep it. If "Yes", Erase and continue.

Screen	Description
SYSTEM STOPPED? NO	Yes or No. If No and Enter, go back to Path Options. Must be Yes to advance.
USB INSERTED? NO	Yes or No. If No and Enter, go back to Path Options. Must be Yes to advance.

If USB Inserted is Yes, but No USB Detected.

Screen	Description
NO USB DETECTED (PRESS ENTER)	Press Enter and go back to Setup Group - Path
SELECT FILE PATH1.way	+ or – to scroll through the files on the USB drive - looking for file extensions of "*.way" - File name must be 8 characters maximum. If "Error, File Not Found", check file name. Must end with ".way"
LOAD FROM USB? No	Yes or No, If No and Enter, go back to Path Options. Yes and Enter, begin downloading file from USB.
LOADING PATH1 *	Indicates the Path is being Loaded.
WAYPOINTS LOADED 04 (Enter)	Once file is loaded, the number of Waypoints is shown. Press Enter, go to next screen if Interm Points. If no Interm Waypoints then return to Setup Groups.
ORGANIZING PATH1	If 1 or more Intermediate Points, then Organize them. Waypoints are organized in order of distance from Home. - the 1st point in the file is always Home (Rev end) and the last point is always the Fwd (farthest point) - any points in between are organized by distance from Home. - the From and To points to use for navigation will then be determined by Rover distance from Home.
ORGANIZING PATH1 DONE (Enter)	Press Enter to Continue. Go to Setup Groups.
CYLCE POWER TO NAV BD (Enter)	You will need to power the Nav Board Off and back On to load the correct waypoints for Navigation.

If SAVE TO USB:

- An existing path can be saved to the USB for viewing of lat-long waypoints on a computer or to be Loaded back into the Rover if Restore Settings is to be used.

Screen	Description
PATH OPTIONS SAVE TO USB	SAVE TO USB = save a Path in the Nav Bd to the USB.
PATH TO SAVE PATH1	Enter which Path to Save. - Path1, Path2, Path3 or Path4.
NO PATH (Press Enter)	If no path exists on the Navigation Board for the one selected, press Enter and return to Path Options.
SYSTEM STOPPED? NO	Yes or No. If No and Enter, go back to Path Options. Must be Yes to advance.

USB INSERTED? NO

Yes or No. If No and Enter, go back to Path Options. Must be Yes to advance.

If USB Inserted is Yes, but No USB Detected.

NO USB DETECTED (PRESS ENTER)

Press Enter and go back to Setup Group - Path

SAVE FILE 02021017

File is saved to the USB
- **Example "02021017.way"**
- 2nd .way file saved on the USB on Feb. 10, 2017.

Example "02021017.WAY" file Saved to USB:

PATH1 BASE	40.5512630999	-98.2941304099	
LAT	LONG	DISTANCE	WP
40.538831700	-98.295098999	0.0000000000	Home
40.543693699	-98.295070199	1774.2950079542	Int 1
40.548012999	-98.295041100	3350.5464347996	Int 2
40.551100599	-98.295021800	4477.3089915631	Fwd

If REVIEW PATH (Added in HL54):
- An existing path can be reviewed at the Rover.

PATH OPTIONS REVIEW PATH

PATH TO REVIEW PATH1

Enter which Path to Review.
- Path1, Path2, Path3 or Path4.

Use "+" and "-" keys to scroll back and forth through Path waypoints.
Start with Home Reverse, then any Intermediate Points, then Forward.
Press ENTER to return to Setup Group – Path.

REV 40.1234567N 098.1234567W

INT1 40.1234567N 098.1234567W

INT2 40.1234567N 098.1234567W

FWD 40.1234567N 098.1234567W

10.8. DATALOGGING Setup Group

This may be used to log information from the Rover such as Rover Lat-Long, XTE, etc. to a USB.
NOTE: If power is lost while data logging, the data logging will resume at next power up.
Note in HL56, added operation that if Datalogging is on and power is cycled and then "Error Creating File" for more than 1 minute like it can't find the USB when powered back up, then Datalogging is set to Stop and the Error message no longer appears.

Use "TEST/ENTER" key to enter through the Data Logging setup and Start or Stop Logging.
Use "+" or "-" keys to scroll settings for each item to be set.

Screen	Description
SETUP GROUP DATALOGGING	
DATA LOGGING START	Start or Stop Data Logging If Start, advance to next screen.
USB INSERTED? NO	Make sure USB is inserted and recognized.
NO USB DETECTED (PRESS ENTER)	If USB Inserted is Yes, but No USB detected, press Enter. Will go back to Setup Group Data Logging.
LOGGING INTERVAL 01 SECS	Set data logging interval. (01 to 60 secs)
LOGGING ACTIVE (Enter)	Press Enter to exit to Main Dir Screen. **- Logging will start only after Enter is pressed.**

Start Logging – regardless of System Switch - Run Stop Position.

Press "SETUP" key to return to the Setup Groups.
Press "SETUP" key again to return to the main base screen.

At the Main Dir Screen, the Direction and word "Log" will flash alternately when Data Logging.

FWD PLC RTK P1 XTE 00.4" TO LFT	**LOG** PLC RTK P1 XTE 00.4" TO LFT

File Format for Data Logging to USB:
- File will be saved with number & date for filename and ".log" ext
  - Example: 01072816.log = 1st file logged on July 28, 2016
- Transfer from the USB to a computer, open in Notepad to review.
  - or the file can opened in Excel (Delimited, Tab Delimited, General Column Format)

To **Stop Logging**, access the Data Logging Setup Group and Stop and press Enter.

Screen	Description
SETUP GROUP DATALOGGING	
DATA LOGGING STOP	Start or Stop Data Logging Change to Stop and press Enter.

Rover Log File Information:

Log Parameters – Used with PLC Linear

Firmware HL54												
BASE	40.5778099	-98.2681457										
HOME	40.5385572	-98.2950991										
FWD	40.5511006	-98.2950193										
DATE	**TIME**	**AM/PM**	**DIR**	**ROVER LAT**	**ROVER LONG**	**FROM LAT**	**FROM LONG**	**TO LAT**	**TO LONG**	**XTE**	**SIDE**	**CURR DIST**
3/6/2020	12:53:46	PM	FWD	40.5409288	-98.29508442	40.540047	-98.2950899	40.5413	-98.2951	0.06	TO RIGHT	853

DIST FROM HOME	DIST TO NEXT	DIST FROM LAST	GGA QUAL	# SATS	UTC	RUN	LAT/LON CNG	WP1	WP2	XTE TIME	PATH	GGA AGE
865	122	322	4	19	191444	1	119	1	2	298	3	1

10.9. SAFETY HISTORY Setup Group

Use "TEST/ENTER" key to enter through the Safety History setup.
Use "+" or "-" keys to scroll settings for each item to be set.

SETUP GROUP
SAFETY HISTORY

SAFETY HISTORY
IS EMPTY

If no Safeties have occurred yet, the Safety History will be empty.

SETUP GROUP
VIEW

Select **View** or **Save**.

If **View**

S1 RECVR RESPOND
06/24/16 11:22A

Etc. last 8 saved.

If **Save** to USB :

USB INSERTED
NO

If No, back to Safety History

NO USB DETECTED
(PRESS ENTER)

If No USB, back to Safety History

SAVE SFTY HIST
(PRESS ENTER)

Press Enter to save Safety History to the USB.

File Format for Save Safety History to USB:

- File will be saved with number & date for filename and ".saf" ext.
- Copy from the USB to a computer and change to ".txt" extension, or open in Notepad.
  Example = 01062417.saf (1st file saved on June 24, 2017)

No.	Message	Date	Time
S1	RECVR NO RESPOND	06/24/17	11:22A

10.10. GPS RECVR INFO Setup Group (below is for Hemisphere)

Use "TEST/ENTER" key to enter and scroll through the GPS Receiver Information.

	- GPS Engine	Ex.	DF5r	
	- Bd Serial No.		111111	
	- Bd Fleet No.		1	
	- Bd Hdwre Version		07	
	- Bd Prodctn Code		02122009	
	- Subscrpt Begin		01/01/1900	
	- Subscrpt Expir		01/01/3100	
	- Software Version		1.2Qq06t	
	(Following added in HC38)			
	- RTK Subscrpt		Yes	
If Base	- RTK TX Status		22	
	- P300 Correction		None or SBAS	
	- RTK Accuracy		0	Accuracy Status
	- L1,L2,L5,G1,G2		10,10,0,7,7	Number of Satellites from GPS & Glonass
	- B1,B2,E1B,E5A,E5B		0,0,2,2,2	Number of Satellites from BeiDou & Galileo P326 & P34 (P300 & P306 = No Subscription)
	- QL1, QL2		0,0	Number of Satellites from Japan system P326 & P34 (P300 & P306 = No Subscription)

RTK Board	Engine	Software	Ship Dates
Red P300 =	DF3r	1.2Qe	04/14-11/15
Red P306 =	DF4r	1.2Qf9b	11/15-03/18
Green P326 =	DF5r	1.2Qq06t	04/18-
Black P34 =	DF5r	1.2Qr04ta	?? -

Subscrpt Expir last 4 digits:
- 3100 = Base or Rover
- 3104 = Base ONLY
- 3108 = Base or Rover
- 4104 = P326 Base or Rover
- 4140 = P34 Base or Rover

Press "SETUP" key to return to the Setup Groups.
Press "SETUP" again to return to the main screen.

10.11. DATE/TIME Setup Group

Use "TEST/ENTER" key to access the Date / Time settings.

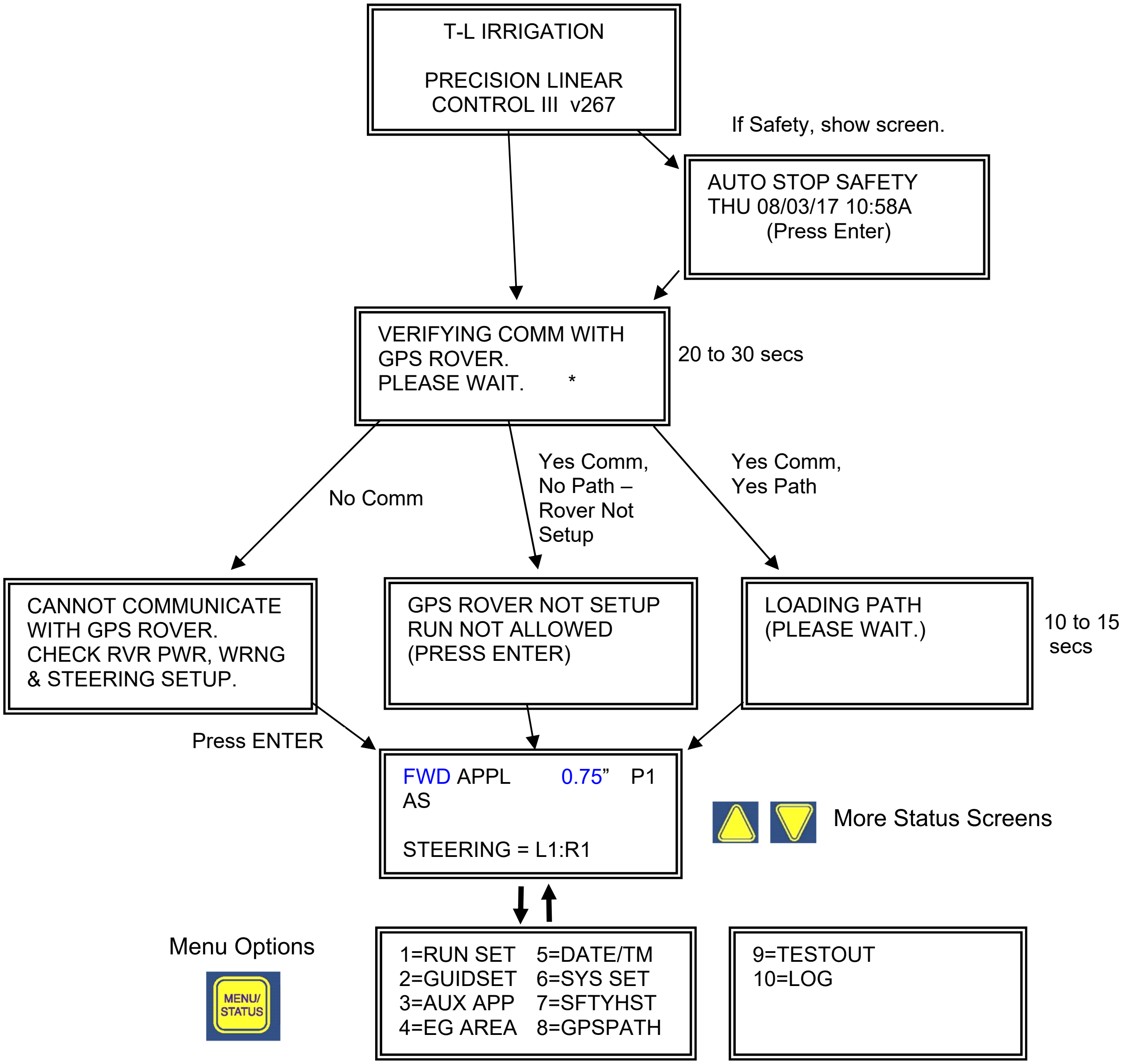
```
08/03/20   MON
03:15      PM
```

- Use "+ and "-" keys to set each item,
  - then press Enter after each is set.

Press "SETUP" key to return to the Setup Groups.
Press "SETUP" key again to return to the main screen.

11. PLC III SCREENS & MENUS

PLC III Power Up - GPS Navigation:



STATUS Screens (STT & ULTRA G2)

Status Screen 1
"Main Direction Screen"

("Enter" toggles Screen 6)

```
FWD APPL        0.75"    P1
AS
EG L R
STEERING = L1:R1      AO
```

Line 1 = Direction, Appl Rate, Path if GPS Nav.
Line 2 = Auto Stop (AS) or Auto Rev (AR) for that direction as set in 1=RUN Setup.
Line 3 = End Guns On (if used).
Line 4 = Steering Status.
(AO shown if in Anti-Oversteer)

Status Screen 2

```
CURR DIST =    02475 ft
TO FWD DIST= 02600 ft
SET IPM  018
        01:42:16 TO STOP
```

Line 1 = Current Position of system.
NOTE: HL54 Curr Dist = Center of Twr Base
See Sec. 11.1 1=RUN SETUP for more info.
Line 2 = Direction & Set Distance to AS or AR.
Line 3 = Set Travel Speed.
Line 4 = Time to reach Set Distance at Set Spd.

Status Screen 3
(If sensors enabled)

```
WATER PRESSURE
                  38  PSI
WATER FLOW
                 723  GPM
```

Line 2 = Water Pressure (If installed).

Line 4 = Water Flow (If installed).

Status Screen 4

```
SET IPM            018

LEFT SENSOR              *
RIGHT SENSOR             *
```

Line 1 = Set Travel Speed.

Lines 3 & 4: Asterisk will flash when respective speed sensor is operational.

Status Screen 5

```
LEFT SET IPM     012
LEFT ACT IPM     014   -
RGHT SET IPM     024
RGHT ACT IPM     022   +
```

Set Speed for each end shown.
- per the Steering Ratio.

Actual Speed for each end shown.
Speed Changes displayed with "+" & "-".
- speed changes occur every 10 seconds for 6 times, then every 30 seconds.

Status Screen 6 – GPS Navigation
(At Status Screen 1, "Enter" key toggles to Status Screen 6 and back.)

```
FWD XTE          01.5"
"0"
TOWARD LEFT END
SET018   L018   R018  AO
```

Direction, Cross Track Error Dist off Path.
"0" if Single Tower Tractor, "0" or "180" if Ultra G2 as set in 1=RUN Setup.
Side toward which the GPS Antenna is off.
Set Travel Speed, Set L & R End Speeds based on Steering Ratio.
AO shown if in "Anti-Oversteer" condition.
- per settings in Guidance Setup.

Additional Status Screens if "GPS Navigation"

```
ANTENNA DISTANCE
To FWD       04000Ft
Frm HOM      02487Ft
```

Same as Rover Test Screen 3.

```
No. of Satellites        19
RTK                     Yes
Overwater Timer        1563
```

Number of Satellites being received.
RTK Fix (from Rover) Yes or No.
Overwater Timer in Seconds.
- Uses setting from 1=Run Setup if Enabled.
- Additional 300 seconds added at Startup.

Other Possible Status Screens #1:

Rover Not Setup	If PLC III System Toggle set to Run, show this screen. GPS ROVER NOT SETUP RUN NOT ALLOWED (SET RUN SWITCH BACK TO STOP)	Rover Nav Board Screen if Not Setup. FWD PLC P1 ROVER NOT SETUP
Waiting for Fix (No RTK Fix)	If PLC III System Toggle set to Run, show this screen. - At startup, will not move. - If running with a Fix, then loses Fix, will continue to run. - RTK Fix Safety after Delay time set in 2=GUID Setup. - XTE Safety not active when Waiting. WAITING FOR FIX	Rover Nav Board Screen if Not Setup. FWD PLC P1 WAITING FOR FIX
Speed Disabled (GPS Nav)	FWD SPD DISABLED AS RIGHT ETSEN FIX TRCK DSABL L052 R043 IPM 06:23	ETSEN FIX TRACK SFTY is Disabled in Run Setup. DIR button to change direction and toggle which end to manually adjust speed. “+” “-“ to adjust speed when in Run mode. Actual speed displayed if Sensors working. Time left before automatic shutdown in this mode. Timer can be reset when toggling direction.
Auto Rev Delay Timer. (Ditch Feed)	AR DELAY 03 min TIME LEFT 02:25 02599 ft.	Auto Reverse Timer Count down screen if Ditch Feed and AR Delay. Current Distance location of system in field.
Aux Application Zone	FWD AUX1 0.75” P1 AS EG L R STEERING = L1:R1 AO	APPL and AUX Area flashing every second Example: APPL-AUX1-APPL-AUX1-etc. Use Aux Appl Table set for Path being used.

Single Tower Tractor - Pivot Mode Screens

Screen 1	CCW SPD 122 IPM AS 13 Hr/Rv STEERING = PIVOT	Set Direction & Travel Speed. (End Tower Speed can be set up to 1.7 times the Max Linear Speed in 6=Sys Setup.) Hr/Rev calculated using Dist to End Tower in Run Setup.
Screen 2	SET IPM 122 LEFT SENSOR * RIGHT SENSOR *	
Screen 3	LEFT SET IPM 000 LEFT ACT IPM 000 RGHT SET IPM 122 RGHT ACT IPM 120 +	Up/Down Arrow keys scroll these screens.

ULTRA G2 Pivot Mode Screens

Screen 1 with Encoder Encoder used for angle.	CCW SPD 45 IPM AS 0.55” 48 Hr/Rv EG R 182.5 deg PIVOT TO: 180.0 deg	Set Direction & Travel Speed when pivoting. DIR button to change CW to CCW. Number keys or “+” “-“ to enter application. (End Tower speed can be set up to 1.7 times the Max Linear Speed in System Setup.) - Current Angle and Angle to Pivot To shown on screen.
Screen 1 no Encoder Encoder not used.	CCW SPD 45 IPM AR 0.55” 48 Hr/Rv STERING = PIVOT	Set Direction & Travel Speed when pivoting. DIR button to change CW to CCW. Number keys or “+” “-“ to enter application. (End Tower speed can be set up to 1.7 times the Max Linear Speed in System Setup.) AS/AR: Will display AR if set in Run Setup.
Screen 2	SET IPM 122 LEFT SENSOR * RIGHT SENSOR *	
Screen 3	LEFT SET IPM 000 LEFT ACT IPM 000 RGHT SET IPM 122 RGHT ACT IPM 120 +	
Screen 4	OVERWATER TIMER Failure to Move 1 Deg 1054 secs	
Auto Reverse Delay Timer. (Pivot Mode – No Encoder)	AR DELAY 03 min TIME LEFT 02:25	Timer Count down screen, shows time left. - Auto Reverse in Pivot Mode Time set in 1=Run Setup

ULTRA G2 Rotate Tractor Screen

Rotate Tractor	CW ROTATE AS LEFT ETSEN PRIM REF DSABL L000 R000 IPM 10:00	Set Direction to Rotate Tractor. DIR button to change CW to CCW and to access Reset Timer screen. “+” “-“ keys to adjust Left End (Tractor) speed.

PLC III Menus:
GPS Nav

1=RUN SET	5=DATE/TM	9=TESTOUT
2=GUIDSET	6=SYS SET	10=LOG
3=AUX APP	7=SFTYHST	
4=EG AREA	8=GPSPATH	

11.1. 1 = RUN SETUP

LINEAR MODE Screens – Order of RUN SETUP Screens if STT

FWD AS =	04488 ft.
REV AS=	00012 ft.
CURR DIST	02700 ft
AR DELAY	01 Min

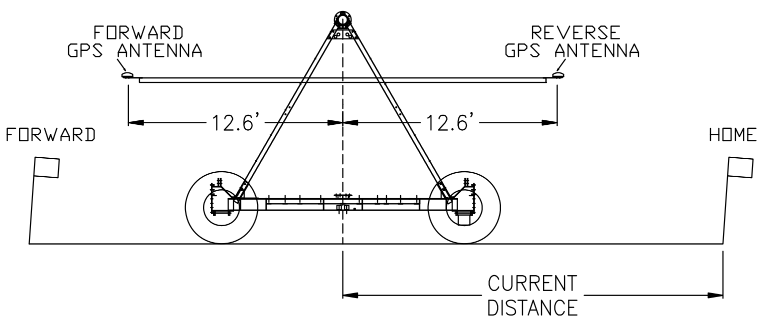
NOTE: Max Fwd Dist = Distance Home to Fwd Lat Lon points – 12 ft.
Min Rev Dist = 12 ft (changed in v264)
Hose Drag can only be set to “AS” (Auto Stop).
Ditch Feed can be set to “AS” or “AR” (Auto Reverse).
CURR DIST: HL54 = Home to Center of Tractor or Tower Base
- Cannot be changed.
AR DELAY only shown if Ditch Feed.

IMPORTANT: If using FWD or REV Distances to Auto Stop or Reverse, make sure the linear system does not travel out of the field and into an obstruction.
- Field Ends MUST HAVE BARRICADES.
- Program the FWD & REV Distances 12 ft in from the HOME & FWD locations.
- v264 has taken this into account.

NOTE HL54: CURRENT DISTANCE IS NOW THE CENTER OF THE TRACTOR/TOWER BASE.

- Previously it was the Distance to either the Forward or Reverse GPS Antennas and would change by 26 ft when changing direction. With HL54 it should not change now.
- The Reverse and Forward GPS Antennas are still used for Navigation, but 12.6 ft is either added or subtracted from their location to calculate the center of the Tractor which is now the Current Distance (or Tower Base if End Feed with GPS at Center).
- Intermediate Waypoint transitions still occur at the GPS Antenna distance from Home.

Fig 23.2

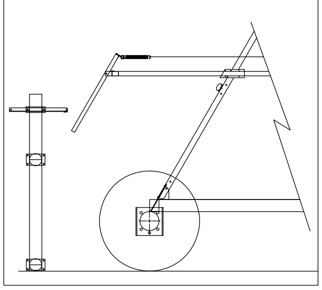


<u>CURR DIST is used for:</u>
Auto Stop/Reverse
End Gun Areas
Aux. Appl. Areas

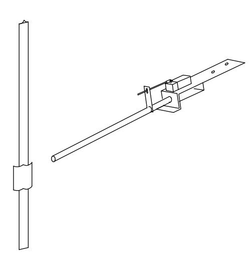
Typical Stop or Reverse methods:

Fig 23.3

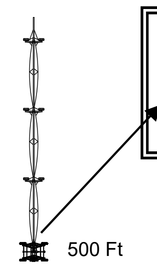
End Tower Barricade Switch
- At Field Ends



Position Stop Switch
- For Hose Changes



FWD or REV AS/AR
- Ditch Feed systems, in field operation



FWD AS =	04488 ft.
REV AR=	00500 ft.
CURR DIST	00500 ft
AR DELAY	01 Min

DISTANCE EXPLANATION

NOTE HL54: CURRENT DISTANCE IS NOW THE CENTER OF THE TRACTOR/TOWER BASE.

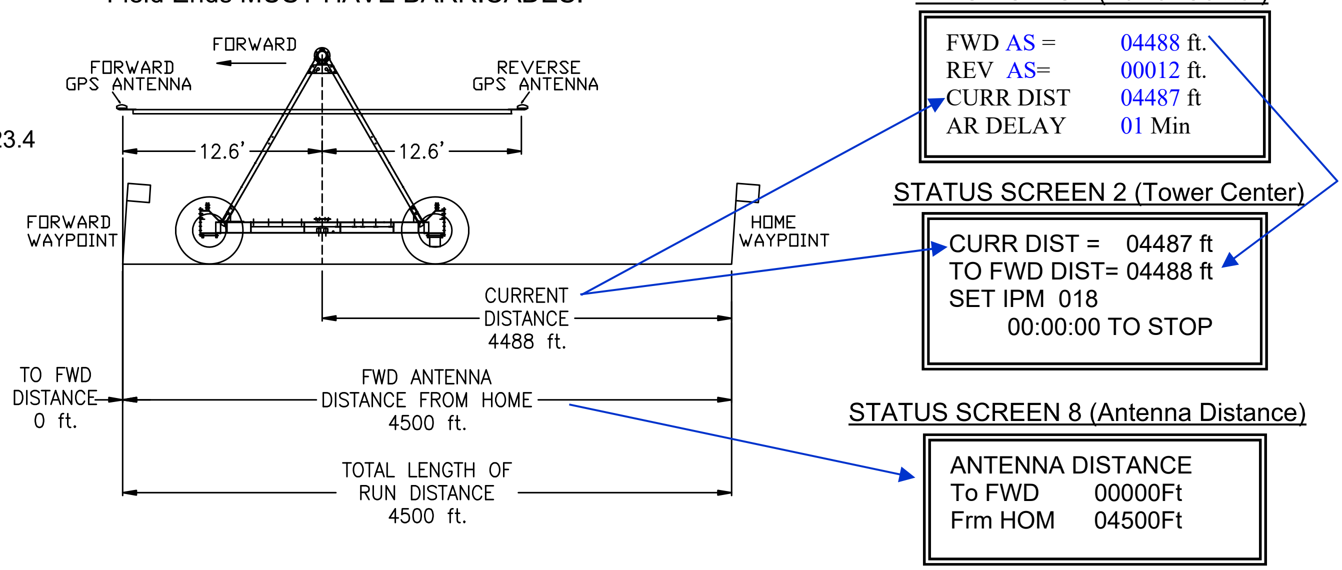
CURR DIST is used for: Auto Stop/Reverse
End Gun Areas
Aux. Appl. Areas

IMPORTANT: If using FWD or REV Distances to Auto Stop or Reverse, make sure the linear system does not travel out of the field and into an obstruction.
- Field Ends MUST HAVE BARRICADES.
- Program the FWD & REV Distances 12 ft in from the HOME & FWD locations.

DISTANCE EXAMPLE IN FORWARD AT FORWARD END

Program the FWD AS Distance 12 ft in from FWD end.
Field Ends MUST HAVE BARRICADES.

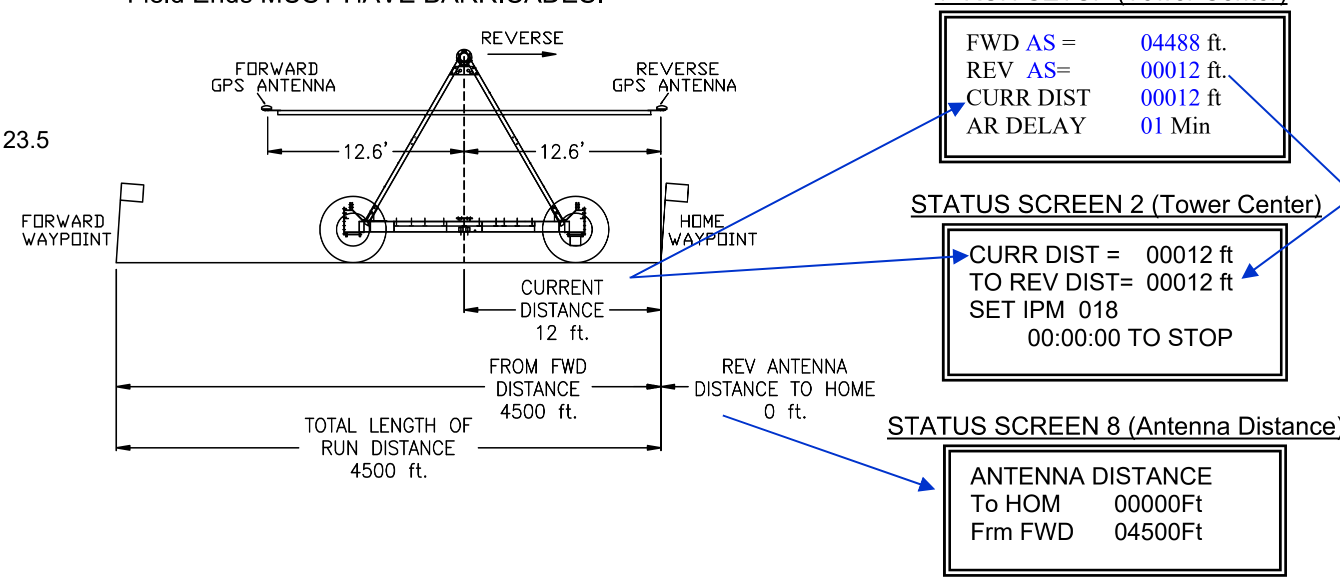
Fig 23.4



DISTANCE EXAMPLE IN REVERSE AT REVERSE END

Program the REV AS Distance 12 ft in from REV end.
Field Ends MUST HAVE BARRICADES.

Fig 23.5



ET SENSOR SAFETY 60 sec ETSEN FIX TRACK SFTY ENABLE

ET Sensor Safety time delay = time reading 0 ET Speed for Left or Right before ET Sensor Safety occurs.
ETSEN FIX TRACK SFTY Disables the following:

- Left & Right ET Sensor Safety.
- Automatic Speed Control
- RTK Fix (Waiting for Fix)
- XTE (Track Error) Safety

If ETSEN FIX TRACK SFTY set to DISABLE

MFC VALVES ADJUSTING WAIT 45 SECONDS

Then to Status Screen 1.

OPERATING MODE LINEAR

Linear or Pivot Mode

LOW POWER SAFETY ENABLE

GPS PATH TO USE PATH1 (Cycle Power if changing Paths.)

Path1 or Path2 or Path3 or Path4.
IMPORTANT: If selecting a different path, cycle power on the PLC III & Rover Nav Bd.

WARNING PATH1 NOT SET (Enter)

If select a Path, and that Path not Setup & stored in the Rover, alert the operator.

TRACTOR OVRWTR TIMER DISABLE

Enable or Disable

If Enable

OVERWATER SETTINGS TRACTR DISTNCE 20 FT TIME 30MIN

<u>PIVOT MODE Screens – Order of RUN SETUP Screens if STT</u>

Set to Pivot Mode, exit Run Setup, then enter 1=RunSetup:

OPERATING MODE PIVOT DIST TO END TOWER 1000 ft

END TOWER STOP SET? No

LOW POWER SAFETY ENABLE

LINEAR MODE Screens – Order of RUN SETUP Screens if ULTRA G2

OPERATING MODE
LINEAR
SPAN POSITION
“0” Zero

FWD AS =	09000 ft.
REV AS=	00000 ft.
CURR DIST	02700 ft
AR DELAY	01 Min

Important for HL54, see Current Distance explanation for STT.

ET SENSOR SAFETY
60 sec
ETSEN FIX TRACK SFTY
ENABLE

MFC VALVES ADJUSTING
WAIT 45 SECONDS

LOW POWER SAFETY
ENABLE

ROTATE ULTRA TRACTOR
90 DEGREES
DISABLE

WHEELS SET TO ROTATE
YES
2 GEARS DISENGAGED
YES

SWITCH TRACTOR COILS
YES
TRACTOR GUIDANCE OK
YES

READY TO ROTATE
YES

GPS PATH TO USE
PATH1

WARNING
PATH1 NOT SET
(Enter)

TRACTOR OVRWTR TIMER
ENABLE

OVERWATER SETTINGS
TRACTR DISTNCE 20 FT
TIME 30MIN

PIVOT MODE Screens – Order of RUN SETUP Screens if ULTRA G2

OPERATING MODE
PIVOT
DIST TO END TOWER
1000 ft

Encoder for Pivot = NO

ENCODER FOR PIVOT
NO
MUST SET STOPS
Press Enter

PIVOT MODE AS/AR
CW AS
CCW AR
AR DELAY 01 min

Encoder for Pivot = YES

ENCODER FOR PIVOT
YES
ENTER DEGREES
Press Enter

CURRENT ANGLE
000 deg
ANGLE TO PIVOT TO
180 deg

OVERWATER TIMER
ENABLE
TIME TO MOVE 1 DEG.
60 min

ET SENSOR SAFETY
60 sec

LOW POWER SAFETY
ENABLE

11.2. 2 = GUID SETUP

Screen	Description
GUIDANCE TYPE GPS NAVIGATION	STT = GPS Navigation ULTRA = GPS Nav at Ult Trctr or GPS Nav at Ult Cntr

Once the GPS Navigation method selected, then PLC III will establish communication with Rover:

Screen
VERIFYING COMM WITH GPS ROVER. PLEASE WAIT. *

After trying to verify communication with the Rover, one of two messages will appear:

Screen	Description
CANNOT COMMUNICATE WITH GPS ROVER. CHECK RVR PWR, WRNG & STEERING SETUP.	If can't communicate to Rover: - note to check power to Rover, comm wiring, and Steering Setup of Rover set to "PLC Linear". Press Enter, or Menu/Status – Return to Main Menu screen.
or	
COMMUNICATION OK TO GPS ROVER. (Enter to Continue)	If communication is okay, that means the GPS Rover is set to "PLC Linear".

Screen	Description
STEERING SIG TIME 10 secs.	Amount of time the XTE level must be at or above (or below) set steering level before steering is activated.
2:1 XTE STR LVL 0.5 Inches 3:1 XTE STR LVL 1.5 Inches	Cross Track Error level at which 2:1 and 3:1 steering should occur in PLC. 2:1 = 0.5" to 2.0" in 0.5" increments 3:1 = 1.0" to 3.0" in 0.5" increments
ANTI OVRSTR OUT 2.0 Inches ANTI OVRSTR IN 1.5 Inches	XTE distance the antenna must move away from the path before looking at Anti-Oversteer In. 0.5 to 10.0" in 0.5" increments XTE distance the antenna must move back in towards the path before activating Anti-Oversteer (1:1 steering) 0.5 to 10.0" in 0.5" increments
ANTI OVRSTR BACK OUT 0.5 Inches	XTE distance the antenna must move back out if Anti-Oversteer is already activated, to re-activate steering. 0.5 to 10.0" in 0.5" increments
TRACK ERROR ZONE 12 Inches RTK FIX DELAY 05 Minutes	If the XTE distance off the path is equal to or greater than the Track Error Zone for 10 seconds, cause a PLC III "XTE Safety". 5 to 30 inches (10 second delay initial & run). RTK Fix Delay: Initial and Run Time delay. If RTK Fix not established in this time, cause a PLC III Safety. This is the time after "Waiting for Fix" appears on the PLC III. 1 to 10 minutes. At Start, the system will not move when "Waiting for Fix". Once running with a Fix and Fix is lost, the system will continue to run and will safety after this time setting.

11.3. *3 = AUX APPLICATION Areas*

- Auxiliary Application Areas can be set to apply a different amount of water than what is set on the Main Direction Status Screen 1.
- The Auxiliary Application areas use the "Current Distance" from Home for activation.

AUX APPL DISABLE	Enable or Disable Disable = Aux Application will not be used.

If ENABLED:

Select Path for which to set Aux Application.

AUX APPL ENABLE PATH TO SET AUX APPL PATH1	Enable Set Path1 or Path2 or Path3 or Path4

FORWARD = Two different application rates and distance ranges can be set for each Path.

Path 1	Path 2	Path 3	Path 4
FWD AUX1 APPL 0.75" 016 IPM FWD AUX1 APPL RANGE 00500 ft TO 01000 ft	FWD AUX1 APPL 0.75" 000 IPM FWD AUX1 APPL RANGE ----- ft TO ----- ft	FWD AUX1 APPL 0.75" 000 IPM FWD AUX1 APPL RANGE ----- ft TO ----- ft	FWD AUX1 APPL 0.75" 000 IPM FWD AUX1 APPL RANGE ----- ft TO ----- ft
FWD AUX2 APPL 0.50" 024 IPM FWD AUX2 APPL RANGE 01600 ft TO 02200 ft	FWD AUX2 APPL 0.75" 000 IPM FWD AUX2 APPL RANGE ----- ft TO ----- ft	FWD AUX2 APPL 0.75" 000 IPM FWD AUX2 APPL RANGE ----- ft TO ----- ft	FWD AUX2 APPL 0.75" 000 IPM FWD AUX2 APPL RANGE ----- ft TO ----- ft

REVERSE NOTE = *Reverse Distance Ranges must be set from high to low distance number.*

Path 1	Path 2	Path 3	Path 4
REV AUX1 APPL 0.75" 016 IPM REV AUX1 APPL RANGE 02200 ft TO 01600 ft	REV AUX1 APPL 0.75" 000 IPM REV AUX1 APPL RANGE ----- ft TO ----- ft	REV AUX1 APPL 0.75" 000 IPM REVAUX1 APPL RANGE ----- ft TO ----- ft	REV AUX1 APPL 0.75" 000 IPM REV AUX1 APPL RANGE ----- ft TO ----- ft
REV AUX2 APPL 0.50" 024 IPM REV AUX2 APPL RANGE 01000 ft TO 00500 ft	REV AUX2 APPL 0.75" 000 IPM REV AUX2 APPL RANGE ----- ft TO ----- ft	REV AUX2 APPL 0.75" 000 IPM REV AUX2 APPL RANGE ----- ft TO ----- ft	REV AUX2 APPL 0.75" 000 IPM REV AUX2 APPL RANGE ----- ft TO ----- ft

11.4. 4 = EG AREAS

- The End Gun Areas use the "Current Distance" from Home for activation.

```
EG AREAS
DISABLE
```

If ENABLED

```
EG AREAS
ENABLE
PATH TO SET EG
PATH1
```

Path 1

```
RT EG PATH1 AREA 1
----- ft    TO    ----- ft
RT EG PATH1 AREA 2
----- ft    TO    ----- ft
```

```
LF EG PATH1 AREA 1
----- ft    TO    ----- ft
LF EG PATH1 AREA 2
----- ft    TO    ----- ft
```

Path 2

```
RT EG PATH2 AREA 1
----- ft    TO    ----- ft
RT EG PATH2 AREA 2
----- ft    TO    ----- ft
```

```
LF EG PATH2 AREA 1
----- ft    TO    ----- ft
LF EG PATH2 AREA 2
----- ft    TO    ----- ft
```

Path 3

```
RT EG PATH3 AREA 1
----- ft    TO    ----- ft
RT EG PATH3 AREA 2
----- ft    TO    ----- ft
```

```
LF EG PATH3 AREA 1
----- ft    TO    ----- ft
LF EG PATH3 AREA 2
----- ft    TO    ----- ft
```

Path 4

```
RT EG PATH4 AREA 1
----- ft    TO    ----- ft
RT EG PATH4 AREA 2
----- ft    TO    ----- ft
```

```
LF EG PATH4 AREA 1
----- ft    TO    ----- ft
LF EG PATH4 AREA 2
----- ft    TO    ----- ft
```

Note: STT can be pivoted, but cannot run with water when pivoting so don't allow EG angle areas to be set while pivoting if STT.

If ULTRA G2 Tractor Type set in 6=System Setup,
and 1=Run Setup set to Pivot Mode
and "Encoder for Pivot" set to YES,
Then allow 2 EG Areas to be set by angle the Encoder is reading:

```
RT END GUN     AREA 1
--- deg   TO   --- deg
RT END GUN     AREA 2
--- deg   TO   --- deg
```

11.5. 5 = DATE & TIME

```
RUN TIMES        HR:M:S
CURR        00013:29:41
USER        00106:07:13
TOTAL       01890:52:58
```

```
RESET USER RUN
TIME TO ZERO
NO
```

```
THU
08/03/17
12:52:43 P
```

```
SET DATE      08/03/17
                   THU
SET TIME         11:00
SET AM/PM            P
```

11.6. 6 = SYSTEM SETUP

```
UNITS
English
IRRIGATED WIDTH
01285           Ft
```

```
WATER SUPPLY
Hose Drag
WATER FLOW
0800            GPM
```

```
LEFT (TRACTOR) GEAR
75:1
LEFT (TRACTOR) TIRE
23.5        In Roll Radius
```

```
RIGHT GEAR
68:1
RIGHT TIRE
23.5        In Roll Radius
```

```
MAX SPD         060 IPM
MIN APPL        0.15 in
```

```
SPD ADJ RATE LEFT
0.016  sec/IPM
SPD ADJ RATE RIGHT
0.014  sec/IPM
```

```
ANALOG WATER PRESS
Yes
IF YES, FULL SCALE
100  PSI
```

```
LOW WATER PRESS
12 PSI          0:30 m:s
INITIAL TIME
15:00
```

```
WATER FLOW METER
Yes
IF YES
1.25   GAL/PULSE
```

```
LINEAR TYPE
Single Tower Tractor
```

11.7. 7 = SAFETY HISTORY

```
SAFETY HISTORY
IS EMPTY
```

```
SAFETY HISTORY
VIEW
```

If View

```
S1   XTE SAFETY
     03/27/17  11:22A
S2   DISTANCE STOP
     02/19/17  8:48P
```

Etc., Enter or Arrow Down. Can view last 8 Safeties.

If Save to USB

```
SAFETY HISTORY
SAVE TO USB
USB INSERTED
NO
```

```
NO USB DETECTED
(PRESS ENTER)
```

```
SAVE SAFETY HISTORY
(PRESS ENTER)
```

Save Format on USB:

- File will be saved with number & date for filename and ".saf" ext

  Example = 01062417.saf (1st file saved on June 24, 2017)

No.	Message	Date	Time	Curr Dist
S1	XTE SAFETY	03/27/17	11:22A	00512 ft
S2	DISTANCE STOP	02/19/17	08:48P	02000 ft

11.8. 8 = GPS PATH

11.8.1 GPS Path Waypoints Overview

If PLC SYSTEM TOGGLE in RUN, no access.

- Each Path must have a Home & a Forward Point.
- Home & Forward Waypoints MUST BE SET FIRST.
- Then set Intermediate Waypoints if used.
- Waypoints are entered through the PLC III, stored in the GPS Rover.
- Home & Forward Waypoints can be altered by going back to "Key In".
- Complete Path can be erased in the ROVER.

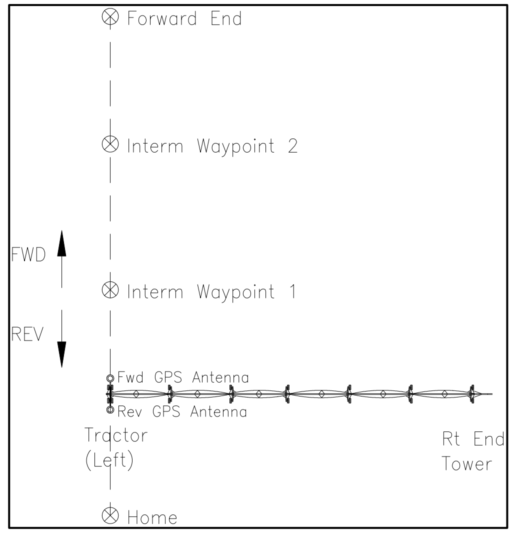


Figure 24. GPS Path Waypoints Overview

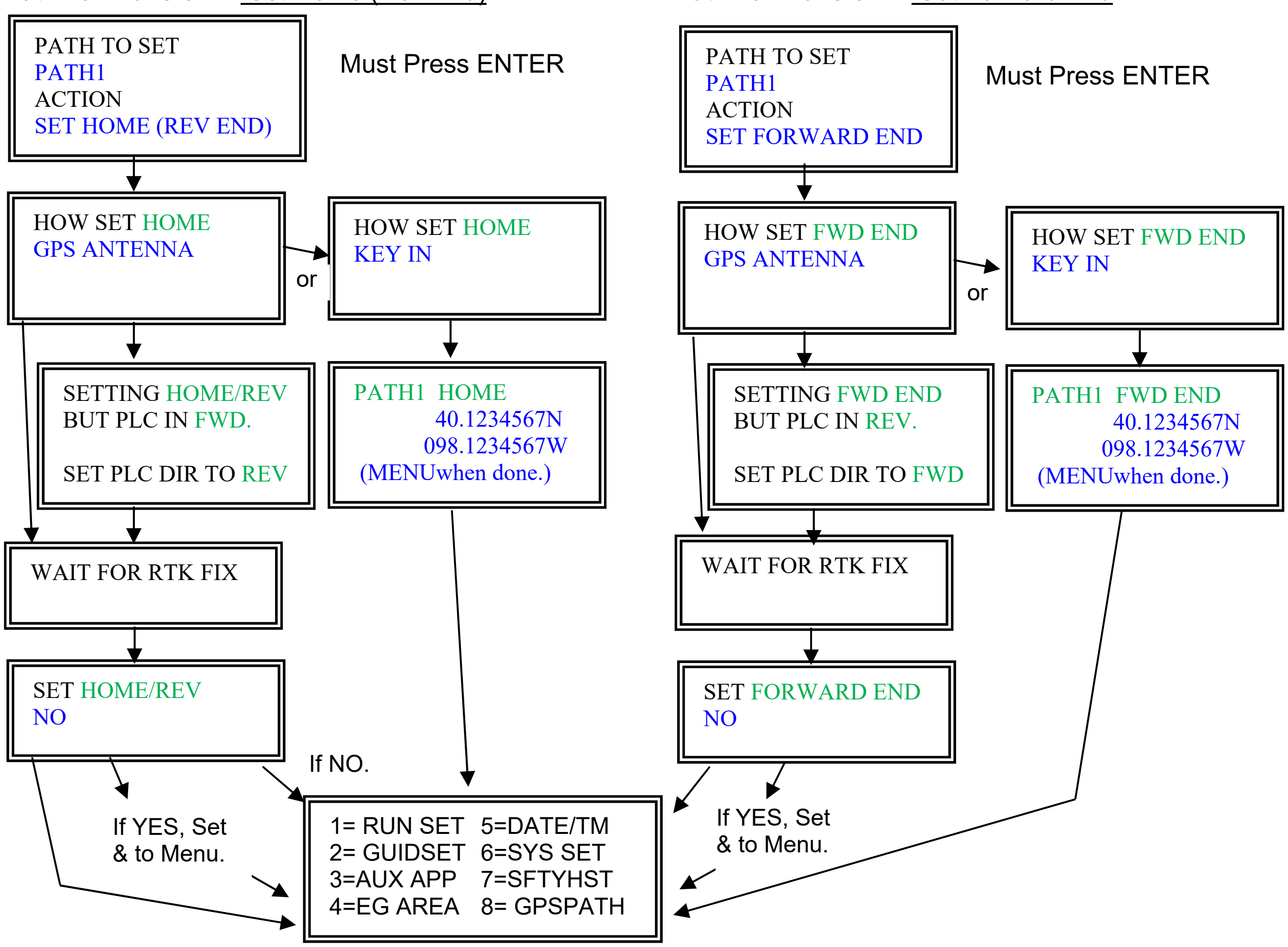
11.8.2 ACTION Options for PATH TO SET

Set Home (Rev End)
Set Forward End
Set Interm Waypoint
Erase Int Waypoint
Review Path

11.8.3 SET HOME and SET FORWARD END Flowchart

Important: If Home or Forward points are changed, you must go back through 1=RUN SETUP to change the Fwd Distance set point.

Path1 or 2 or 3 or 4: <u>Set Home (Rev End)</u>   Path1 or 2 or 3 or 4: <u>Set Forward End</u>



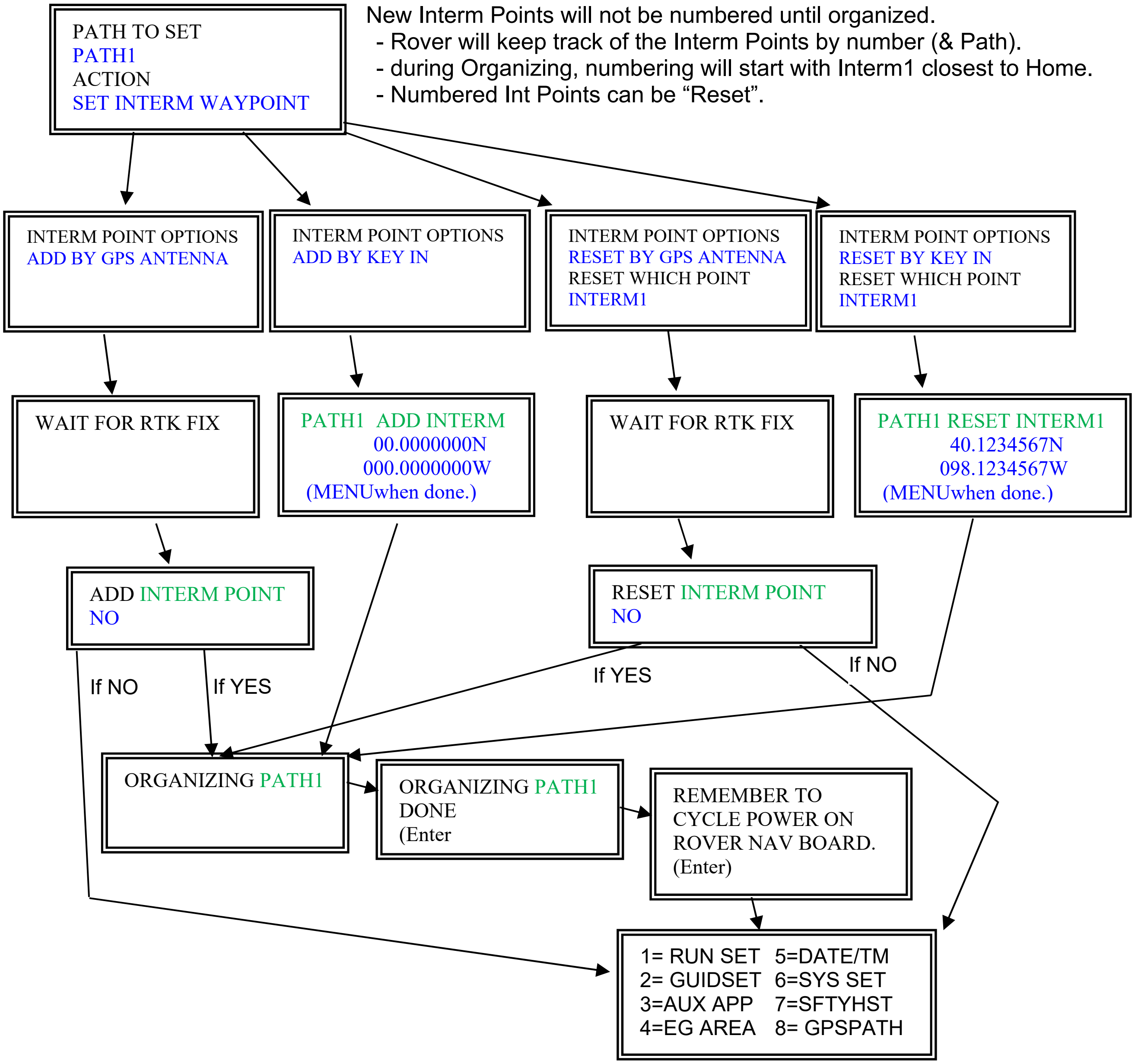
11.8.4 SET INTERM WAYPOINT Flowchart

NOTE: Home & Forward Waypoints MUST BE SET FIRST.
Entered through the PLC III, organized & stored in the GPS Rover Nav Bd.

Intermediate Waypoints, if set by GPS Antenna, will use the Antenna for whichever direction the PLC is set for.

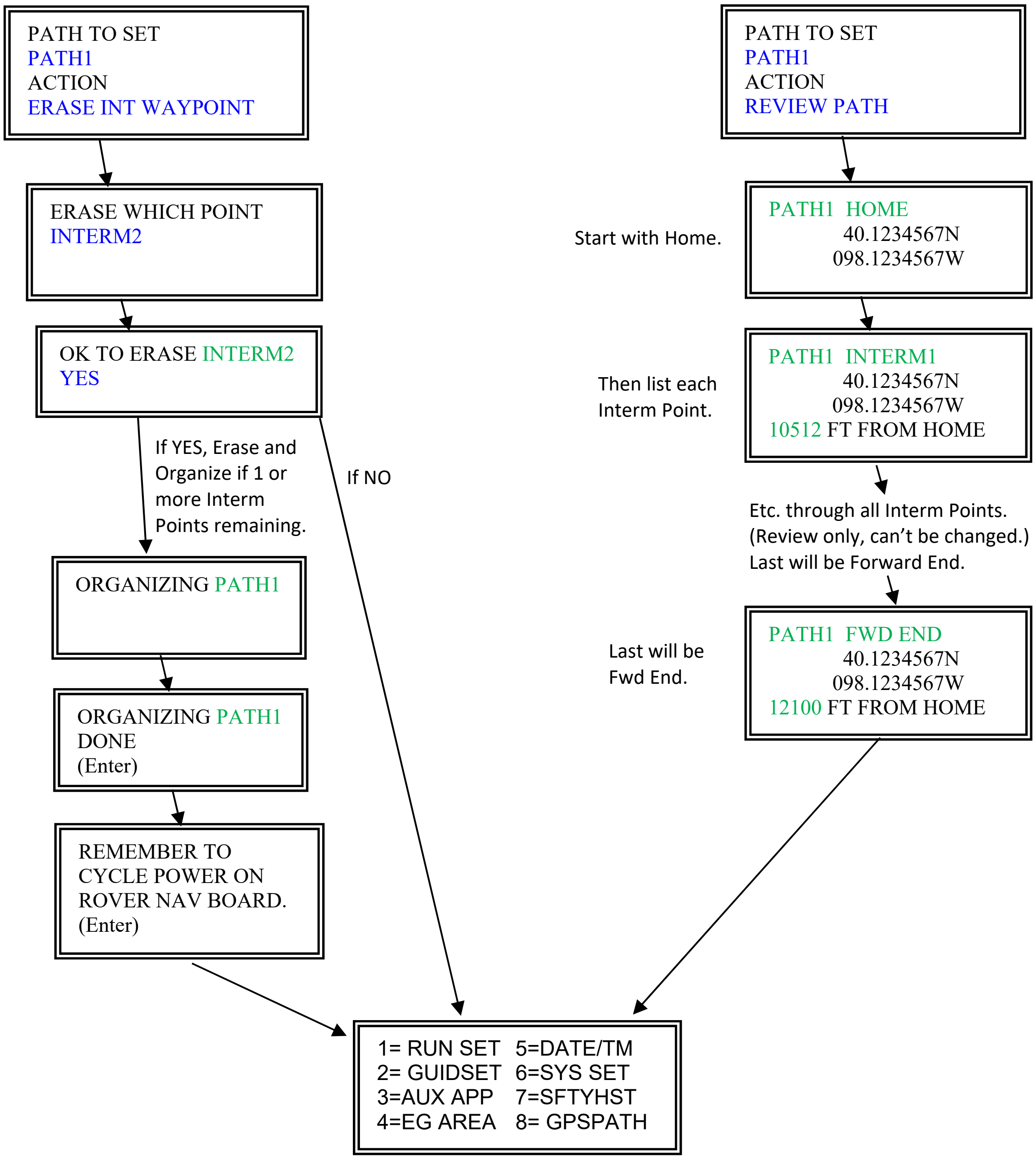
- Existing Points can be "Reset" by either GPS Antenna or Key-In

New Interm Points will not be numbered until organized.
- Rover will keep track of the Interm Points by number (& Path).
- during Organizing, numbering will start with Interm1 closest to Home.
- Numbered Int Points can be "Reset".



Repeat for Path2, Path3, & Path4

11.8.5 ERASE INT WAYPOINT and REVIEW PATH Flowchart



Repeat for Path2, Path3, & Path4

11.9. 9 = TEST OUTPUTS

If System Toggle in RUN:

```
SYSTEM RUNNING
        (Press MENU)
```

If System Toggle in STOP:

```
TEST OUTPUT
FWD
OFF
```

```
TEST OUTPUT
REV
OFF
```

```
TEST OUTPUT
END GUN
OFF
```

```
TEST OUTPUT
+ SPEED
OFF
```

```
TEST OUTPUT
- SPEED
OFF
```

```
TEST OUTPUT
SAFETY RELAY
OFF
```

```
COM LOOP TEST
NO
```

11.10. 10 = LOGGING

Allows data logging of several operational parameters. Must have a USB inserted on the PLC III Board.
NOTE: If power is lost while data logging, the data logging will resume at next power up.

To **START** Logging:

DATA LOGGING START USB INSERTED NO	Start or Stop Data Logging If Start, show "USB INSERTED" Make sure USB is inserted and recognized.
NO USB DETECTED (PRESS ENTER)	If USB Inserted is Yes, but No USB detected, press Enter. Will go back to Setup Group Data Logging.
LOGGING INTERVAL 05 SECS	Set data logging interval. (01 to 60 secs) - default is 5 secs
LOGGING ACTIVE (Enter)	Press Enter to exit to Main Dir Screen. **- Logging will start only after Enter is pressed.**

Start Logging – regardless of System Switch - Run Stop Position.

At Status Screen 1, the Direction and word "Log" will flash alternately when Data Logging.

FWD APPL 0.75" AS EG L R STEERING = L1:R1 AO	**LOG** APPL 0.75" AS EG L R STEERING = L1:R1 AO

To **STOP** Logging, access the Data Logging Setup Group and Stop and press Enter.

9=TESTOUT 10=LOG	
DATA LOGGING STOP	Start or Stop Data Logging

File Format for Data Logging to USB:
- File will be saved with number & date for filename and ".log" ext
  - Example: 01072816.log = 1st file logged on July 28, 2016
- Transfer from the USB to a computer, open in Notepad to review.
  - or the file can opened in Excel (Delimited, Tab Delimited, General Column Format)

PLC III used with GPS Navigation
- Items to Log: (most of this is from the PLC Status Screens)

DATE	TIME	DIR	APPL	CURR DIST	REV SET DIST	FWD SET DIST	LFT SET IPM	LFT ACT IPM
RT SET IPM	RT ACT IPM	ANGLE/mV/XTE	END TOWARDS	OVRWTR TIMER	UTC			

12. STEERING OPERATION - STT

12.1. Steering Operation Overview for Single Tower Tractor

- Left End of Linear System will always be the Left End, regardless of direction of travel.
- in Forward, the Forward Antenna will be used, from RF Relay.
- in Reverse, the Reverse Antenna will be used, from RF Relay.

See the Next Page for corresponding PLC III Status Screen 6 information for Examples 1 to 4 below.

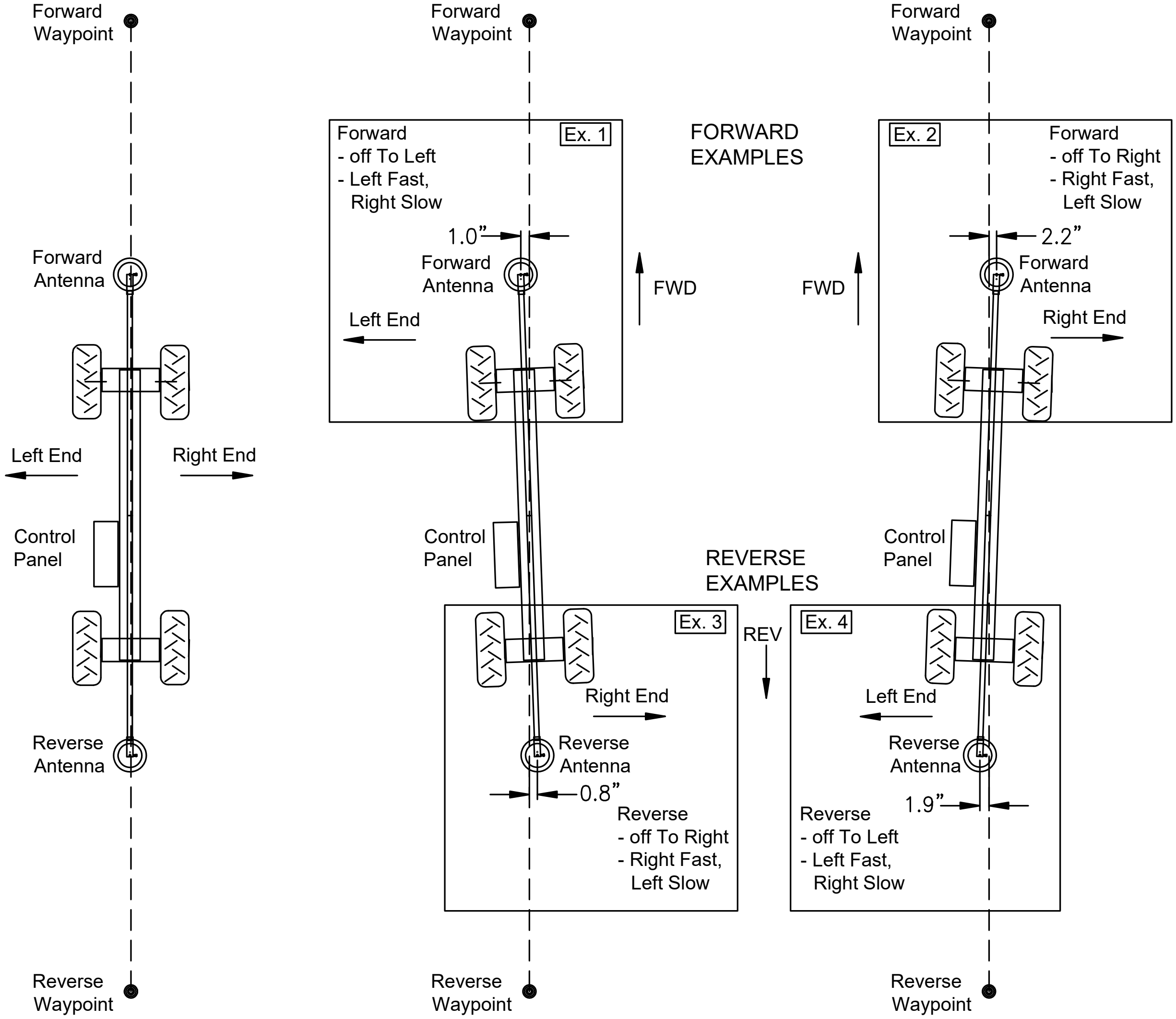


Figure 25. GPS Navigation Steering - STT.

12.2. Steering Examples

PLC / GPS Nav Steering:

- the side the GPS Antenna is off towards will always be the faster side for steering.

Steering Example 1

- System moving Forward
- Steering Setpoints 2:1 = 0.5”, 3:1 = 1.5”
- Forward Antenna is to the Left by 1.0”
- Left End Fast & Right End Slow by 2:1 (L2:R1)
- Continue Steer Right until under 0.5” or AntiOversteer

```
FWD XTE          01.0”

TOWARD LEFT END
SET018  L024  R012
```

Steering Example 2

- System moving Forward
- Steering Setpoints 2:1 = 0.5”, 3:1 = 1.5”
- Forward Antenna is to the Right by 2.2”
- Left End Slow & Right End Fast by 3:1 (L1:R3)
- Continue Steer Left until under 0.5” or AntiOversteer

```
FWD XTE          02.2”

TOWARD RIGHT END
SET018  L009  R027
```

Steering Example 3

- System moving Reverse
- Steering Setpoints 2:1 = 0.5”, 3:1 = 1.5”
- Reverse Antenna is to the Right by 0.8”
- Left End Slow & Right End Fast by 2:1 (L1:R2)
- Continue Steer Left until under 0.5” or AntiOversteer

```
REV XTE          00.8”

TOWARD RIGHT END
SET018  L012  R024
```

Steering Example 4

- System moving Reverse
- Steering Setpoints 2:1 = 0.5”, 3:1 = 1.5”
- Reverse Antenna is to the Left by 1.9”
- Left End Fast & Right End Slow by 3:1 (L3:R1)
- Continue Steer Right until under 0.5” or AntiOversteer

```
REV XTE          01.9”

TOWARD LEFT END
SET018  L027  R009
```

12.3. RF Relay Operation

- The RF Relay selects which GPS Antenna will be used by the GPS RTK Receiver Board.

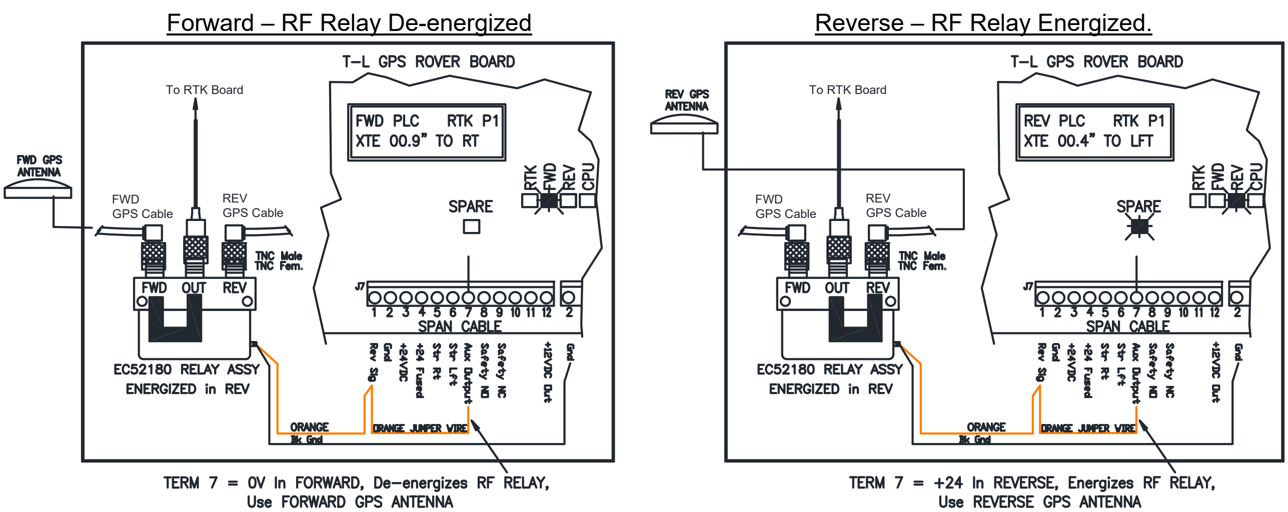


Figure 25.1 RF Relay Operation.

13. STEERING OPERATION – ULTRA G2

13.1. Steering Operation for ULTRA G2 – GPS NAV AT TRACTOR -“0” Span Position

The side the GPS Antenna is off towards will always be the faster side for steering.

- Examples: Off to the Left = L2:R1 or L3:R1. Off to the Right = L1:R2 or L1:R3.

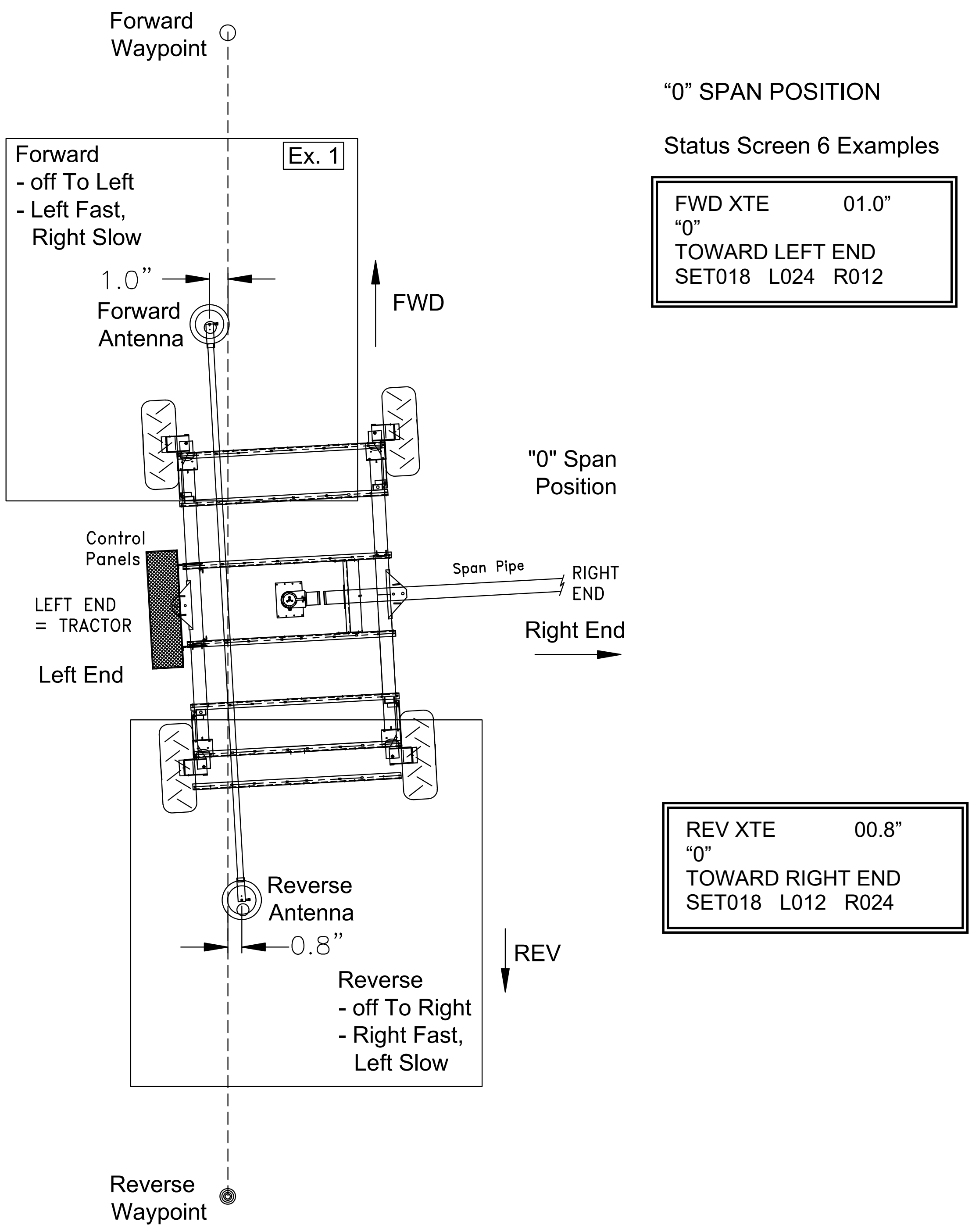


Figure 26. GPS Navigation Steering – Ultra G2 “0” Span Position.

13.2. Steering Operation for ULTRA G2 – GPS NAV AT TRACTOR – “180” Span Position

IMPORTANT: ULTRA G2, GPS NAV AT TRACTOR, 180 Span Position
Steering Response is different than “0” Position.
PLC III Status Screen 6 will show the End off Towards opposite of Rover Screen.

The side the GPS Antenna is off towards as shown on the PLC Screen will always be the faster side.

- Examples of PLC Screen:
  Off to the Left = L2:R1 or L3:R1.
  Off to the Right = L1:R2 or L1:R3.

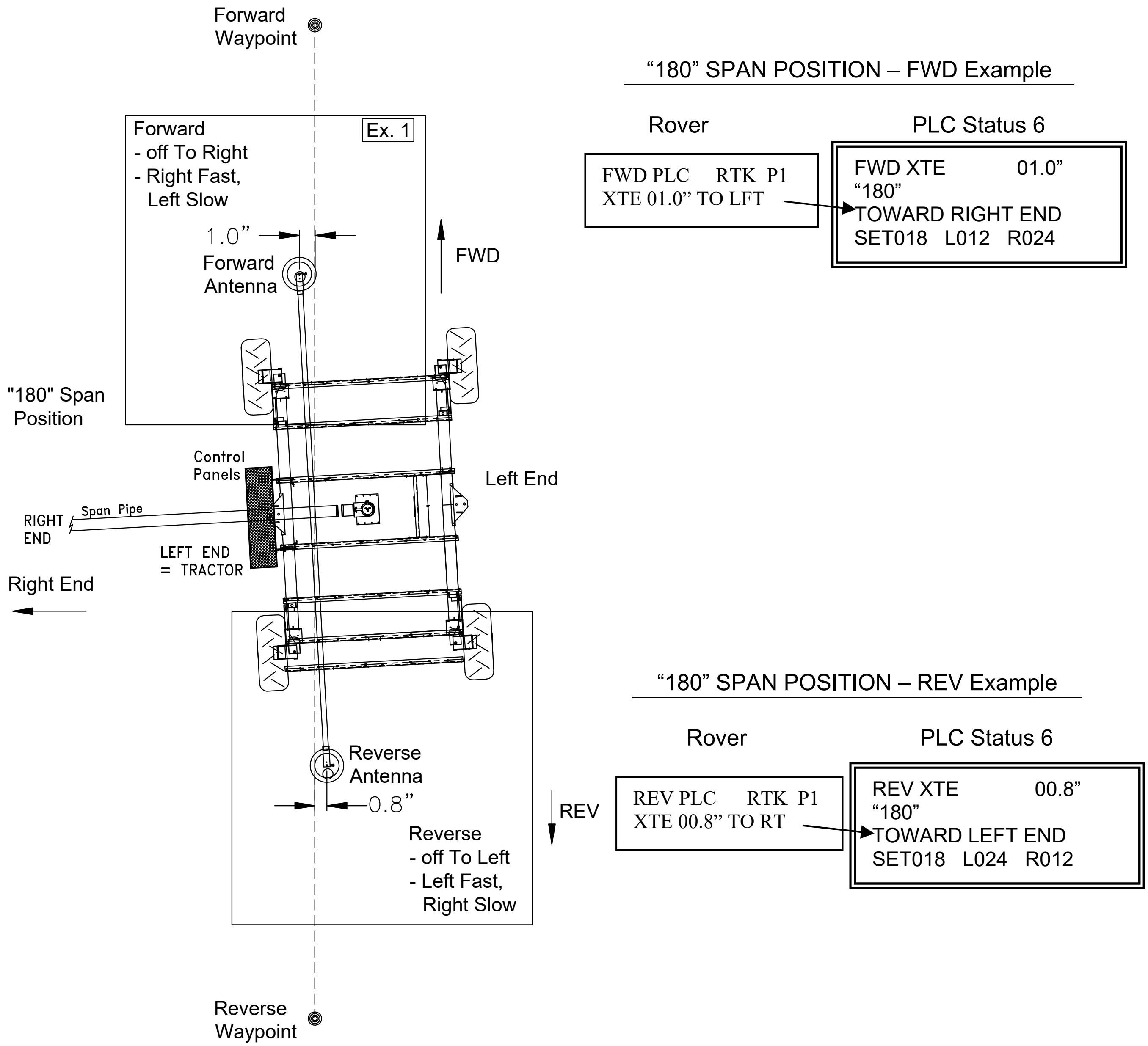


Figure 27. GPS Navigation Steering – Ultra G2 “180” Span Position.

13.3. Steering Operation for ULTRA G2 – GPS NAV AT CENTER – “0” vs. “180”

Need 2 Paths: Path 1 = “0” Position, Path 2 = “180” Position

180 Span Position – PLC Side off Towards will be opposite of the Rover.

- Rover Direction LED will be opposite of what is shown on the screen.

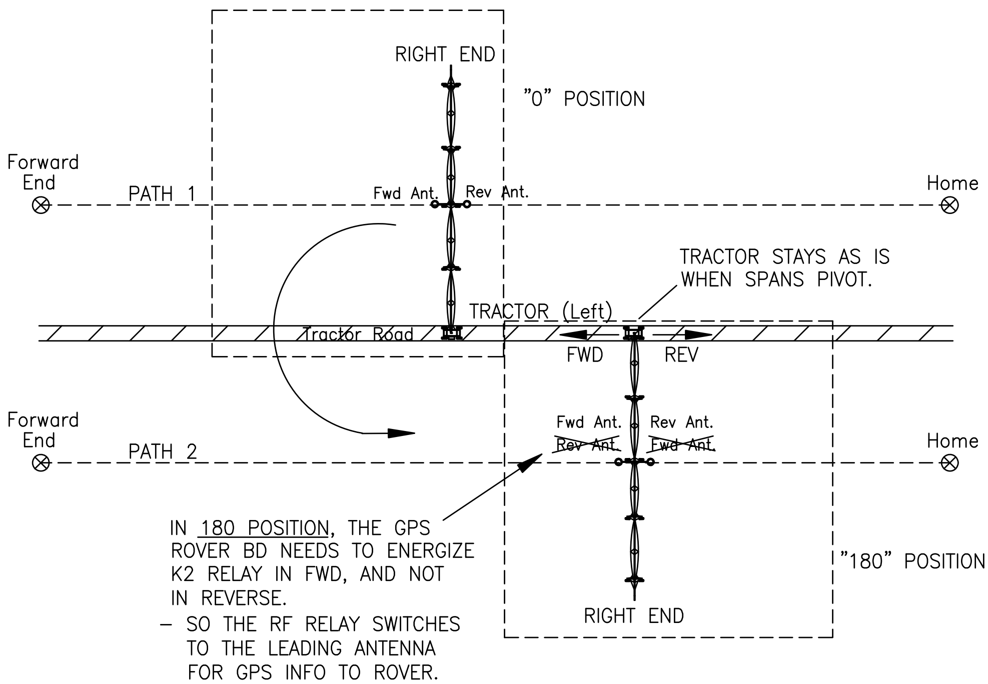


Figure 28. GPS Navigation Steering – Ultra G2 with GPS Navigation at Center.

ULTRA 2 - GPS NAV AT CENTER – Examples of Span Pos., Steering and GPS Antenna used:

Span Pos & Dir.	PLC III Board Main Dir.	PLC Status Screen #6	GPS Nav Board Screen	K2 Relay	Nav Bd LED
0 Fwd	FWD APPL 0.75” P2 AS EG L R STEERING = L2:R1	FWD XTE 01.5” “0” TOWARD LEFT END SET018 L024 R012	FWD PLC RTK P1 XTE 01.5” TO LFT	Off	FWD
0 Rev	REV APPL 0.75” P2 AS EG L R STEERING = L2:R1	REV XTE 01.5” “0” TOWARD LEFT END SET018 L024 R012	REV PLC RTK P1 XTE 01.5” TO LFT	On	REV

NOTE: In the **180 Position**, Side off To is opposite & the Direction LED on the Rover is opposite.

Span Pos & Dir.	PLC III Board Main Dir.	PLC Status Screen #6	GPS Nav Board Screen	K2 Relay	Nav Bd LED
180 **Fwd**	FWD APPL 0.75” P2 AS EG L R STEERING = L2:R1	FWD XTE 01.5” “180” TOWARD RIGHT END SET018 L024 R012	FWD PLC RTK P1 XTE 01.5” TO LFT	On	**REV**
180 **Rev**	REV APPL 0.75” P2 AS EG L R STEERING = L2:R1	REV XTE 01.5” “180” TOWARD RIGHT END SET018 L024 R012	REV PLC RTK P1 XTE 01.5” TO LFT	Off	**FWD**

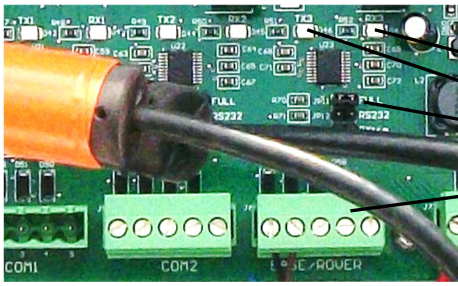
14. DATA RADIO OPERATION

The GPS Base Board needs to communicate and send correction information to the GPS Rover Board on the Linear. Data Radios are used to wirelessly send this information.
MUST BE LINE OF SIGHT BETWEEN BASE RADIO ANTENNA AND ROVER RADIO ANTENNA.

14.1. Communication LED's on Navigation Board for RTK Corrections

Transmit (TX) and Receive (RX) LED's will flash on the respective boards when communicating.

Base	Rover	Description
TX3 →	RX3	Slow blink every couple secs - Base sending RTK corrections to Rover GPS Bd for RTK Fix. - Base Radio transmits to Rover Radio. - Base Radio must have good communication to the Rover Radio.



Rover = RX3
Base = TX3
Jumper on FULL

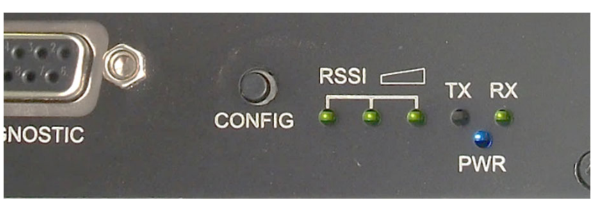
Measure the voltage on the two Comm wires (Rd & Bn) for Base/Rover – should be fluctuating between 3.3 and 0.0 VDC.

14.2. Base Radio LED's under Normal Operation

RSSI = Not Lit (Receive Signal Strength Indicator)
TX = Red Flashing (indicates the radio is transmitting)
RX = Not Lit
PWR = Solid Blue (radio has power)

14.3. Rover (Tractor) Radio LED's under Normal Operation

RSSI = Receive Signal Strength Indicator
Starts flashing furthest to left. As the received signal strength increases, the number of LED's lit will increase and become solid.
TX = Not Lit
RX = Solid Green (indicates the radio has received valid packets from the Base Radio)
PWR = Solid Blue (radio has power)



15. TROUBLESHOOTING BASE & ROVER

15.1. Base Safety Message Overview

If a Safety from the Base GPS Navigation Board has shut the system down, it will be displayed at start up the next time the Navigation Board is powered up. Possible safety message is:

Message:	DelayTime	Description
Recvr No Respond	300 seconds	Occurs if the GPS Nav Board cannot communicate with the GPS Receiver Bd in the first 300 seconds after power up.

15.2. Rover Safety Message Overview

If a Safety from the Rover GPS Navigation Board has shut the system down, it will be displayed at start up the next time the Navigation Board is powered up. Possible safety messages are:

Message:	DelayTime	Description
Recvr No Respond	300 seconds	Occurs if the GPS Nav Board cannot communicate with the GPS Receiver Bd in the first 300 seconds after power up. - Stops Communication to the PLC III if this safety occurs in the Rover. - Power Rover off and back on. Press Test/Enter to clear the safety in Rover. PLC III will show no comm to Rover if not cleared.
Lat Lon Chg	2 minutes	The Rover Lat-Long numbers have not changed for 2 minutes with the PLC Run-Stop System switch in RUN. - communicated to the PLC III and also displayed on the PLC III screen.
XTE Change Sfty (added in L39)	5 Minutes	The XTE value to 2 decimal places has not changed when in Run for 5 minutes. This safety will lock up the Rover Nav Board with no communication, and then the PLC will shut down with “Rover Com Lost”.
Recvr Comm Safety (added in HL41)	90 secs	Loss of Communication from the GPS Nav Board to Receiver Board after initial “Waiting for Receiver” 300 sec countdown and when in RUN. This safety will lock up the Rover Nav Board with no communication, and then the PLC will shut down with “Rover Com Lost”.

15.3. System Running in a Different Track than Previous

A. Check the Base Lat-Lon coordinates to see if they have changed. If they have changed, manually set the Base Lat-Lon back to the previous.
B. Check that the Base GPS Antenna has not been physically moved.

15.4. System Running in a Different Track in Fwd vs. Rev

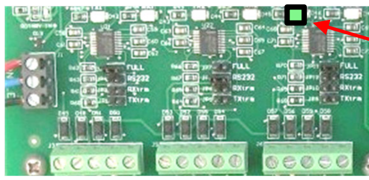
A. Adjust either the Fwd or the Rev GPS Antennas.

15.5. “No RTK Fix” Safety

1. Try cycling complete system power so the PLC and Rover are both rebooted.

2. See Section 14.1 to measure voltages on Com Port Terminals.

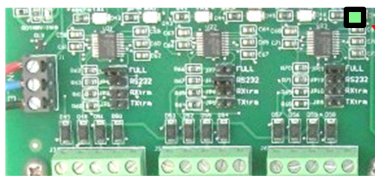
3. At the Base, check that the TX3 LED is flashing on the GPS Nav Board.
    a. If not flashing, check the number of satellites being received at the Base (Test Screen).



Base
TX3

4. At the Base, in Setup, check “GPS Receiver Info” to verify the GPS RTK Receiver Board is operating. This will ensure information is being transmitted from the small GPS RTK Receiver Board to the larger GPS Nav Board.

5. At the Rover, check that the RX3 LED is flashing on the GPS Nav Board.
    a. If not flashing, check the number of satellites being received at the Rover (Test Screen)
    b. If the system has not been powered for a couple of weeks or more it may take a couple minutes to start flashing. If powered recently, the RX3 should start flashing in about 30 seconds after power up.

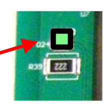


Rover
RX3

6. At the Rover, in Setup, check “GPS Receiver Info” to verify the GPS RTK Receiver Board is operating. This will ensure information is being transmitted from the small GPS RTK Receiver Board to the larger GPS Nav Board.

7. Check that the “P300 LED” is flashing intermittently on the GPS Nav Board at both Base & Rover.

P300 LED



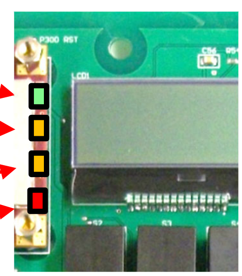
8. Check the LED’s on the GPS Receiver Board (under the shield).

`Green Flashing = Using Differential Lock.
Green Solid = Position is within a set limit.

Yellow Flashing = Getting Diff (WAAS) Lock
Yellow Solid = Has Differential Lock

Yellow = Antenna connected, has position (not RTK)

Red Solid = Power to Receiver Board



9. Check GPS Antenna and GPS Coax Cable Connections at Base & Rover.

10. Check Base and Rover Data Radios (Section 14).

11. Check the 2032 coin battery on the PLC Board, it should have at least 2.8VDC to 3.3VDC.
    b. A low battery can cause a longer time to obtain an RTK Fix, on both the Base & Rover.

12. At the Rover, strong wind inside the box with the lid open can cause a temperature change on the GPS Receiver Board which can cause a loss of Fix.

13. RTK Receiver Board Issue (Green Board and/or Red Board issue).
    a. Review Section 8.12 Set Correction at Base.
    b. If using P326 green boards, make sure firmware in them is 1.2Qq06t for Base & Rover.

16. TROUBLESHOOTING PLC III

16.1. Safety Messages caused by the External Safeties Wired to PLC III Board.

Safeties caused by switches or components wired into PLC Panel. The system must be moving with the SYSTEM Toggle Switch in RUN for the safety to be logged to the Safety History.

NOTE: Engine may not stay running, even with PLC Panel in OFF if one of these are safetied.

Safety	Cause	Checks
SECONDARY	Not used for GPS Navigation	
HYDR OIL LVL	A. Hydr Oil Level Switch in Hydr. Tank	A.1. Check for Gnd on Term 2 of SAFETY CONNS Block.
HYD OIL PRESS	A. Hydr Oil Press Switch in Hydr. Tank (1500 psi)	A.1. Check for Gnd on Term 3 of SAFETY CONNS Block. A.2. Check Hydr Press Run-Start Sw.
WATER PRESS	A. Water Press Sw. Gauge	A.1. Check for Gnd on Term 4 of SAFETY CONNS Block. A.2. Check Water Press Run-Start Sw.
ENGINE OIL	A. Engine Oil Pressure Gauge	A.1. Check for Gnd on Term 5 of SAFETY CONNS Block.
ENGINE TEMP	A. Engine Temp Gauge	A.1. Check for Gnd on Term 6 of SAFETY CONNS Block.
POSITN STOP	A. Position Stop Switch	A.1. Check for Gnd on Term 7 of SAFETY CONNS Block.
AUXILIARY	A. Tractor Guidance Ultra G2	A.1. Gnd on Be/Bk Wire from Tractor Pnl to Term 8 of SAFETY CONNS block. A.2. Primary or Proc Safety from Logic Bd. A.3. Check if Prox Switches seeing metal. A.4. Check Primary LED on Logic Board.
	B. Auxiliary Safety Switch	B.1. Check for Gnd on Term 8 of SAFETY CONNS Block.
AUTO STOP	A. AR/AS Switch	A.1. Check for +24 VDC on Term 7 of LEFT & RIGHT END TOWER block. A.2. Check End Tower Reversing Switch.

16.2. Existing Safety Messages caused by the PLC Board

Safeties caused by the PLC Panel, Set point exceeded or reached.

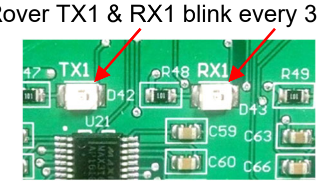
Safety	Cause	Checks
LEFT ET SENS	A. No pulses from Left ET Sensor Motor for nn sec. (Adjustable 30 to 120s)	A.1. Check Circuit Bd in ET Manifold for SENS PWR & SPD SIG. A.2. Check Term 5 of LEFT END TOWER Block in PLC for pulses. A.3. Check back of PLC Bd for LED SPD PULSE.
RIGHT ET SENS	A. No pulses from Right ET Sensor Motor for nn sec. (Adjustable 30 to 120s)	Same as LEFT ET SENS above, - except check Right side.
ANALOG WATER PRESS (if used)	A. Low Analog Water Press. (0-100 psi Sensor)	A.1. Check settings in SYSTEM SETUP. A.2. Check Water Pressure reading on STATUS SCREEN 3. A.3. Check Water Press Run-Start Sw.
PRIMARY	Not used for GPS Navigation	
REFERENCE	Not used for GPS Navigation	
GUID BD COM FAULT	Not used for GPS Navigation	
ETSEN PRI REF TIMEOUT	Not used for GPS Navigation	
LOW VOLTAGE	A. Engine Battery voltage dropped below 10.5 VDC with SYSTEM toggle in RUN (10 second delay).	A.1. Check "+12 IN" Orange wire of J12 Power Connector to PLC Bd.
DISTANCE STOP	A. CURR DIST = SET DIST	A.1. Check settings in RUN SETUP.

If using GPS Navigation:

Safety	Cause	Checks
NO RTK FIX	A. RTK Fix not established in time set in 2=Guid. Setup. (5 min. typical) - At start, and when running. - Will not move at Start when "Waiting for Fix". - Once running and Fix lost, will continue to run for Delay time in 2=Guid Setup.	A.1. Check RTK Fix Delay Time setting. in Guid Setup. A.2. Try cycling power to the Rover Board. A.3. Verify good Comm Base to Rover.
XTE SAFETY (Track Error)	A. XTE reading has exceeded the setting in 2=Guid Setup for 10 secs. - XTE Safety not active when "Waiting for Fix".	A.1. Check for proper steering from PLC. A.2. Check for field condition that may affect steering, such as deep wheel tracks or poor system alignment. A.3. Check that the Base Coordinates have not changed.
ROVER COM LOST	A. Communication cannot be established with the Rover. (70 sec. delay v257, was 10)	A.1. Check Comm 1 TX and RX LED's at both the PLC III and the Rover. B.1. Check the Rover Screen to see if it has a "Lat Lon Chg" or other Safety. If so, press Enter to clear the Rover safety then cycle power to the PLC III and try to run the system.

PLC RX1 changes every 3 secs.

Rover TX1 & RX1 blink every 3 secs.



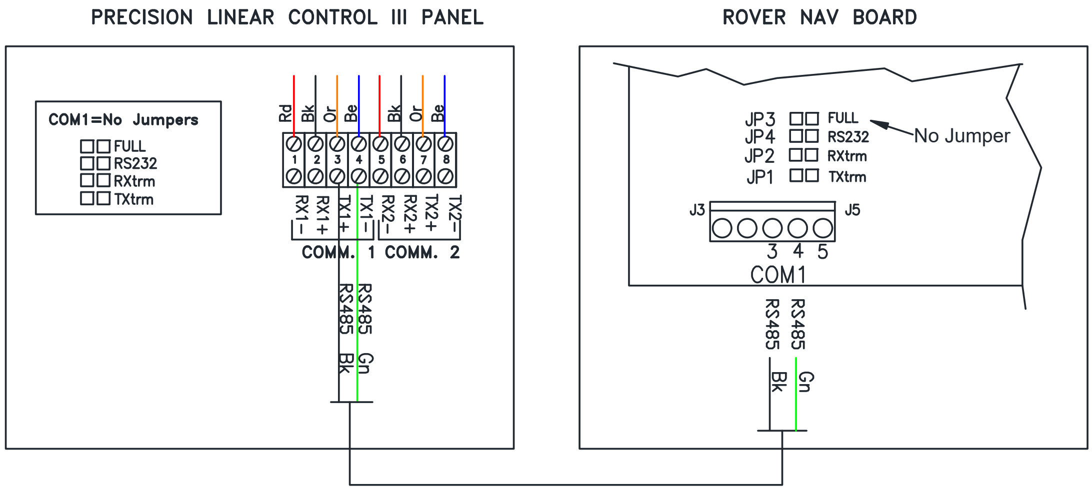
Safety	Cause	Checks
TRACTR OVRWTR (GPS Nav)	A. Tractor has not moved the set distance in the set time from #1=RunSetup.	A.1. Check if Tractor is stuck. A.2. Check settings, lengthen out? - 10 IPM or less set to 10 minutes, 40 ft.
ETSEN FIX TRACK SFTY	A. ETSEN TRACK FIX SFTY Disabled in RUN SETUP with 10 minute timeout.	A.1. 10 min. time out once disabled. (but timer can be reset.) See Operation.
LAT LON CHG	A. From Rover	A.1. See Rover Troubleshooting, Sec. 15.2.

Additional Safeties if Ultra G2 Tractor Type:

Safety	Cause	Checks
ENCODER COM FAULT (If Encoder used Pivot Mode)	A. No Comm with Encoder in Coll. Ring.	A.1. Check wiring of Encoder. A.2. Check COM3 jumper set to FULL on back of PLC Panel.
OVERWATER SAFETY (If Encoder used Pivot Mode)	A. Pivot Mode – Angle did not increase or decrease 1 deg. in set time.	A.1. Check setting in RUN SETUP. A.2. Check if End Tower is stuck. A.3. Check speed of system vs. setting. A.4. Check for loose connections, if encoder is actually rotating.

16.3. Communication Issue Between PLC III and Rover

Step 1: Verify Wiring and Jumper Settings on Com 1

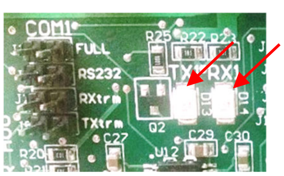


Step 2: Verify Firmware in both PLC III and Rover Nav Board are up to date.
- Check Dealer Point for latest firmware.
- If update needed then Reprogram, Restore Settings, etc. per Sections 6 & 7.

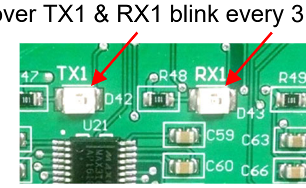
Step 3: See Section 9.2 and follow the Step by Step instructions. Follow <u>exactly</u>.

Step 4: Verify Comm 1 LED action at PLC III and Rover Boards.
- About 20 secs after power up, the PLC III TX1 LED will blink every sec. or so a few times.
- Then the LED's should blink as follows:

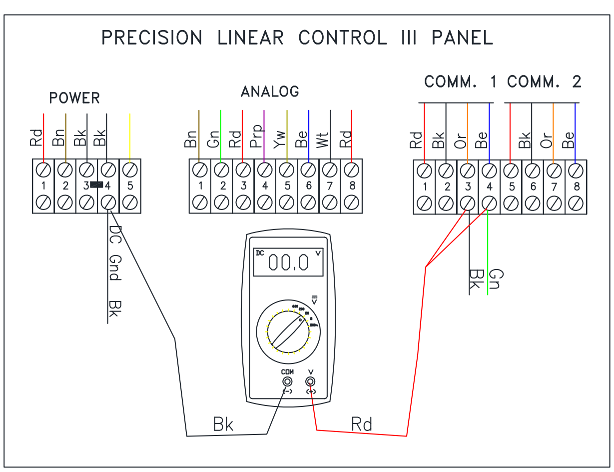
PLC TX1 solid, RX1 blinks every 3 secs.



Rover TX1 & RX1 blink every 3 secs.



Step 5: Verify voltage readings on Terminals 3 & 4 of Com 1 at PLC III and Rover.
- Cycle Power to the PLC III which should also cycle power to the Rover.
- Measure voltages while boards are powering up and after.



<u>Cycle Power and Measure Voltages</u>

<u>Voltage on Term 3 (PLC or Rover)</u>

1st 20 secs after powered	~3.3VDC
~20 secs and thereafter	~0.7VDC
Then every 3 secs	~0.9 - 1.5 VDC Pulses higher.

<u>Voltage on Term 4 (PLC or Rover)</u>

1st 20 secs after powered	~0.0VDC
~20 secs and thereafter	~0.7VDC
Then every 3 secs	~0.9 - 1.5 VDC Pulses higher.

<u>Unplug Com 1 at Rover, or unwire 3 & 4 at PLC III:</u>
Pins 3 & 4 on Rover = 0.9VDC, pulses higher.
Pins 3 & 4 on PLC III = 0 VDC, does not pulse.

17. PLC III SUMMARY SCREENS

1 = RUN SETUP (Single Tower Tractor set in 6=System Setup)

Item	Setting	Choices/Units
Linear Mode		
FWD (AS or AR and distance)	______	(AS or AR) at ________ ft (Ditch only AR)
REV (AS or AR and distance)	______	(AS or AR) at ________ ft (Ditch only AR)
CURR (Current) Distance	______	ft
AR Delay Timer	______	min. (Ditch only)
ET Sensor Safety Time	______	sec.
ETSEN FIX TRACK Sfty	______	Enable or Disable
MFC Valves Adjusting if Disabled		
Operating Mode	Linear	Linear or Pivot Mode
Low Power Safety	______	Enable/Disable
If GPS Navigation:		
GPS Path to Use	______	Path1, Path2, Path3, or Path4
Tractor Overwater Timer	______	Enable or Disable
If “Enable”, settings:	______	ft (or m) (distance)
	______	min (time)
Pivot Mode		
Operating Mode	Pivot	Linear or Pivot Mode
Dist to End Tower	______	Ft
End Tower Stop Set?	______	Yes or No
Low Power Safety	______	Enable/Disable

1 = RUN SETUP (ULTRA G2 Tractor set in 6=System Setup)

Item	Setting	Choices/Units
LINEAR Mode:		
Operating Mode	Linear	Linear or Pivot Mode
Span Position	______	0 or 180
FWD (AS or AR and distance)	______	(AS or AR) at ________ ft (Ditch only AR)
REV (AS or AR and distance)	______	(AS or AR) at ________ ft (Ditch only AR)
CURR (Current) Distance	______	ft
AR Delay Timer	______	min. (Ditch only)
ET Sensor Safety Time	______	sec.
ETSEN FIX TRACK Sfty	______	Enable or Disable
MFC Valves Adjusting if Disabled		
Low Power Safety	______	Enable/Disable
Rotate Ultra Tractor 90 Degrees	______	Disable or Enable
If Enable Rotate Tractor 90 Deg:		
Wheels Set to Rotate	______	Yes or No
2 Gears Disengaged	______	Yes or No
Switch Tractor Coils	______	Yes or No
Tractor Guidance Okay	______	Yes or No
If GPS Navigation at Ult Trctr or Ult Cntr:		
GPS Path to Use	______	Path1, Path2, Path3, or Path4
Tractor Overwater Timer	______	Enable or Disable
If “Enable”, settings:	______	ft (or m) (distance)
	______	min (time)
PIVOT Mode:		
Operating Mode	Pivot	Linear or Pivot Mode
Dist to End Tower	______	ft
Encoder for Pivot	______	Yes/No
If No, “Must Set Stops”		
CW AS/AR	______	AS or AR
CCW AS/AR	______	AS or AR
AR Delay	______	min.
If Yes, “Enter Degrees”		
Current Angle	______	deg
Angle to Pivot To:	______	deg
Overwater Timer	______	Enable or Disable
Time to Move 1 Deg	______	min
ET Sensor Safety Time	______	sec.
Low Power Safety	______	Enable/Disable

2 = GUIDANCE SETUP (suggested)

Item		Example	Setting	Choices/Units
Guidance Type	If STT		______	GPS Navigation
	If ULTRA G2		______	GPS Nav at Ult Trctr or GPS Nav at Ult Cntr

If STT GPS NAVIGATION or ULTRA GPS NAV AT TRACTOR or ULTRA GPS NAV AT CENTER:

Waiting for Comm, then Communication OK to GPS Rover or Cannot Communicate with GPS Rover

Item	Example	Setting	Choices/Units
Steering Sig Time	*10.0*	______	sec
2:1 XTE Steering Level	*0.50*	______	in.
3:1 XTE Steering Level	*1.50*	______	in
Anti Ovrstr Out	*2.00*	______	in
Anti Ovrstr In	*1.50*	______	in
Anti Ovrstr Back Out	*0.50*	______	in
Track Error Zone	*12*	______	in
RTK Fix Delay	*05*	______	min

3 = AUX APPL

FOR STT OR ULTRA G2, IF ANY GPS NAVIGATION METHOD:

AUX APPL ___________ Enable/Disable

Path1

Fwd Aux1 _____ in. _______ ft To _______ ft
Fwd Aux2 _____ in. _______ ft To _______ ft
Rev Aux1 _____ in. _______ ft To _______ ft
Rev Aux2 _____ in. _______ ft To _______ ft

Path2

Fwd Aux1 _____ in._______ ft To _______ ft
Fwd Aux2 _____ in._______ ft To _______ ft
Rev Aux1 _____ in._______ ft To _______ ft
Rev Aux2 _____ in._______ ft To _______ ft

Path3

Fwd Aux1 _____ in. _______ ft To _______ ft
Fwd Aux2 _____ in. _______ ft To _______ ft
Rev Aux1 _____ in. _______ ft To _______ ft
Rev Aux2 _____ in. _______ ft To _______ ft

Path4

Fwd Aux1 _____ in._______ ft To _______ ft
Fwd Aux2 _____ in._______ ft To _______ ft
Rev Aux1 _____ in._______ ft To _______ ft
Rev Aux2 _____ in._______ ft To _______ ft

4 = EG AREAS

FOR STT OR ULTRA G2, IF ANY GPS NAVIGATION METHOD:

Item		
EG AREAS	______	Enable/Disable
Path to Set EG	______	Path1, Path2, Path3, Path4

Path1

RT EG PATH1 Area1 _______ ft To _______ ft
RT EG PATH1 Area2 _______ ft To _______ ft
LF EG PATH1 Area1 _______ ft To _______ ft
LF EG PATH1 Area2 _______ ft To _______ ft

Path2

RT EG PATH2 Area1 _______ ft To _______ ft
RT EG PATH2 Area2 _______ ft To _______ ft
LF EG PATH2 Area1 _______ ft To _______ ft
LF EG PATH2 Area2 _______ ft To _______ ft

Path3

RT EG PATH3 Area1 _______ ft To _______ ft
RT EG PATH3 Area2 _______ ft To _______ ft
LF EG PATH3 Area1 _______ ft To _______ ft
LF EG PATH3 Area2 _______ ft To _______ ft

Path4

RT EG PATH4 Area1 _______ ft To _______ ft
RT EG PATH4 Area2 _______ ft To _______ ft
LF EG PATH4 Area1 _______ ft To _______ ft
LF EG PATH4 Area2 _______ ft To _______ ft

If Ultra Tractor and set to Pivot Mode WITH ENCODER:

RT End Gun Area 1 _____ deg TO _____ deg
RT End Gun Area 2 _____ deg TO _____ deg

5 = DATE/TIMES

Item		Setting	Choices/Units
Screen 1		N/A	Display the Run Times
Record Times:	CURR	______	HH:MM:SS
	USER	______	HH:MM:SS
	TOTAL	______	HH:MM:SS
Screen 2	Reset User Time to Zero	______	Yes or No
Screen 3		N/A	Display current date & time, Hr:Min:Sec
Screen 4		______	Enter current date & time.

6 = SYSTEM SETUP (suggested)

Item	Example	Setting	Choices/Units
Units	*English*	______	English or Metric
Irrigated Width	*1285*	______	Ft
Water Supply	*Hose*	______	Hose Drag or Ditch Feed
Water Flow	*800*	______	GPM
Left (Tractor) Gear	*75:1*	______	75:1. 68:1, 24:1
Left (Tractor) Tire	*23.5*	______	11.0 = 20.0, 14.9 = 23.5, 16.9=24.0, 11.2x38=27.0, 12.4x38R = 28.5
Right Gear	*68:1*	______	75:1. 68:1, 24:1
Right Tire	*23.5*	______	same as Left
Max Speed	*60*	______	IPM
Min Appl.	0.15”	______	in. (calculated)
Spd Adj Rate Left	*.016*	______	0.001 to 0.050 sec/IPM
Spd Adj Rate Right	*.014*	______	0.001 to 0.050 sec /IPM
Analog Water Pressure	*Yes*	______	Yes/No
Full Scale	*100*	______	0 to 200 (Max Pressure of 4 to 20 mA)
Low Water Press	*12*	______	psi
	00:30	______	mm:ss
Initial Delay	*15:00*	______	mm:ss
Water Flow Meter	*Yes*	______	Yes/No
GPM/Pulse	*1.25*	______	
Linear Type		______	Single Twr Tractor or Ultra G2

7 = SAFETY HISTORY

8 = GPS Path (GPS Navigation)

Item	Setting	Choices/Units
Path to Set	______	Path 1, Path 2, Path 3 or Path 4
What to Set	______	Home or Forward End or Interm Waypoint
Home or Forward Waypoints		
How Set?	______	GPS Antenna or Key In
GPS Antenna		
Setting Home but PLC in Fwd or Setting Fwd but PLC in Rev		
Wait for Fix		
Set Home or Fwd End	______	Yes or No
“Key In”, enter coordinates:		

Path1 Home & Fwd			Path2 Home & Fwd		
Home Lat Long	______________	N or S	Home Lat Long	______________	N or S
	______________	E or W		______________	E or W
Fwd End Lat Long	______________	N or S	Fwd End Lat Long	______________	N or S
	______________	E or W		______________	E or W
Path3 Home & Fwd			**Path4 Home & Fwd**		
Home Lat Long	______________	N or S	Home Lat Long	______________	N or S
	______________	E or W		______________	E or W
Fwd End Lat Long	______________	N or S	Fwd End Lat Long	______________	N or S
	______________	E or W		______________	E or W

8 = GPS Path (continued)

Intermediate Waypoints

Path to Set	______	Path 1, Path 2, Path 3 or Path 4
What to Set	______	Interm Waypoint
Interm Point Options	______	Add by GPS Ant, Add by Key In, Reset by GPS Ant, Reset by Key In, Review or Erase
Add By GPS Antenna		
Wait for Fix		
Set Interm Point?	______	Yes or No (If Yes, then to Organizing.)
Add By Key In		
Enter Lat Long coordinates		Then to Organizing.
Reset By GPS Antenna		
Reset Which Point	______	Interm 1, Interm 2, etc up to Interm 9
Wait for Fix		
Reset Interm Point?	______	Yes or No (If Yes, then to Organizing.)
Reset By Key In		
Reset Which Point	______	Interm 1, Interm 2, etc up to Interm 9
Edit Lat Long coordinates		Then to Organizing.
Review		
nn Interm Points	Path1 complete Lat Long listing: Home Point Interm1 to Interm9 Points, *nnnnn* Ft from Home Fwd End Point, *nnnnn* Ft from Home	
Erase		
Erase Which Point	______	Interm 1, Interm 2, etc up to Interm 9
OK to Erase	______	Yes or No (If Yes, then to Organizing)

Path1 Interm Points

Interm 1 Lat Long	________________	N or S
	________________	E or W
Interm 2 Lat Long	________________	N or S
	________________	E or W
Interm 3 Lat Long	________________	N or S
	________________	E or W
Etc. up to Interm 9		

Path2 Interm Points

Interm 1 Lat Long	________________	N or S
	________________	E or W
Interm 2 Lat Long	________________	N or S
	________________	E or W
Interm 3 Lat Long	________________	N or S
	________________	E or W
Etc. up to Interm 9		

Path3 Interm Points

Etc. Intem 1 to Interm 9

Path4 Interm Points

Etc. Intem 1 to Interm 9

Complete Path Records

Path1

Home	____________	N or S
	____________	E or W
Interm1	____________	N or S
	____________	E or W
Interm2	____________	N or S
	____________	E or W
Interm3	____________	N or S
	____________	E or W
Interm4	____________	N or S
	____________	E or W
Interm5	____________	N or S
	____________	E or W
Interm6	____________	N or S
	____________	E or W
Interm7	____________	N or S
	____________	E or W
Interm8	____________	N or S
	____________	E or W
Interm9	____________	N or S
	____________	E or W
Fwd End	____________	N or S
	____________	E or W

Path2

Home	____________	N or S
	____________	E or W
Interm1	____________	N or S
	____________	E or W
Interm2	____________	N or S
	____________	E or W
Interm3	____________	N or S
	____________	E or W
Interm4	____________	N or S
	____________	E or W
Interm5	____________	N or S
	____________	E or W
Interm6	____________	N or S
	____________	E or W
Interm7	____________	N or S
	____________	E or W
Interm8	____________	N or S
	____________	E or W
Interm9	____________	N or S
	____________	E or W
Fwd End	____________	N or S
	____________	E or W

Path3

Home	____________	N or S
	____________	E or W
Interm1	____________	N or S
	____________	E or W
Interm2	____________	N or S
	____________	E or W
Interm3	____________	N or S
	____________	E or W
Interm4	____________	N or S
	____________	E or W
Interm5	____________	N or S
	____________	E or W
Interm6	____________	N or S
	____________	E or W
Interm7	____________	N or S
	____________	E or W
Interm8	____________	N or S
	____________	E or W
Interm9	____________	N or S
	____________	E or W
Fwd End	____________	N or S
	____________	E or W

Path4

Home	____________	N or S
	____________	E or W
Interm1	____________	N or S
	____________	E or W
Interm2	____________	N or S
	____________	E or W
Interm3	____________	N or S
	____________	E or W
Interm4	____________	N or S
	____________	E or W
Interm5	____________	N or S
	____________	E or W
Interm6	____________	N or S
	____________	E or W
Interm7	____________	N or S
	____________	E or W
Interm8	____________	N or S
	____________	E or W
Interm9	____________	N or S
	____________	E or W
Fwd End	____________	N or S
	____________	E or W

9 = TEST OUTPUTS

Fwd
Ref
End Guns
+ Speed
- Speed
Safety Relay
Com Loop Test

10 = LOG

Item	Setting	Choices/Units
Data Logging	______	Start or Stop
USB Inserted	______	Yes or No
Logging Interval	______	secs

18. GPS BASE INFORMATION

SYSTEM NAME: ______________________________

BASE Position

Lat-Long: ______________________N or S ______________________W or E

Correction Type __________ (RTCM3)

__________ Galileo Yes/No

Comments & Notes: